DATA MINING PROJECT

Master’s in data science and advanced Analytics

NOVA Information Management School

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1. EXECUTIVE SUMMARY

This report presents the results of a customer segmentation analysis conducted on AIAI's airline loyalty dataset. Building on insights from Phase 1, multiple segmentation perspectives of the customer were explored, including behavioral, value-based, and demographic. Arriving at a final solution that integrates the most meaningful insights into one segmentation framework that supports targeted marketing, loyalty program optimization, and personalized service strategies.

The analysis identifies 18 final customer segments, ranging from loyal travelers that flight consistency thought the year and only with companions on certain months to new travelers that only flight on holidays; each with distinct behavioral patterns, revenue potential, and strategic opportunities. These segments enable AIAI to better allocate resources, improve customer lifetime value, and enhance loyalty program effectiveness, enabling the creation of campaigns that do upsell to the frequent travelers that can afford it, awareness benefits for the newcomers, and provide point incentives to encourage travel during unusual flight patterns.

2. INTRODUCTION

In the highly competitive airline industry, customer retention and loyalty optimization are critical drivers of long-term profitability. Loyalty programs generate large volumes of customer data, which, when properly analyzed, can support advanced segmentation strategies and personalized decision-making.

Following the exploratory and data understanding phase (Phase 1), where it was combined the customers information with the flight's transaction data, into one dataset, see Annex1. It contains information regarding 16,572 customers, with variables that can be grouped into the following perspective:

- **Value-Based:** the overall value generated by the customer, categorizing customers according to their overall economic contribution, loyalty status, tenure, if they reengaged, recency, total number of flights, total distance traveled, percentage of flights with companions and total dollar points remaining.
- **Demographics:** categorizing customers by attributes like gender, education, income, location and marital status.
- **Behavioral:** purchasing habits and travel behaviors by months and by year.

This next phase focuses on applying clustering techniques to uncover meaningful customer segments within AIAI's loyalty program. The objectives of this phase are threefold:

1. Preprocess the data, compare multiple clustering approaches and do profiling within each perspective.
2. Decide on the best clustering approach within each perspective.
3. Integrate insights into a unified segmentation framework to create data-driven marketing campaigns tailored to each group of customers.

It is necessary to have in mind the following innings that were found on the Phase 1: some of the variables had different scales, so the use of a scaler to convert the different variable scales into one was necessary, because we want each one of the features to have the same importance when we implement the calculation of the distance for each cluster's solution; it was shown that some variables appear to have some extreme distributions (one dimensional outliers) so, will need to be take care of that also; and finally, since we created new variables the moment we joined the datasets on the first phase, we need to be careful when we select which variables to use on this second phase guaranteeing that they don't carry the same information (to prevent biasing our solution).

3. METHODOLOGY

It was established that for each perspective (value-based, demographic and behavioral) it was needed to preprocess the data, then test multiple clustering solutions, selecting only the best one at the end, that will be merged into the final unified framework that will serve as a base for creating the segmentation strategies.
In general terms, for to handle the outliers, we decided to apply a lower and upper limit on three times the IQR (interquartile range), that are considered extreme outliers, for the variables that do not have upper and lower bound, meaning that if the observation surpasses that limit their value will be change to 3xIQR. For missing values, they are imputed with 0 because they mean that there is no record of that variable, or those observations were grouped into a specific cluster of 'No_transactions' customers. Finally, the scaler that was used was a minmax scaler.

To test different clusters hyperparameters, we did the following:

- **Hierarchical:** it was using ward linkage, the dendrogram was plotted and decided the number of clusters based on the distance where the clusters were grouped but different to one another.
- **K-Means:** it was used the elbow method (seeing the evolution of inertia, SSW (total variance within clusters)), also seeing other metrics like the silhouette score, SSB (total variance between clusters), and R2 to decide.
- **K-Prototypes:** it was used the elbow method (seeing the evolution of cost, the total dissimilarity (or error) calculated using a mix of Euclidean distance for numerical data and Hamming distance), also seeing the silhouette score.
- **C-means:** a grid search was used looking at the FPC score (Fuzzy Partition Coefficient) and the FPE (Fuzzy Partition Entropy), testing different numbers of clusters (C), fuzziness values (M).

After deciding on the hyperparameters the clusters solutions receive their respective profiling, a two-dimension visualization was performed using UMAP on the merge dataset (using only the variables that were used for those clusters), and metrics like the SSW, SSB, R2, Silhouette score, Calinski-Harabasz score, and Davies-Bouldin score were compared to decide on the best clustering solution.

There were cases where some dimensionality reduction techniques were used to see if those improved the results, specifically it was tested:

- **PCA (principal component analysis)**: First it was founded the number of components that maintained at least one 80% of the explained variance, interpreted the new components, and then we build and compare clustering solutions using as input the PCA transformation of the features.
- **Autoencoders:** a grid search optimization was used to find the best architecture for the autoencoder, it was only tried 1 hidden layer and the code/bottleneck layer, using cross-validation (CV), stooping conditions to have the best result based on the validation and learning rate schedules to reduce the learning rate when the MSE is not improving, and then we build and compare clustering solutions using as input the Autoencoder transformation of the features.

Now, especially for each perspective, the following was done.

3.1 Value-Based Segmentation

Will consists of segmenting the customers on their general behavior and contribution to the company, see Annex1, the variables that were not going to be use because they are redundant that are from this perspective are: EnrollmentDateOpening (because it has the same information as the Days_with_us variable), Re_engaged_people, CancellationDate and EnrollmentType (because they have the same information as the Days_until_reengagement variable). Using only the numerical variables for clustering (not using the LoyaltyStatus because it is a categorical feature).

Exists 16,373 missing values on Days_until_reengagement, they were imputed with 0. Also exists 1471 missing values on the variables Total_NumFlights, Total_DistanceKM, Dollar_Cost_Points_Remaining, Recency_Days, they were isolated into a separate cluster labeled “No_transactions”.

To handle the outliers, the 3xIQR was applied to the variables: Customer Lifetime Value. For the Recency we decided to apply the upper bound when a customer had not brought in more than a year (365 days). Finally, there was an inconsistency problem with the variable Dollar_Cost_Points_Remaining since it has negative values, something that is impossible because on overall terms a customer cannot spend more points than the total points that they received, so those observations were transformed to have 0 Dollar_Cost_Points_Remaining. Then, it was used minmax scaler (features range –1:1) and finally the clustering algorithms used on this perspective were the K-means and the Hierarchical.

3.2 Demographic Segmentation

Will consists of segmenting the customers on their demographic characteristics, see Annex1, the numerical variables that we will use from this perspective are: Income, Longitude and Latitude; and the categorical variables are: Gender, Marital Status, and Education.

Exists 0 missing values and 0 extreme outliers, and the clustering algorithms used in this perspective were K-means (only with numerical variables), Hierarchical (only with numerical variables), and K-Prototypes (with numerical and categorical variables). It was used minmax scaler (features range 0:1), for all the features to contribute the same since on this perspective it was used K-Prototypes and for the categorical features it used hamming distance that is bounded between 0-1.

3.3 Behavioral Segmentation

Since information about yearly behavior and monthly behavior exists, this perspective is going to be split into two, to analyze individually the customer's behavior in different years and on different months.

3.3.1 Yearly Behavioral Segmentation

Will consists of segmenting the customers on their yearly behavior and it will be using all the yearly variables, see Annex1, excluding all the variables from the year 2019 because, as it was seeing on the Phase 1, it has a perfect correlation with the variables of 2020, so to prevent redundant information to enter into our model we will drop them, see Annex2.

Exists 1471 missing values on all the variables; they were isolated into a separate cluster labeled “No_transactions” and exists 0 extreme outliers. Then, it was used minmax scaler (features range –1:1) and finally the clustering algorithms used on this perspective were the K-means and the Hierarchical.

3.3.1 Monthly Behavioral Segmentation

Will consists of segmenting the customers on their monthly behavior and it will be using all the monthly variables, see Annex1. Exists 1471 missing values on all the variables; they were isolated into a separate cluster labeled “No_transactions”, and to handle the outliers, the 3xIQR was applied to the variables: *Dollar_Cost_Points_Remaining_Per_Month_X* (to all the months). Then, it was used minmax scaler (features range –1:1).

Finally, the clustering algorithms used in this perspective were the K-means and the Hierarchical, but were also compared with the K-means and the Hierarchical obtained after a PCA transformation, this was implemented to see if reducing the number of original features (that were 48) could improve the final metrics and the structure of the UMAP.

3.4 Merged Solution

To merge all the perspective into one, it was necessary the apply all the preprocessing steps like they were done on each of the perspective since it was needed to use all the variables to arrive at a final solution. So, all the numerical variables that were used on the individual perspective for clustering were used now (removing the year 2019), the outliers were handling the same way as in the individual perspective, then all the missing values were imputed with 0 (including the ones with no transactions). Then, it was used minmax scaler (features range –1:1); and finally, it was grouped the observations by each cluster label perspective combination and calculated the centroids of each group, centroids that are going to be used as the input for our final unified cluster solution.

The clustering algorithms used in this unified part were the K-means and the Hierarchical, but these solutions are going to be biased, benefiting the travels information, since it is represented in a monthly, yearly and general/total way. For that reason, to still maintain the importance of other features like the Customer Lifetime Value, Recency, Income, etc, an Autoencoder was used to learn a latent representation of all the features, to remove this bias problem. Implementing afterwards the K-means, the Hierarchical and Fuzzy C-means clustering on the centroids that are now on the new latent representation. And finally, we will compare the results (profiling) and metrics of the different clusters with and without the use of the Autoencoder to decide on the final clusters (on this entire part the profiling will be using all the features, numerical and categorical, to analyze final clustering results).

4. RESULTS & VALIDATION

It is time to see the number of clusters that were decided on each perspective and cluster method, how they performed, and what is going to be the final cluster of each client, with their respective unique characteristics.

4.1 Value-Based Segmentation

Starting with the Hierarchical clustering method, after seeing the dendrogram, Annex3, it was decided that a solution with 5 or 7 clusters could be the best one using this method, it was analyzed the different clusters profiles that were formed and plotted on two dimensions using an UMAP, see Annex4, they achieve a satisfactory separation seeing the UMAP, with unique value-based characteristics between groups.

Continuing into the K-means, it was analyzed the SSW, SSB, Silhouette Score and R2, see Annex5 ,and decided that the best number of clusters after seeing at all the metrics were 6 and 8, because it was when the Silhouette increased in comparison to their alternatives. It was analyzed the different clusters profiles that were formed and plotted on two dimensions using an UMAP, see Annex6, and they achieved a satisfactory separation seeing the UMAP, with unique value-based characteristics between groups.

To know which clustering solution we are going to use on our Value-Based segmentation perspective, the SSW, SSB, Silhouette Score and R2 were compared on these 4 clustering methods, see Annex7, and it was concluded that the best one was the K-means with 8 clusters because it achieve the bets results across all the metrics in comparison with the other alternatives. These 8 clusters were added to the cluster of clients that did not have made any transactions, so there were 9 final clusters for the Value-Based segmentation perspective.

4.2 Demographic Segmentation

Starting with the K-means, the elbow plot was analyzed (SSW) and the Silhouette Score, see Annex8, and decide that the best number of clusters after seeing at all the metrics were 4. It was analyzed the different clusters profiles that were formed, see Annex9.
Following with the K-Prototypes, the elbow plot was analyzed (cost) and the Silhouette Score, see Annex10, and decided that the best number of clusters after seeing at all the metrics were 4. It was analyzed the different clusters profiles that were formed, see Annex11.

Continuing into the Hierarchical clustering method, after seeing the dendrogram, see Annex12, it was decided that a solution with 4 or 7 clusters could be the best ones using this method, it was analyzed the different clusters profiles that were formed, see Annex13.

To know which clustering solution we are going to use on our Demographic segmentation perspective, the Silhouette Score, were compared on these 4 clustering methods, see Annex14, and it was concluded that the best one was the K-means with 4 clusters because it achieve the bets results across the all the metrics Silhouette score, Calinski-Harabasz score, and Davies-Bouldin score in comparison with the other alternatives.

4.3 Behavioral Segmentation

4.3.1 Yearly Behavioral Segmentation

Starting with the K-means, the elbow plot was analyzed, SSW, SSB, Silhouette Score and R2, see Annex15, and decided that the best number of clusters after seeing at all the metrics were 5 and 8, 5 because it was when the SSW stopped improving in the same rate that had before (elbow) and 8 because it was when the Silhouette increased in comparison to their alternatives and . It was analyzed the different clusters profiles that were formed and plotted on two dimensions using an UMAP, see Annex16, and they achieved a satisfactory separation seeing the UMAP, with unique yearly behavioral characteristics between groups.
Continuing into the Hierarchical clustering method, after seeing the dendrogram Annex17, it was decided that a solution with 6 clusters could be the best one using this method, it was analyzed the different clusters profiles that were formed and plotted on two dimensions using an UMAP Annex18 , it achieve a satisfactory separation seeing the UMAP, with unique yearly behavioral characteristics between groups.

To know which clustering solution we are going to use on our Yearly Behavioral segmentation perspective, the SSW, SSB, Silhouette Score and R2 were compared on these 3 clustering methods, Annex19, and it was concluded that the best one was the K-means with 8 clusters because it achieve the bets results across the metrics SSW, SSB, and R2 in comparison with the other alternatives, with a Silhouette Score close to the best one, and because it added more details information about the customers. These 8 clusters were added to the cluster of clients that did not have made any transactions, so there were 9 final clusters for the Yearly Behavioral segmentation perspective.

4.3.2 Monthly Behavioral Segmentation

Starting with the K-means, the elbow plot was analyzed, SSW, SSB, Silhouette Score and R2, see Annex20, and decided that the best number of clusters after seeing at all the metrics were 14, because of the elbow. It was analyzed the different clusters profiles that were formed and plotted on two dimensions using an UMAP, see Annex21, and they don't look separated seeing at the UMAP, but they have unique monthly behavioral characteristics between groups.
Continuing into the Hierarchical clustering method, after seeing the dendrogram Annex22, it was decided that a solution with 7 clusters could be the best one using this method, it was analyzed the different clusters profiles that were formed and plotted on two dimensions using an UMAP, see Annex23, and they

don't look separated seeing at the UMAP, but they have unique monthly behavioral characteristics between groups.

Since the UMAP showed that the groups were not clearly separated, it was decided to use PCA to reduce the dimensionality of the data, given that it had 48 variables as previous inputs, and they had some level of correlation, as it was seen on the Phase1, see Annex24. To see if the results improved (including the groups on the UMAP, profiling, and metrics).

To maintain at least 80% of the explained variance, it was found that there were needed 22 principal components, see Annex25, that managed to maintain 80,7% of the explained variance of the dataset. The original variables were transformed into the new features created by the PCA, originating unique, uncorrelated features, see Annex26, to see the loadings, and interpretation of the new principal components). And it was performed again the analysis on K-means and Hierarchical, using as features the new principal components.

For the K-means, the elbow plot was analyzed, SSW, SSB, Silhouette Score and R2, see Annex27, and decided that the best number of clusters after seeing at all the metrics were 12, because of the elbow. It was plotted on two dimensions using an UMAP, see Annex28, and they also don't look separated seeing at the UMAP that was created using the principal components.

For the Hierarchical clustering method, after seeing the dendrogram Annex29, it was decided that a solution with 7 clusters could be the best one using this method, it was plotted on two dimensions using an UMAP, see Annex30, and they also don't look separated seeing at the UMAP that was created using the principal components. Concluding that the PCA transformation did not improve clusters separation on the UMAP, and there were simply too many features to compress on 2 dimensions.

To know which clustering solution we are going to use on our Monthly Behavioral segmentation perspective, the SSW, SSB, Silhouette Score and R2 were compared on these 4 clustering methods, see Annex31, and it was concluded that the best one was the K-means with 14 clusters, with no use of PCA, because it achieves similar scores across the metrics SSW, SSB, and R2 in comparison with the other alternatives, and it was the one with simpler and more interpretable results in the profiling. These 14 clusters were added to the cluster of clients that did not have made any transactions, so there were 15 final clusters for the Monthly Behavioral segmentation perspective.

4.3 Merged Solution

It started by combining each of the labels coming from each individual perspective, 4 demographic, 9 yearly behavioral, 9 value-based and 15 monthly behavioral; if all combined created an unique group it would have been 4860 unique groups of customers; however when they were merge only 1274 unique combinations were generated (unique centroids of customers), meaning that there are some indications that we could unite the groups into a more compact representation of the customers.

A K-means and Hierarchical were performed with all the features that were used in the individual perspective (67 features). For the Hierarchical clustering method, after seeing the dendrogram Annex32, it was decided that a solution with 15 clusters could be the best one using this method, it was analyzed the different clusters profiles that were formed and plotted on two dimensions using an UMAP (on the centroids and original data), see Annex33, and they don't look separated seeing at the UMAP with the original data, but for the centroids the majority of clusters are well divided and the profiling show that they have unique characteristics between groups. For the K-means, the elbow plot was analyzed, SSW, SSB, Silhouette Score and R2, see Annex34, and decided that the best number of clusters after seeing at all the metrics were 16, because of the elbow. It was analyzed the different clusters profiles that were formed and plotted on two dimensions using an UMAP (on the centroids and original data), see Annex35, and they don't look separated seeing at the UMAP with the original data, but for the centroids the majority of clusters are well divided and the profiling show that they have unique characteristics between groups.

However, as it was discussed these solutions were biased, so Autoencoder was used to learn a compressed version of the original features of the centroids. It was tested using only 1 layer on the encoder and decoder hidden layer, with sizes: 64, 48 and 32; started on 64 because the maximum number of features were 67, and stopped on 32 because the code/bottleneck layer was tested with a maximum number of 32 neuros (that is almost half of the original features), 48 was also used because it is the value that is on the middle of the extremes; the code layer had sizes of 8, 16, 22 and 32. And seeing at the results, see Annex36, the autoencoder network architecture that manages to achieve a low reconstruction error while compressing as much as possible the input space, were the one with 64 neurons on the hidden layer and 22 neurons on the code layer. Afterwards, the autoencoder was trained on all the centroids and then these were compressed into the new latent space (of 22 new features), that were used as input for the merge clusters.

A Hierarchical clustering method was performed using the new features, and after seeing the dendrogram Annex37, it was decided that a solution with 18 clusters could be the best one using this method, it was analyzed the different clusters profiles that were formed and plotted on two dimensions using an UMAP (on the original data and the centroids of the original and new features), see Annex38, and they don't look separated seeing at the UMAP with the original data, but for the centroids the majority of clusters are well divided, they seen even better on the projection of the new latent features and the profiling show that they have unique characteristics between groups.

Also, a K-means clustering method was performed using the new features, the elbow plot was analyzed, SSW, SSB, Silhouette Score and R2, see Annex39, and decided that the best number of clusters after seeing at all the metrics were 14, because of the elbow. It was analyzed the different clusters profiles that were formed and plotted on two dimensions using an UMAP (on the original data and the centroids of the original and new features), see Annex40, and they don't look separated seeing at the UMAP with the original data, but for the centroids the majority of clusters are well divided, they seen even better on the projection of the new latent features and the profiling show that they have unique characteristics between groups.

Finally, a Fuzzy C-means clustering method was performed using the new features, and after testing different values for Fuzziness (m), from 1.1 that is almost hard clustering and 2.0, the one with best FPC score was the Fuzzines 1.1, see Annex41. For that level of fuzziness (1.1), seeing at the FPC and FPE score, the optimal number of clusters are 15, see Annex42. It was established that the centroids with more than 70% certainty of being in a cluster, are going to be considered 'core' (they only have 1 cluster label that is the one with more certainty), below 50% are going to be fuzzy (they should not belong to any cluster since their behavior is not well defined) and in the middle they are standard (can have more than 1 label). Existed 1138 core centroids, 88 standard and 48 fuzzy in data, see Annex43, corresponding to 16048 core customers, 441 standard and 83 fuzzy, and we are going to analyze and do profiling on the core ones, since these are the ones that are well defined. It was analyzed the different clusters profiles of the core customers that were formed and plotted on two dimensions using an UMAP (on the original data of core customers), see Annex44, and they don't look separated seeing at the UMAP with the original data, but the profiling show that they have unique characteristics between groups.

To know which clustering solution we are going to use on our final unified segmentation, since overall they all have a good profiling of customers, the SSW, SSB, Silhouette Score and R2 were compared on these 5 clustering methods, see Annex45**,** and it was concluded that the best one was the Hierarchical with 18 clusters, with the use of an Autoencoder, because it achieves best scores across all the metrics in comparison with the other alternatives.

A clustering solution that managed to maintain information from all perspectives arriving at a solution that captures behavioral patterns (seasonal travel), yearly changes (long-term loyalty and engagement changes), and the general characteristics of the customer, such as the recency and total number of flights, see Annex46.

Final Clustering Profiling, see Annex38:
First, we observe that longitude and latitude lose their significance as they remain almost constant across all clusters, while the other variables used to build the clusters vary significantly depending on the group. Regarding categorical variables, most clusters do not change much; only one or two clusters show behavioral shifts in variables such as Province or State, City, Education, and Marital Status. Here is the profiling for each cluster:

- **The October Loyalls (956 customers):** Medium income, low revenue, long tenure. They have not purchased recently but remain a consistent, engaged group over the years. These are "October people" who accumulate points year-round to use in October when they fly less but with companions.
- **The Pandemic Drop-offs (209 customers):** Medium income, low revenue, moderate tenure. These are lost customers who were active in 2019 and 2020 but stopped almost completely in 2021. They previously traveled with consistent frequency and distance each month.
- **The September Loyalls (2706 customers):** Medium income, low revenue, long tenure. Loyal, very engaged customers who purchased recently. "September people" who save points all year to use in September for flights with companions.
- **The January Average (1045 customers):** Medium income, low revenue, long tenure. Recently active and traveling more now than before (highly engaged). "January people" who use accumulated points for companion travel in January.
- **The Summer-Fall Hybrids (613 customers):** Medium income, medium revenue, moderate tenure. Recently active and showing increased engagement. "July people" who travel with companions in July and tend to travel alone in autumn.
- **The May Planners (1078 customers):** Medium income, medium revenue, moderate tenure. Recently active and now very engaged. "May people" who accumulate points to use during May for companion travel.
- **The Lost Holiday Drifters (565 customers):** Low income, low revenue, short tenure. Lost customers who were never highly engaged; they use points and tend to travel more with companions in December.
- **The Autumn Budget (178 customers):** Low income, low revenue, short tenure. Recently active and becoming more engaged, specifically during the autumn months.
- **The June Loyalls (897 customers):** Medium income, medium revenue, long tenure. Very engaged, loyal customers. "June people" who save points all year for companion travel in June.
- **Holiday Travelers (66 customers):** Medium income, low revenue, short tenure. Recently active and increasing engagement. They travel with companions in February and June, using points in most months except November and December (when they travel the most).
- **The November Loyalls (1029 customers):** Medium income, medium revenue, long tenure. Consistent, engaged group. "November people" who redeem points for companion travel in November.
- **The Dormant Browsers (517 customers):** Low income, low revenue, moderate tenure. Have not purchased recently but show some new engagement; they travel consistently but infrequently across all months and use points.
- **The March Loyalls (1126 customers):** Medium income, medium revenue, long tenure. Consistent, engaged group. "March people" who accumulate points for companion travel in March.

• **VIPs (64 customers):** High income, high revenue, short tenure. Recently active and engaged. They travel in June with companions and are frequent autumn travelers, using points throughout the year except in Q4. Almost 100% Bachelors, that are Divorced or Married, with the more proportion of men's (~65%).
• **The Q4 Boomerang (691 customers):** Medium income, low revenue, short tenure. Re-engaged customers who are now moderately active, particularly in autumn and December with companions.
• **The December Loyalls (949 customers):** Medium income, medium revenue, long tenure. Very engaged, loyal customers. "December people" who redeem points for companion travel in December.
• **The February Loyalls (992 customers):** Medium income, low revenue, long tenure. Very engaged, loyal customers. "February people" who save points for companion travel in February.
• **New travelers (2891 customers):** Medium income, medium revenue, very short tenure (newest). Re-engaged, moderately active customers who use points and travel primarily in autumn.

5. STRATEGIC RECOMMENDATIONS

The following recommendations outline targeted, data-driven strategies designed to maximize customer lifetime value for AIAI. Each initiative aligns marketing actions, service personalization, and loyalty program optimization with the distinct behaviors and needs of each customer segment, aiming to increase retention, flight frequency, and overall revenue while improving customer engagement and satisfaction.
• **Usual Travelers:** Segments include May planners, January average travelers, and loyal travelers in October, September, February, December, June, November, and March. Offer an upgrade to the highest loyalty status tier, guaranteeing flights with companions during their preferred month at 20% off. This initiative is designed to increase revenue through the loyalty program. Additionally, promote premium upsells on existing flights, such as enhanced seating, additional accommodation options, and onboard upgrades, to further increase revenue from these frequent travelers.
• **Autumn Budget Campaign:** Customers who purchase more than one flight between January and July qualify for the *"Free Seats and Priority"* campaign. This benefit includes complimentary seat selection and priority boarding, valid exclusively for travel between September and November, aligning with their typical travel period.
• **Q4 Boomerang Campaign:** Points accumulated between October and December will be doubled and can be redeemed from January through September of the following year. The objective is to encourage travel during lower-demand periods of this group. Additionally, promote premium upgrades on December flights to increase revenue of these customers who already demonstrate strong spending capacity.
• **Lost Holiday Drifters:** Implement targeted win-back campaigns featuring limited-time offers, bonus point accumulation, and flexible redemption incentives, as these customers have not made a purchase for an extended period.
• **Pandemic Drop-Offs:** Launch a broad re-engagement marketing campaign targeting customers who stopped flying after 2021, following the end of pandemic-related travel restrictions, to assess reactivation potential.
• **Dormant Browsers**: Deploy automated *"Price Drop"* alerts for routes customers have previously searched, encouraging conversion through timely and relevant pricing notifications. Since these are low-income individuals, that don't travel much.
• **VIP Customers:** Introduce seasonal destination campaigns tailored to high-value customers to increase flight frequency and overall revenue, leveraging their higher willingness to spend.
• **New Travelers:** Run awareness campaigns highlighting loyalty program benefits and available upgrades. Incentivizing second and third purchases through bonus point offers to accelerate customer engagement.
• **Summer–Fall Hybrids:** Offer an upgrade to the highest loyalty status tier, guaranteeing flights with companions in July at 20% off. Complement this with marketing campaigns promoting fall travel to alternative destinations, to maintain the retention of these customers.

• **Holiday Travelers**: Customers who fly more than once per month will earn double points on subsequent flights, encouraging increased travel frequency of these low-frequency travel customers.

6. CONCLUSION

In conclusion, this study successfully executed a multi-dimensional segmentation analysis encompassing demographic, value-based, and behavioral characteristics. The resulting framework provides AIAI with a data-driven foundation to develop targeted marketing strategies tailored to the specific needs of each identified customer segment.

6.1 Limitations

• **Data Recency Limitation:** Four years have passed since the last known transaction, with 2021 being the most recent year on record, coinciding with the peak impact of the pandemic. As a result, customer behavior from 2022 to 2025 is unknown, making it difficult to determine whether historical travel patterns are still relevant. This uncertainty is critical when designing and implementing current recommendations, as understanding the evolution of travel behavior is essential.

• **Lack of Booking Timing Data:** There is no information on when customers typically make flight reservations. Without this insight, it is not possible to accurately time marketing campaigns or determine the optimal moment to communicate flight options and promotional offers.

• **Missing Pricing Information:** We do not have access to flight prices or loyalty status tier pricing. Consequently, we are unable to quantify the potential revenue impact of customer upgrades or assess the financial return of encouraging increased purchases.

• **Unknown Price Sensitivity:** There is no data indicating whether customers primarily purchased flights during promotional periods. Therefore, we cannot determine their sensitivity to discounts or offers. The closest available proxy is whether customers have historically redeemed loyalty points, which only partially reflects promotional responsiveness.

• **Insufficient Destination Insights:** While all customers are based in Canada, there is no information regarding their travel destinations or preferences. For example, we do not know whether they prefer warm or cold destinations during winter, or whether they favor beaches, mountains, large cities, or other types of locations. This lack of data limits our ability to provide personalized destination recommendations.

• **Lack of Marketing Campaign History:** There is insufficient information regarding previous marketing campaigns. While changes in customer behavior and instances of re-engagement are observed, we do not know which offers were presented or which factors triggered these sudden behavior changes. This limits our ability to replicate successful strategies or avoid ineffective ones.

6.2 Future Research

Addressing the limitations can quantify and generate more accurate and meaningful results, opening the way for a complete data-driven vision of the customer based, that can potentiate in its entirety the growth of the business. And of course, it will be necessary to follow up on a year's basis the evolution of existing customers' behavior and new ones that might arrive, to address their needs as the years pass, and to ensure that the marketing campaigns are having the desired effect.

7. ANNEXES

Annex AI Usage Statement
AI tools (ChatGPT) were utilized for support functions, namely generating code comments, refining sentence structure, proofreading text, and assisting with some code. All insights, conclusions, and recommendations are the result of our independent analysis and critical thinking.

Annex Contribution Statement
Individual student contributions to deliverables.

Luis Mendes (Student ID: 20221949)

- Behavioral month notebook.
- Help with other perspective notebooks.
- Merge notebook clusters.
- Optional part: Autoencoder, C means.
- Report.

Margarida Ourives (Student ID: 20221809)

- Demographic notebook.

Simon Sazonov (Student ID: 20221689)

- Value-based notebook.

Veronica Mendes (Student ID: 20221945)

- Behavioral year notebook.
- Merge cluster notebook.
- Report.
- Video.

All members contributed to collaborative discussions and ideation.

Annex Responsibility Statement
We, Group 12, certify that this report represents our original analytical work and interpretations. While AI tools were utilized as specified above. We collectively take full responsibility for the accuracy, quality, and integrity of this submission.

Annex1

Value Based Variables	Description	Type
Total_NumFlights	Total number of flights.	Numerical
Total_DistanceKM	Total distance of flights on KM	Numerical

Perc_Flights_With_Companions	Percentage of flights with companions.	Numerical
Dollar_Cost_Points_Remaining	Dollar value of points available for redemption.	Numerical
Recency	Days since last Purchase (if did not make any purchase should be Null).	Numerical
LoyaltyStatus	Current tier status in loyalty program (Star > Nova > Aurora).	Categorical
EnrollmentDateOpening	Date when customer joined the loyalty program.	Date
CancellationDate	Date when customer left the program.	Date
Customer Lifetime Value	Total calculated monetary value of customer relationship.	Numerical
EnrollmentType	Method of joining loyalty program.	Categorical
Re_engaged_people	Indicates whether a customers rejoined the loyalty program after a previous cancellation.	Boolean
Days_with_us	Total number of days a customer stayed enrolled in the loyalty program.	Numerical
Days_until_reengagement	Measures how long it took for a previously canceled customer to rejoin the program	Numerical

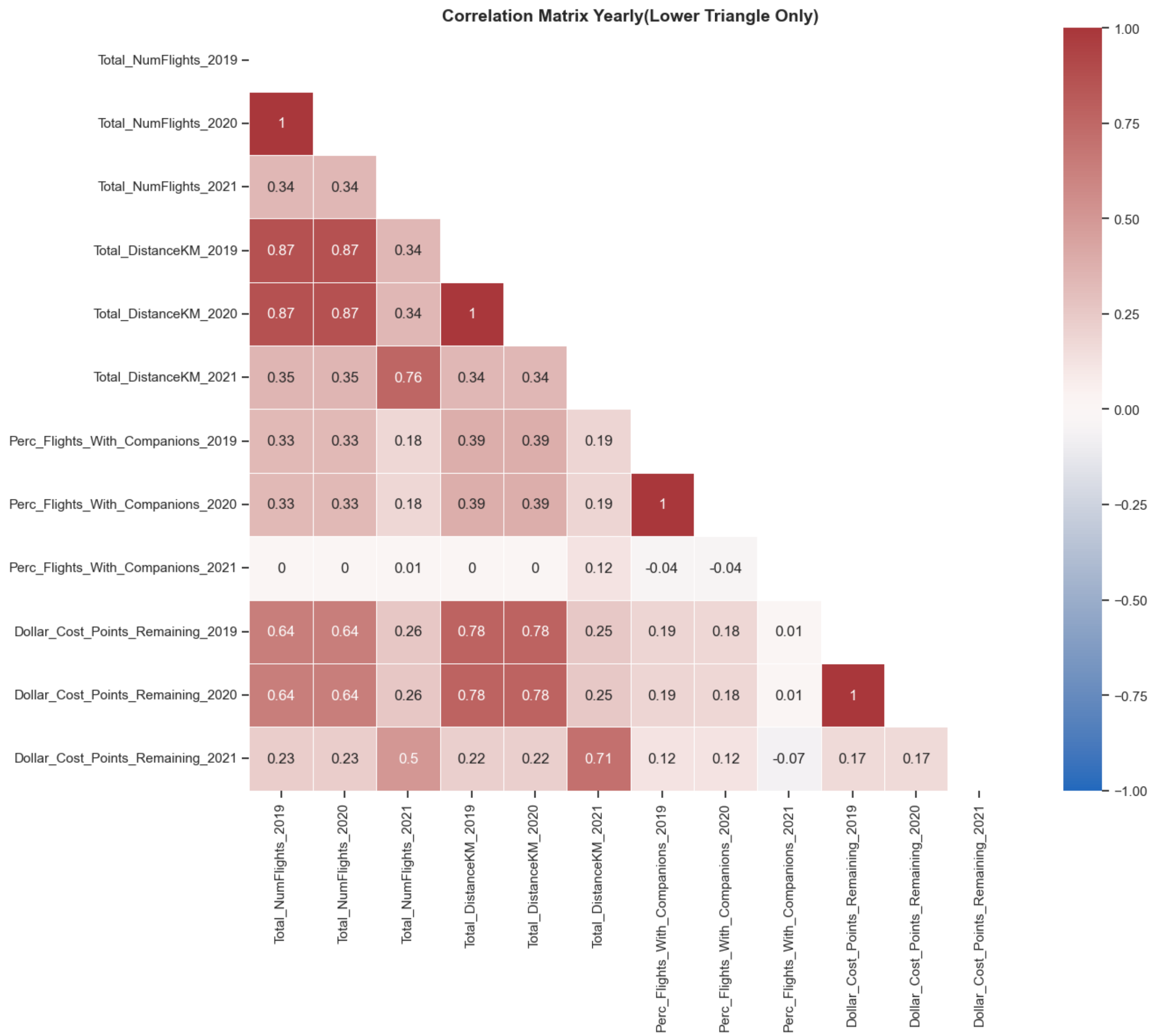
Behavioral - Year	**Description**	**Type**
Total_NumFlights_2019	Total number of flights in 2019.	Numerical
Total_NumFlights_2020	Total number of flights in 2020	Numerical
Total_NumFlights_2021	Total number of flights in 2021	Numerical
Total_DistanceKM_2019	Total distance of flights on KM in 2019	Numerical
Total_DistanceKM_2020	Total distance of flights on KM in 2020	Numerical
Total_DistanceKM_2021	Total distance of flights on KM 2021	Numerical
Perc_Flights_With_Companions_2019	Percentage of flights with companions in 2019	Numerical
Perc_Flights_With_Companions_2020	Percentage of flights with companions in 2020	Numerical
Perc_Flights_With_Companions_2021	Percentage of flights with companions in 2021	Numerical

Dollar_Cost_Points_Remaining_2019	Dollar value of points remains in 2019.	Numerical
Dollar_Cost_Points_Remaining_2020	Dollar value of points remains in 2020.	Numerical
Dollar_Cost_Points_Remaining_2021	Dollar value of points remains in 2021.	Numerical

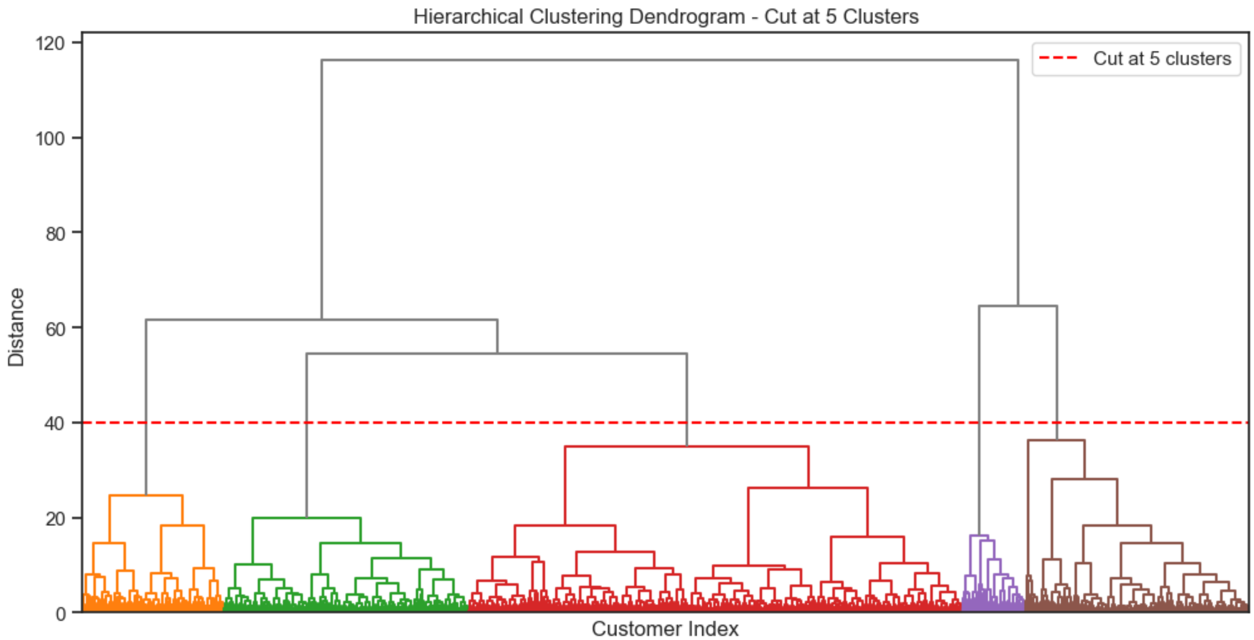
Behavioral - Month	**Description**	**Type**
Total_NumFlights_Per_Month_X	Total number of flights per month (e.g.,Total_NumFlights_month_1).	Numerical
Total_DistanceKM_Per_Month_X	Total distance traveled per month on KM.	Numerical
Perc_Flights_With_Companions_Per_Month_X	Percentage of flights with companions per month.	Numerical
Dollar_Cost_Points_Remaining_Per_Month_X	Dollar value of points remaining per month.	Numerical

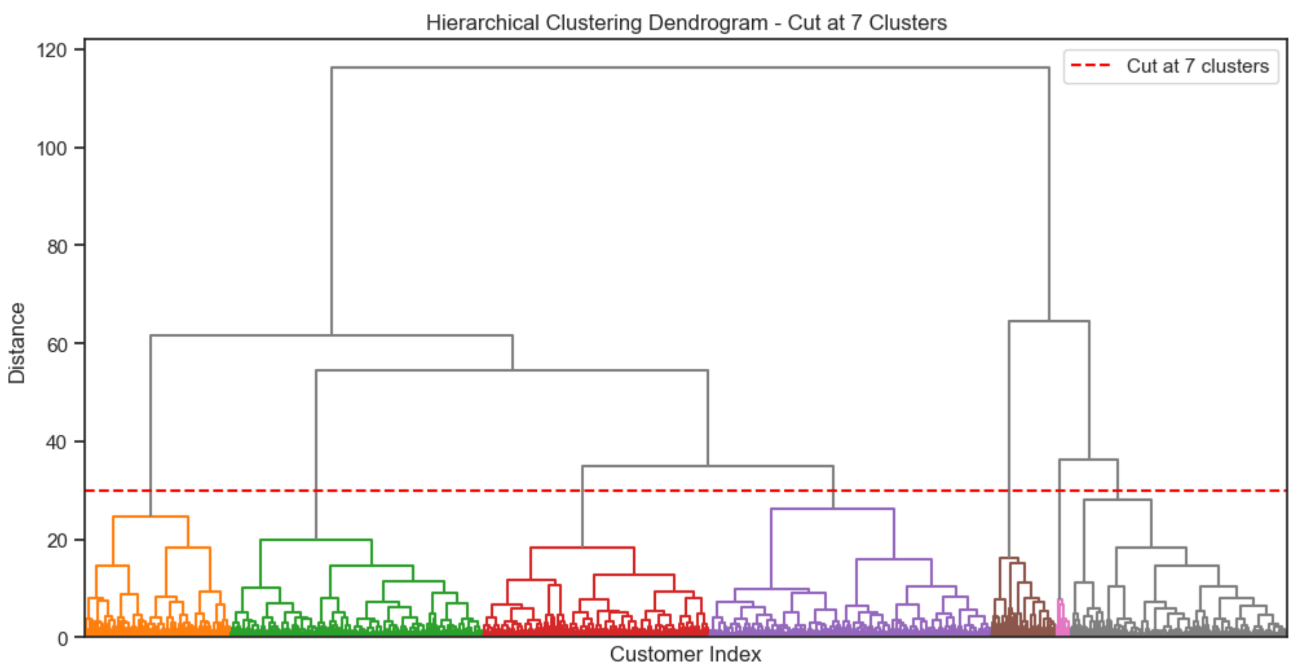
Demographic	**Description**	**Type**
First Name	Customer’s first name	Categorical
Last Name	Customer’s last name	Categorical
Customer Name	Customer’s full name (concatenated)	Categorical
Province or State	Customer’s province or state	Categorical
City	Customer’s city of residence	Categorical
Latitude	Geographic latitude coordinate of customer location	Numerical
Longitude	Geographic longitude coordinate of customer location	Numerical
Postal code	Customer’s postal/ZIP code	Categorical
Gender	Customer’s gender	Categorical
Education	Customer’s highest education level (Bachelor, College, etc.)	Categorical
Location Code	Urban/Suburban/Rural classification of customer residence	Categorical
Marital Status	Customer’s marital status (Married, Single, Divorced)	Categorical
Income	Customer’s annual income	Numerical

Annex2

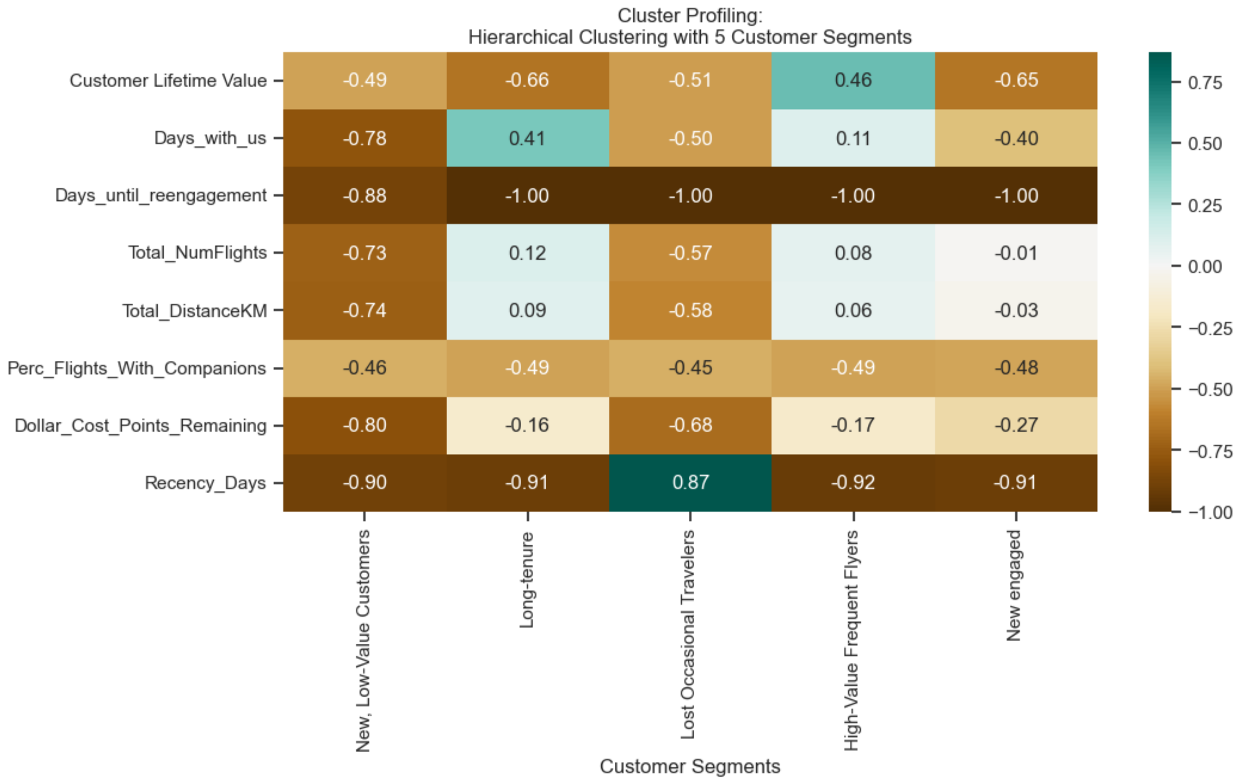

Correlation Matrix Yearly(Lower Triangle Only)
Total_NumFlights_2019
Total_NumFlights_2020
Total_NumFlights_2021
Total_DistanceKM_2019
Total_DistanceKM_2020
Total_DistanceKM_2021
Perc_Flights_With_Companions_2019
Perc_Flights_With_Companions_2020
Perc_Flights_With_Companions_2021
Dollar_Cost_Points_Remaining_2019
Dollar_Cost_Points_Remaining_2020
Dollar_Cost_Points_Remaining_2021
1
0.34 0.34
0.87 0.87 0.34
0.87 0.87 0.34 1
0.35 0.35 0.76 0.34 0.34
0.33 0.33 0.18 0.39 0.39 0.19
0.33 0.33 0.18 0.39 0.39 0.19 1
0 0 0.01 0 0 0.12 -0.04 -0.04
0.64 0.64 0.26 0.78 0.78 0.25 0.19 0.18 0.01
0.64 0.64 0.26 0.78 0.78 0.25 0.19 0.18 0.01 1
0.23 0.23 0.5 0.22 0.22 0.71 0.12 0.12 -0.07 0.17 0.17
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

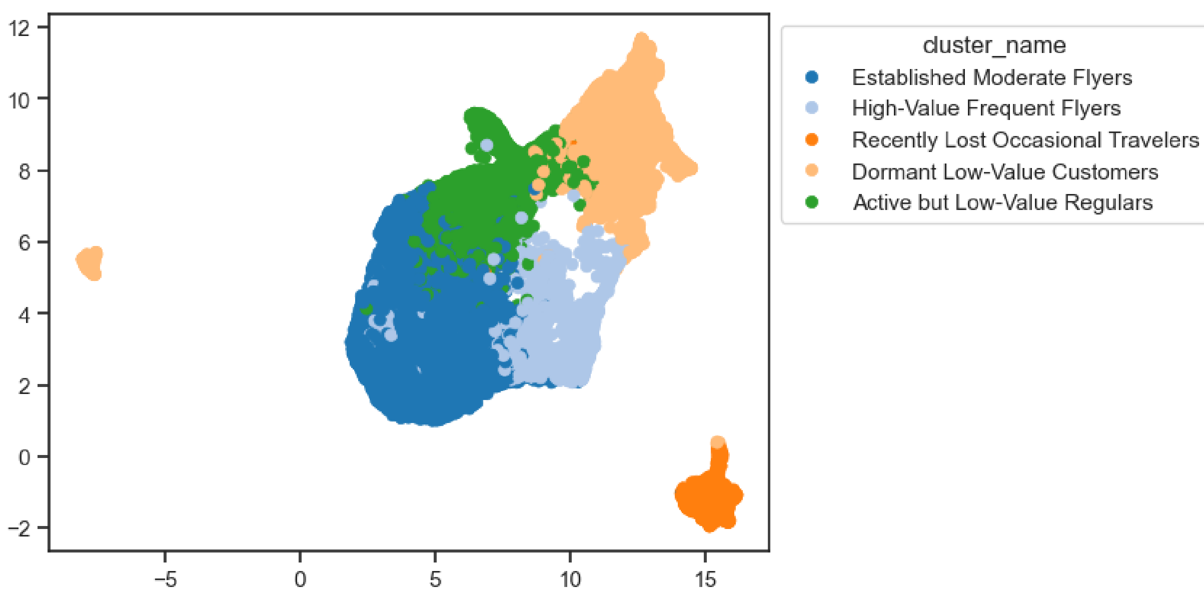
Annex3

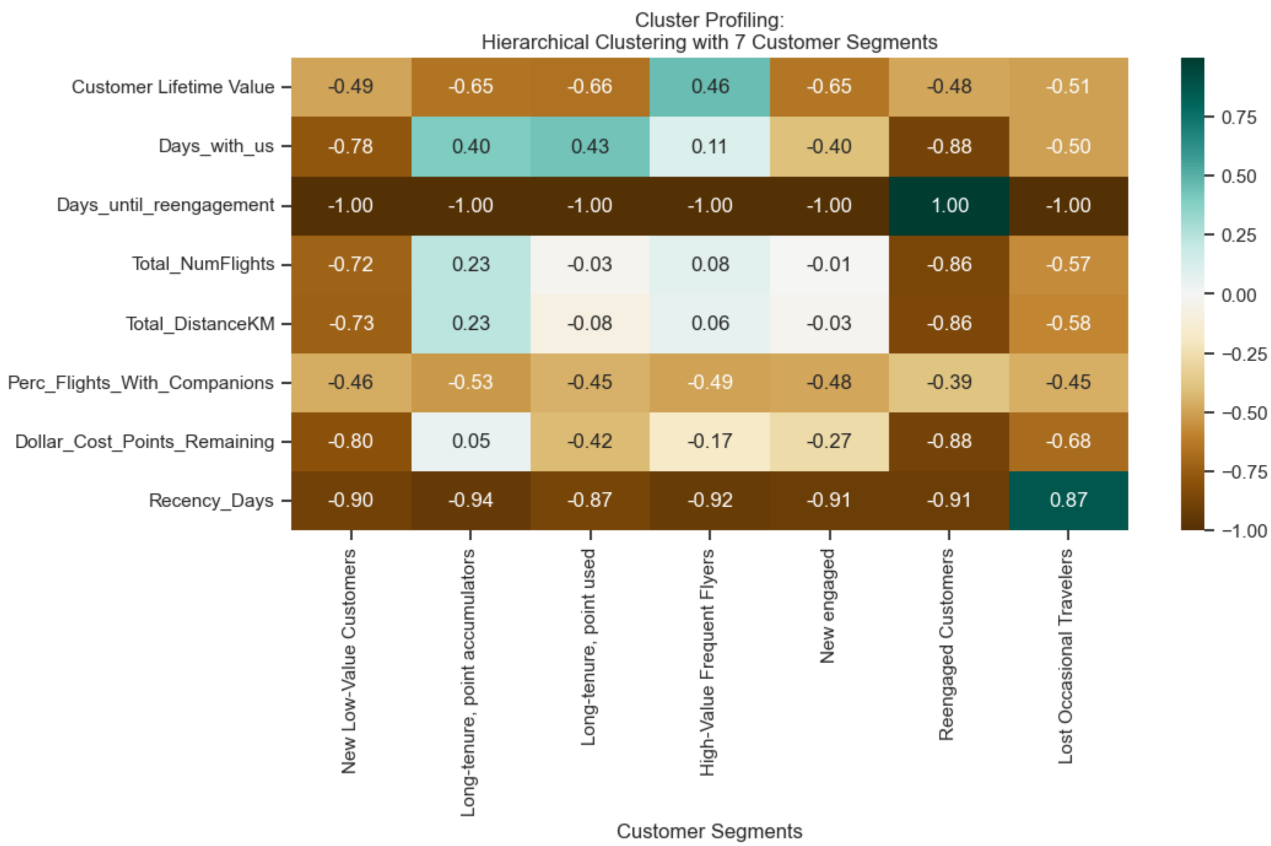

Hierarchical Clustering Dendrogram - Cut at 5 Clusters
Cut at 5 clusters
Distance
120
100
80
60
40
20
0
Customer Index


Hierarchical Clustering Dendrogram - Cut at 7 Clusters
Cut at 7 clusters
Distance
120
100
80
60
40
20
0
Customer Index

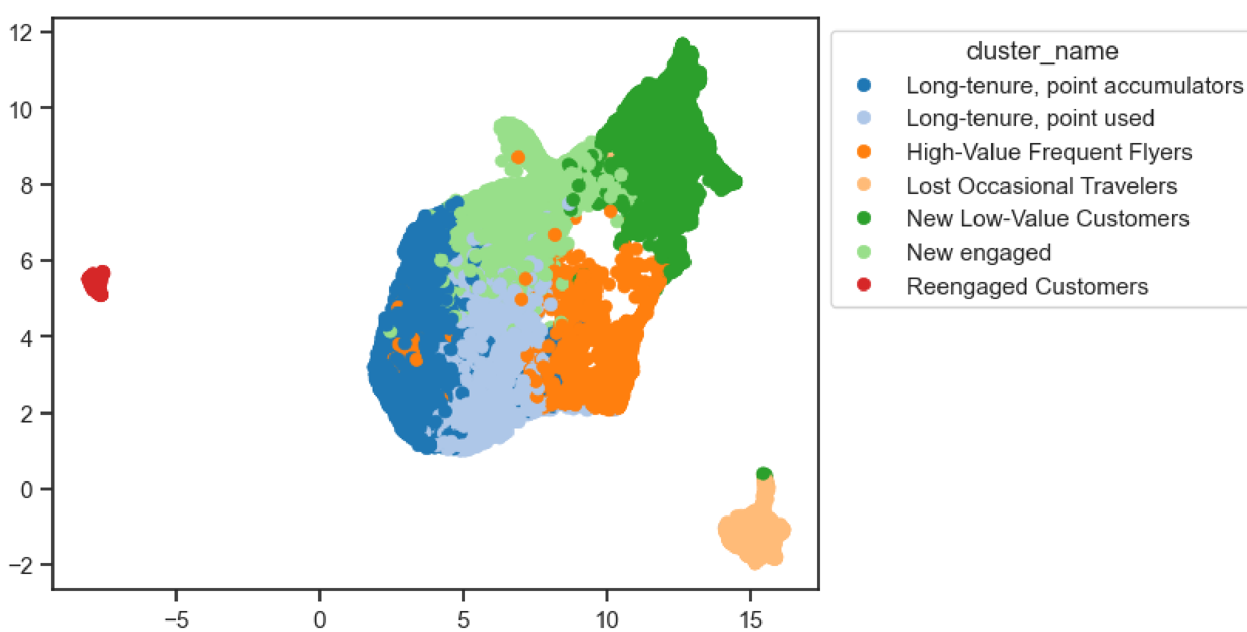
Annex4


Cluster Profiling:
Hierarchical Clustering with 5 Customer Segments
Customer Lifetime Value
Days_with_us
Days_until_reengagement
Total_NumFlights
Total_DistanceKM
Perc_Flights_With_Companions
Dollar_Cost_Points_Remaining
Recency_Days
-0.49 -0.66 -0.51 0.46 -0.65
-0.78 0.41 -0.50 0.11 -0.40
-0.88 -1.00 -1.00 -1.00 -1.00
-0.73 0.12 -0.57 0.08 -0.01
-0.74 0.09 -0.58 0.06 -0.03
-0.46 -0.49 -0.45 -0.49 -0.48
-0.80 -0.16 -0.68 -0.17 -0.27
-0.90 -0.91 0.87 -0.92 -0.91
New, Low-Value Customers
Long-tenure
Lost Occasional Travelers
High-Value Frequent Flyers
New engaged
Customer Segments
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00


cluster_name
Established Moderate Flyers
High-Value Frequent Flyers
Recently Lost Occasional Travelers
Dormant Low-Value Customers
Active but Low-Value Regulars
12
10
8
6
4
2
0
−2
−5
0
5
10
15

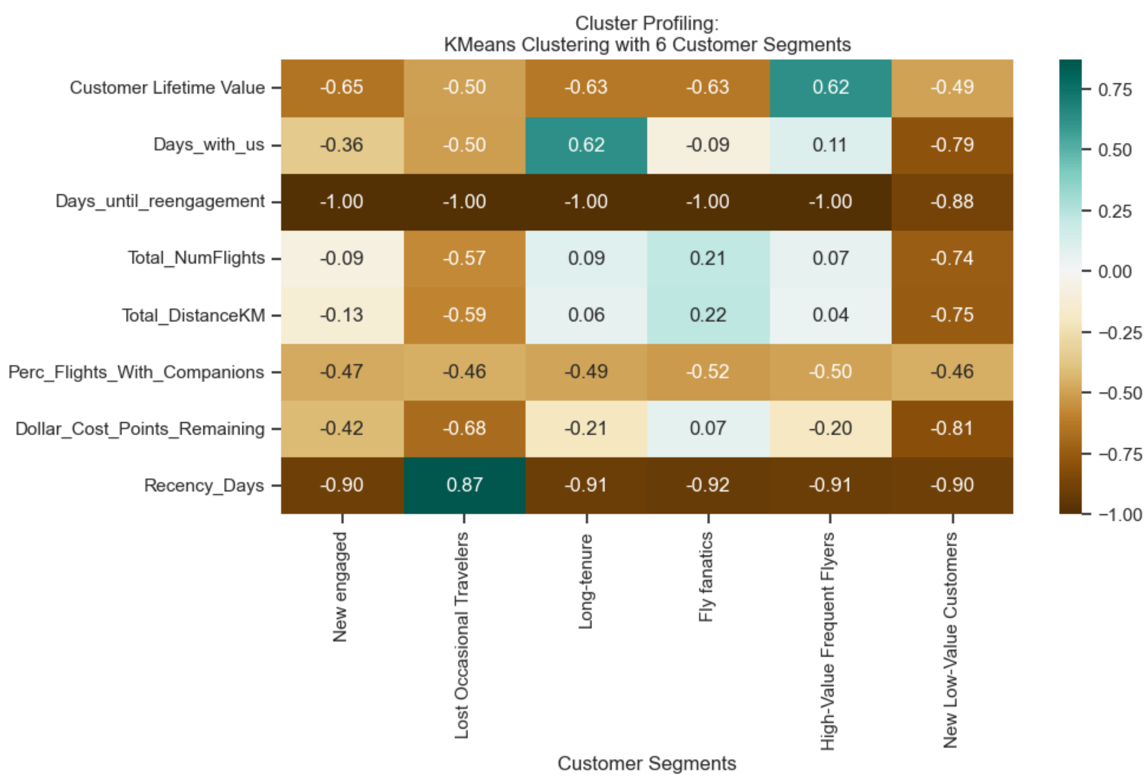


	New Low-Value Customers	Long-tenure, point accumulators	Long-tenure, point used	High-Value Frequent Flyers	New engaged	Reengaged Customers	Lost Occasional Travelers
Customer Lifetime Value	-0.49	-0.65	-0.66	0.46	-0.65	-0.48	-0.51
Days_with_us	-0.78	0.40	0.43	0.11	-0.40	-0.88	-0.50
Days_until_reengagement	-1.00	-1.00	-1.00	-1.00	-1.00	1.00	-1.00
Total_NumFlights	-0.72	0.23	-0.03	0.08	-0.01	-0.86	-0.57
Total_DistanceKM	-0.73	0.23	-0.08	0.06	-0.03	-0.86	-0.58
Perc_Flights_With_Companions	-0.46	-0.53	-0.45	-0.49	-0.48	-0.39	-0.45
Dollar_Cost_Points_Remaining	-0.80	0.05	-0.42	-0.17	-0.27	-0.88	-0.68
Recency_Days	-0.90	-0.94	-0.87	-0.92	-0.91	-0.91	0.87

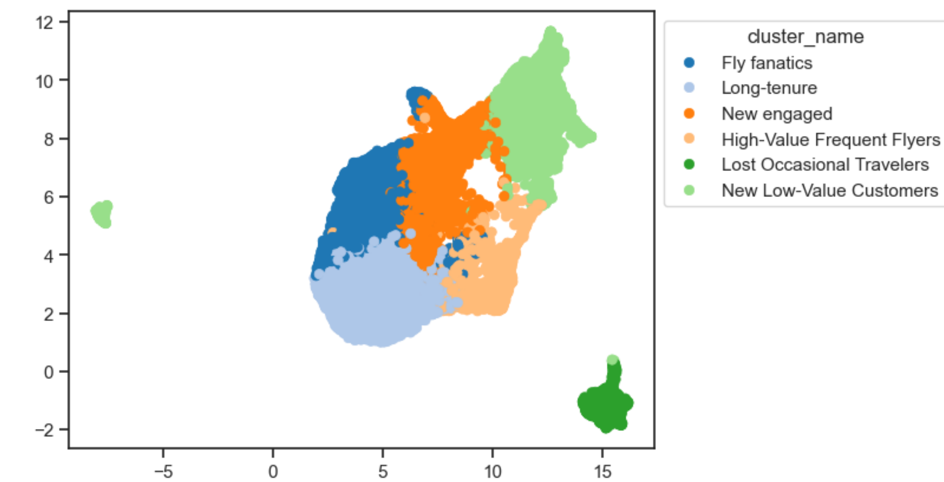


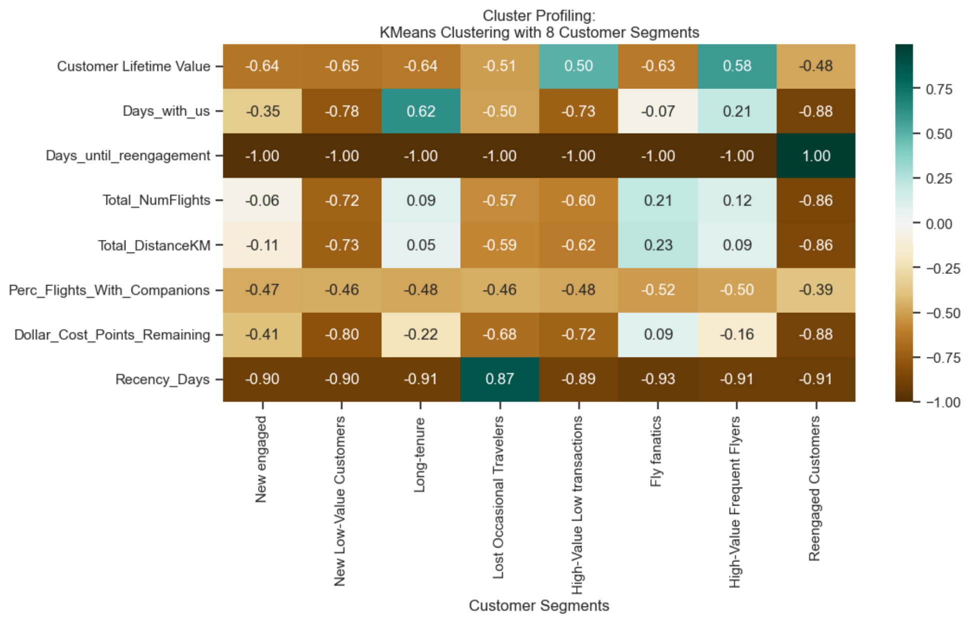
Annex5

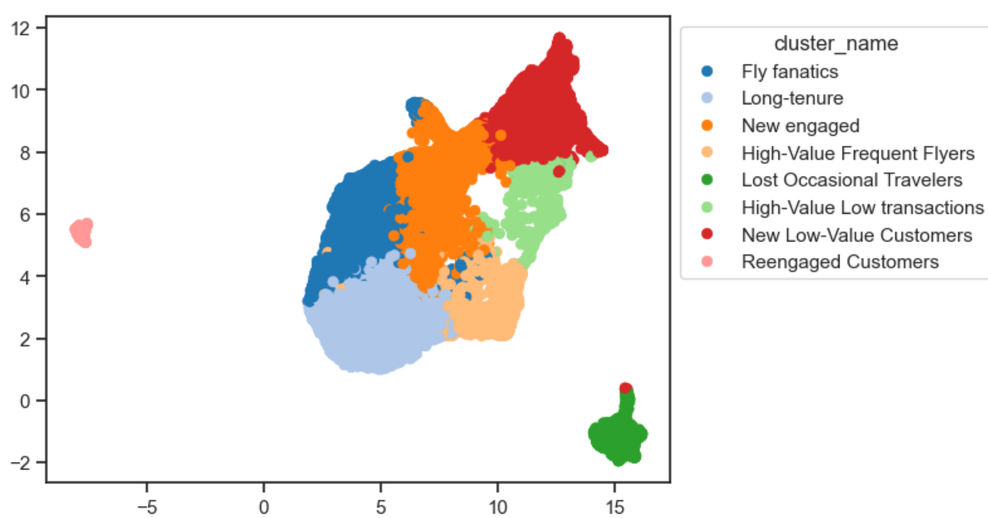
Num Clusters	SSW (Lower is better)	SSB (Higher is better)	Silhouette Score (Higher is better)	R^2 (Higher is better)
2	13278.963583	7162.849631	0.371533	0.350402
3	11432.778908	9009.049377	0.225013	0.440717
4	9687.692123	10754.120097	0.246920	0.526085
5	9120.805925	11321.045180	0.188525	0.553818
6	**7099.304829**	**13342.544617**	**0.236103**	**0.652709**
7	6654.947737	13786.873481	0.211302	0.674445
8	**5957.548430**	**14484.312825**	**0.253790**	**0.708563**
9	5525.746449	14916.077092	0.234027	0.729685
10	5319.091709	15122.725894	0.229096	0.739794



Annex6


cluster_name
Fly fanatics
Long-tenure
New engaged
High-Value Frequent Flyers
Lost Occasional Travelers
New Low-Value Customers

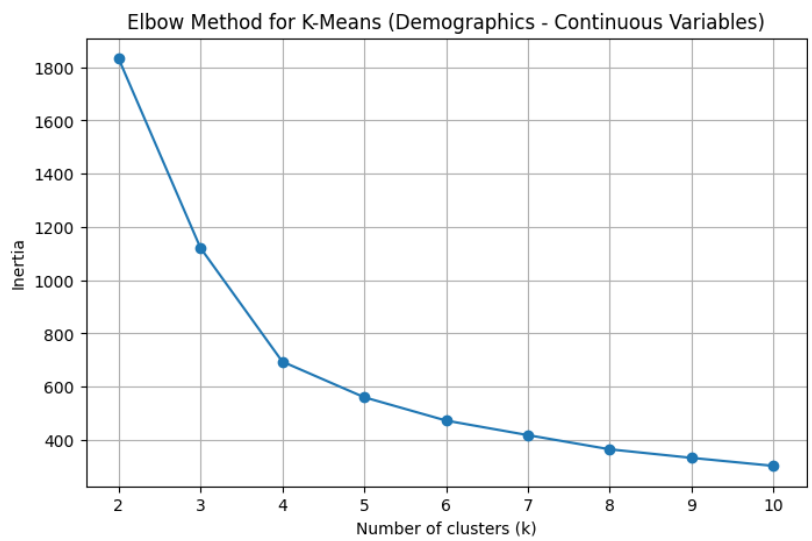

Cluster Profiling:
KMeans Clustering with 8 Customer Segments
Customer Lifetime Value -0.64 -0.65 -0.64 -0.51 0.50 -0.63 0.58 -0.48
Days_with_us -0.35 -0.78 0.62 -0.50 -0.73 -0.07 0.21 -0.88
Days_until_reengagement -1.00 -1.00 -1.00 -1.00 -1.00 -1.00 -1.00 1.00
Total_NumFlights -0.06 -0.72 0.09 -0.57 -0.60 0.21 0.12 -0.86
Total_DistanceKM -0.11 -0.73 0.05 -0.59 -0.62 0.23 0.09 -0.86
Perc_Flights_With_Companions -0.47 -0.46 -0.48 -0.46 -0.48 -0.52 -0.50 -0.39
Dollar_Cost_Points_Remaining -0.41 -0.80 -0.22 -0.68 -0.72 0.09 -0.16 -0.88
Recency_Days -0.90 -0.90 -0.91 0.87 -0.89 -0.93 -0.91 -0.91
New engaged
New Low-Value Customers
Long-tenure
Lost Occasional Travelers
High-Value Low transactions
Fly fanatics
High-Value Frequent Flyers
Reengaged Customers
Customer Segments

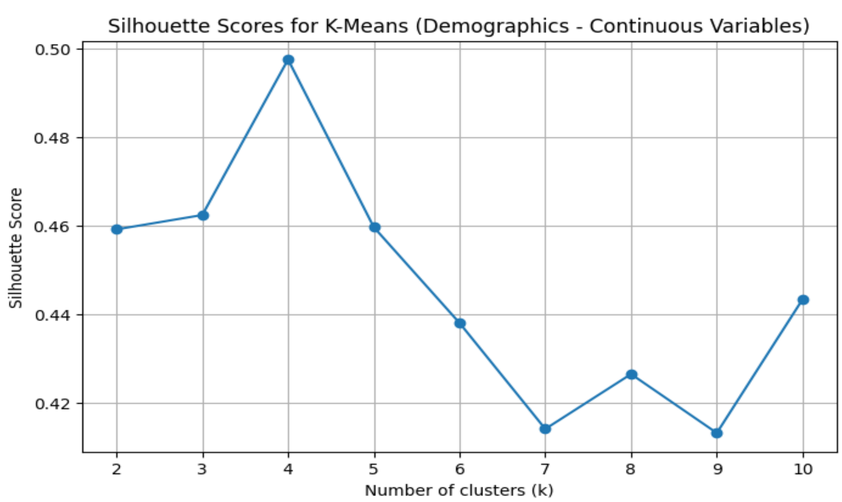

cluster_name
Fly fanatics
Long-tenure
New engaged
High-Value Frequent Flyers
Lost Occasional Travelers
High-Value Low transactions
New Low-Value Customers
Reengaged Customers

Annex7

Method	Num Clusters	R^2 (Higher is better)	Silhouette Score (Higher is better)	SSW (Lower is better)	SSB (Higher is better)
KMeans	6	0.652709	0.236103	7099.304829	13342.544617
KMeans	**8**	**0.708563**	**0.253790**	**5957.548430**	**14484.312825**
Hierarchical	5	0.598007	0.235498	8217.467498	12224.342627
Hierarchical	7	0.660301	0.213959	6944.064774	13497.745351

Annex8

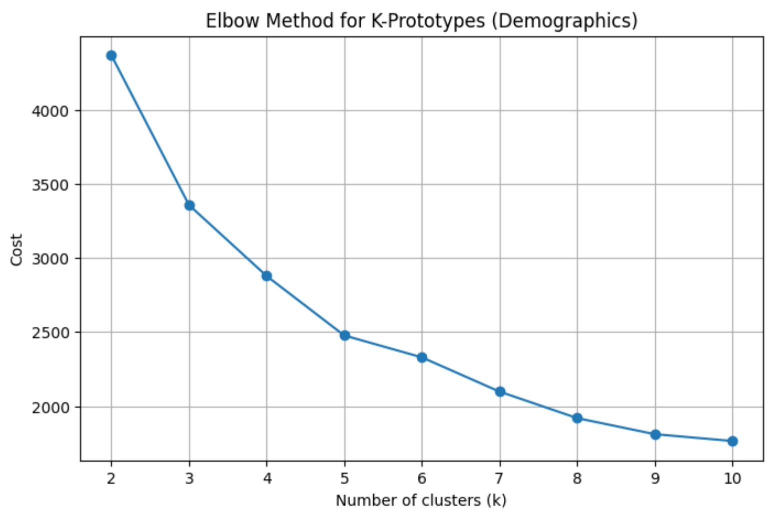


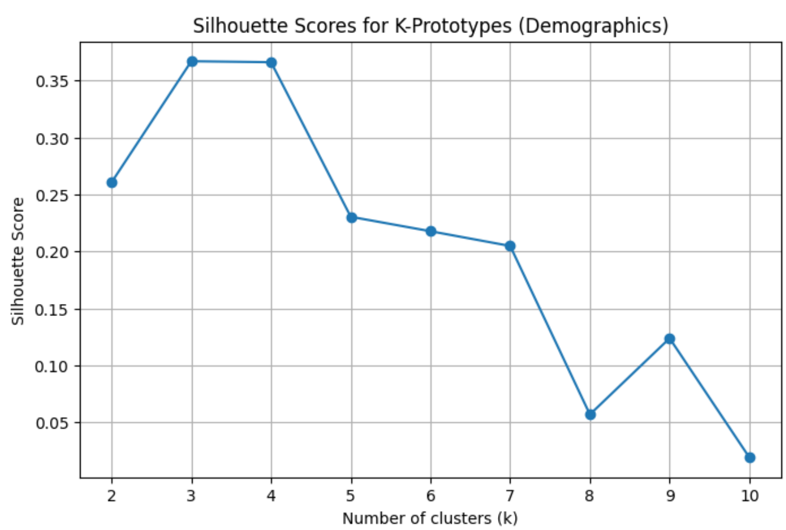


Annex9

Cluster Name	Income	Latitude	Longitude	Female (%)	Male (%)	Divorced (%)	Married (%)	Single (%)	Bachelor (%)	College (%)	Doctor (%)	High School or Below (%)	Master (%)
High-Income East	68,161.63	44.938814	-74.845057	49.79	50.21	15.62	68.03	16.35	100.00	0.00	0.00	0.00	0.00
High-Income West	68,301.87	50.611912	-118.011487	51.70	48.30	15.24	68.14	16.62	100.00	0.00	0.00	0.00	0.00
Low-Income East	15,291.40	44.955724	-75.117118	49.59	50.41	14.69	50.82	34.49	35.18	43.74	7.75	7.96	5.37
Low-Income West	15,086.70	50.637498	-117.907141	50.92	49.08	14.59	51.03	34.38	34.67	44.55	7.34	8.33	5.11

Annex10

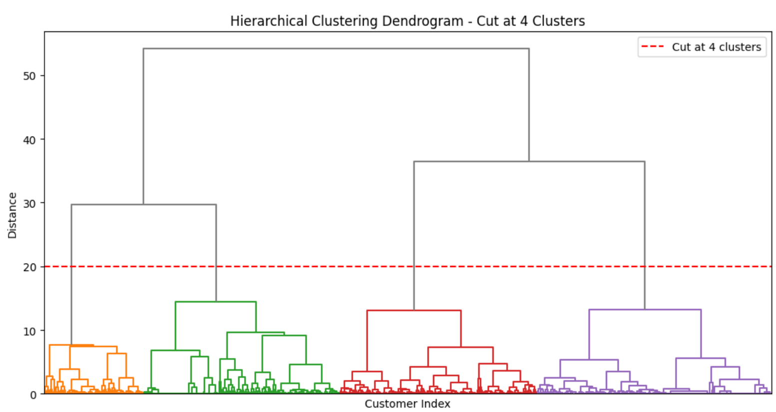


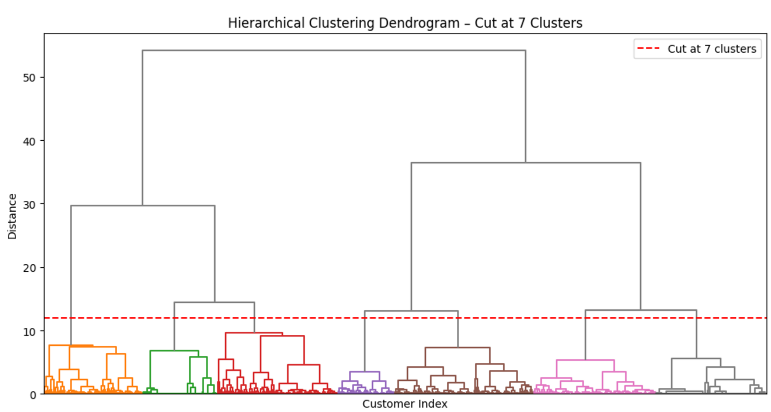


Annex11

Cluster Name	Income	Latitude	Longitude	Female (%)	Male (%)	Divorced (%)	Married (%)	Single (%)	Bachelor (%)	College (%)	Doctor (%)	High School or Below (%)	Master (%)
High-Income East (Bachelor, Male, Married)	55,237.15	50.624480	-118.561607	42.55	57.45	15.49	71.00	13.50	93.38	0.00	2.39	2.81	1.42
High-Income West (Bachelor, Female, Married)	54,751.58	45.070769	-75.490398	57.27	42.73	15.76	71.10	13.14	93.13	0.00	2.49	2.76	1.61
Low-Income East (College, Female, Single)	6,401.41	50.622377	-118.394843	63.61	36.39	13.81	34.94	51.26	7.01	70.97	7.76	8.47	5.78
Low-Income West (College, Male, Single)	6,204.34	45.056257	-75.627014	37.56	62.44	13.79	34.49	51.71	6.35	72.14	7.91	7.86	5.74

Annex12





Annex13

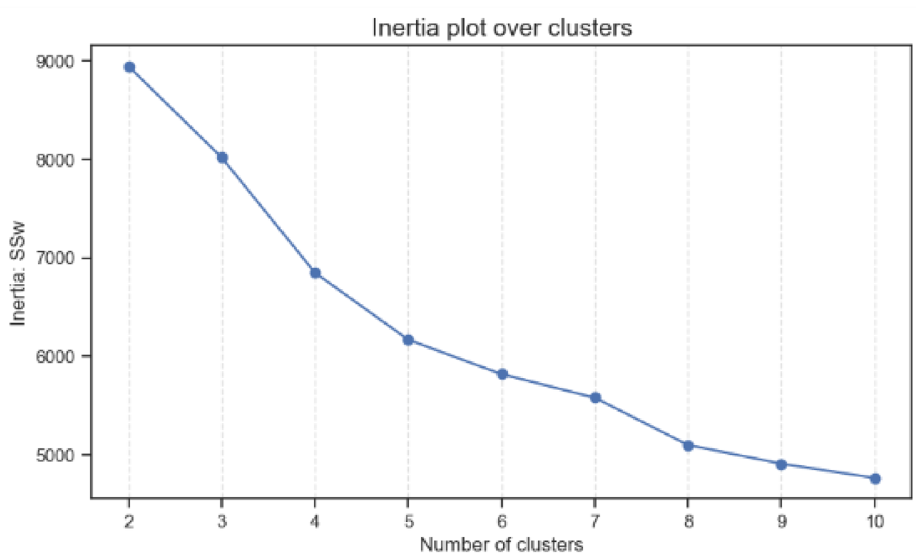
Cluster Name	Income	Latitude	Longitude	Female (%)	Male (%)	Divorced (%)	Married (%)	Single (%)	Bachelor (%)	College (%)	Doctor (%)	High School or Below (%)	Master (%)
High-Income East (Bachelor)	57,364.96	45.009405	-73.867861	49.61	50.39	15.68	67.70	16.62	100.00	0.00	0.00	0.00	0.00
High-Income West (Bachelor)	73,355.63	50.412842	-117.890791	51.99	48.01	14.81	68.81	16.38	100.00	0.00	0.00	0.00	0.00
High-Income2 East (Bachelor)	87,706.52	44.764162	-75.240090	49.23	50.77	16.38	68.39	15.22	100.00	0.00	0.00	0.00	0.00
Low-Income East (Bachelor)	26,088.81	44.688223	-75.560141	51.30	48.70	19.22	63.97	16.80	58.74	0.00	15.13	15.70	10.43
Low-Income East (College)	541.75	44.914415	-74.276288	48.02	51.98	9.22	34.24	56.53	1.15	97.66	0.28	0.44	0.48
Low-Income West (Bachelor)	31,303.30	50.797223	-116.262017	51.21	48.79	18.77	64.09	17.14	69.66	1.12	10.52	11.57	7.12
Low-Income West (College)	0.00	50.304658	-116.542891	50.35	49.65	8.41	34.82	56.76	0.00	100.00	0.00	0.00	0.00

Annex14

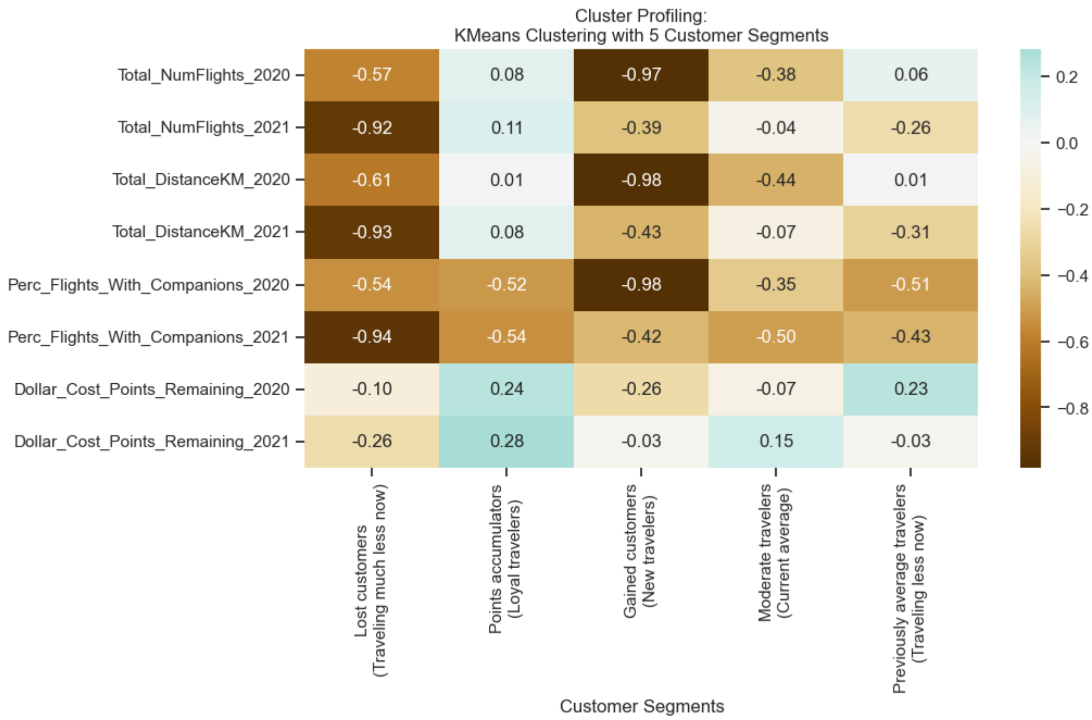
Clustering Method	Silhouette Score (-1:1)	Calinski-Harabasz Score (higher is better)	Davies-Bouldin Score (lower is better)
K-means (K=4)	**0.4975**	**20,818.1160**	**0.7268**
K-Prototypes (K=4)	0.3661	13,335.9157	0.8443
Hierarchical (K=4)	0.4752	19,030.1219	0.7379
Hierarchical (K=7)	0.3838	16,763.3162	0.9059

Annex15

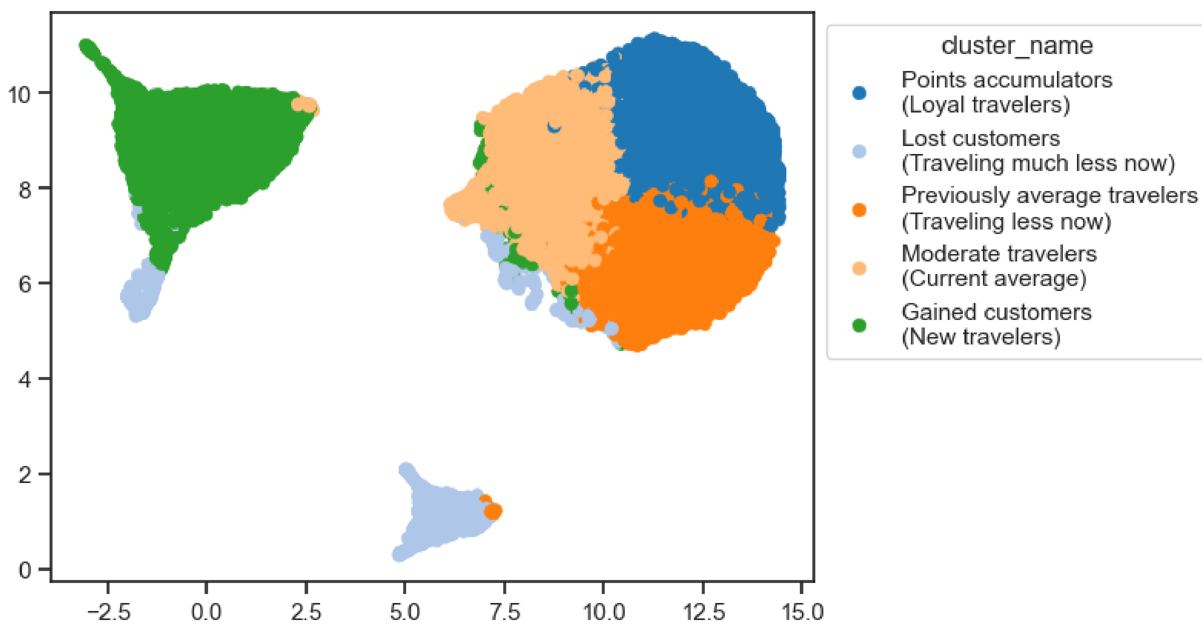
Num Clusters	SSW (Lower is better)	SSB (Higher is better)	Silhouette Score (Higher is better)	R^2 (Higher is better)
2	8949.187994	5488.883353	0.396568	0.380167
3	8021.657726	6416.431143	0.365349	0.444411
4	6851.131082	7586.944656	0.220296	0.525482
5	**6170.571521**	**8267.509176**	**0.192725**	**0.572619**
6	5820.794943	8617.286377	0.174445	0.596845
7	5581.327994	8856.819010	0.164267	0.613435
8	**5100.491436**	**9337.586982**	**0.165211**	**0.646734**
9	4911.352824	9526.720153	0.161703	0.659834



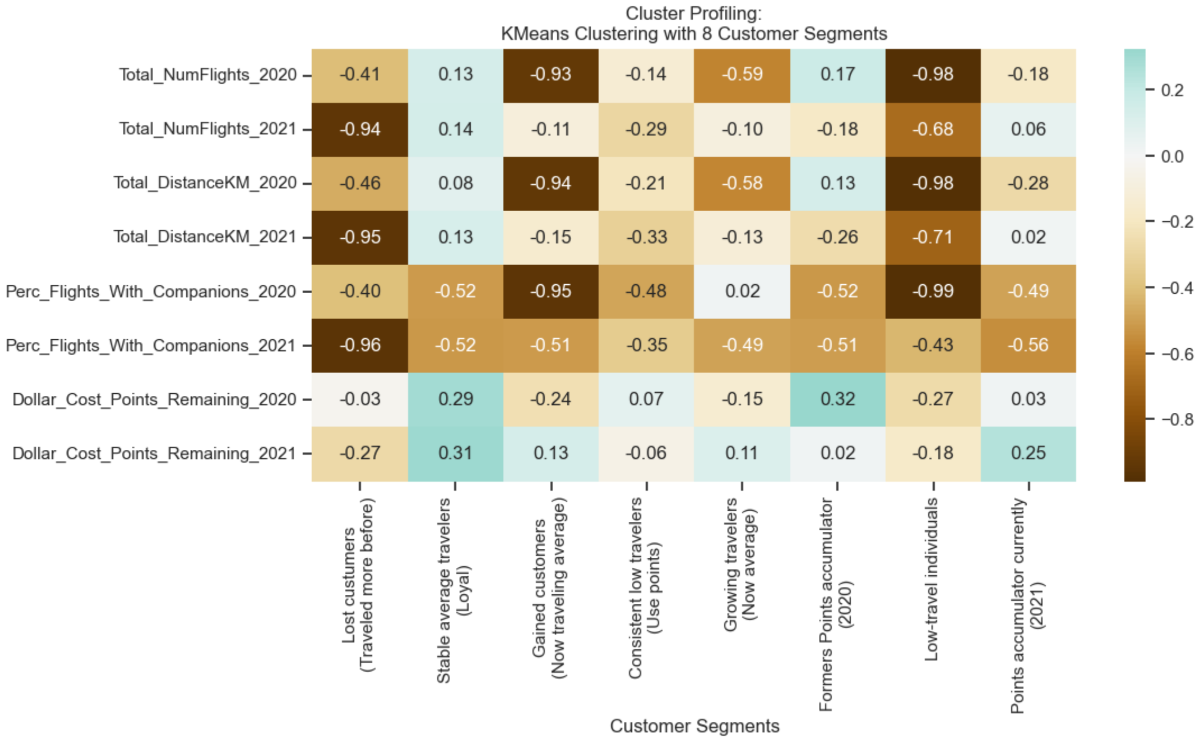
Annex16

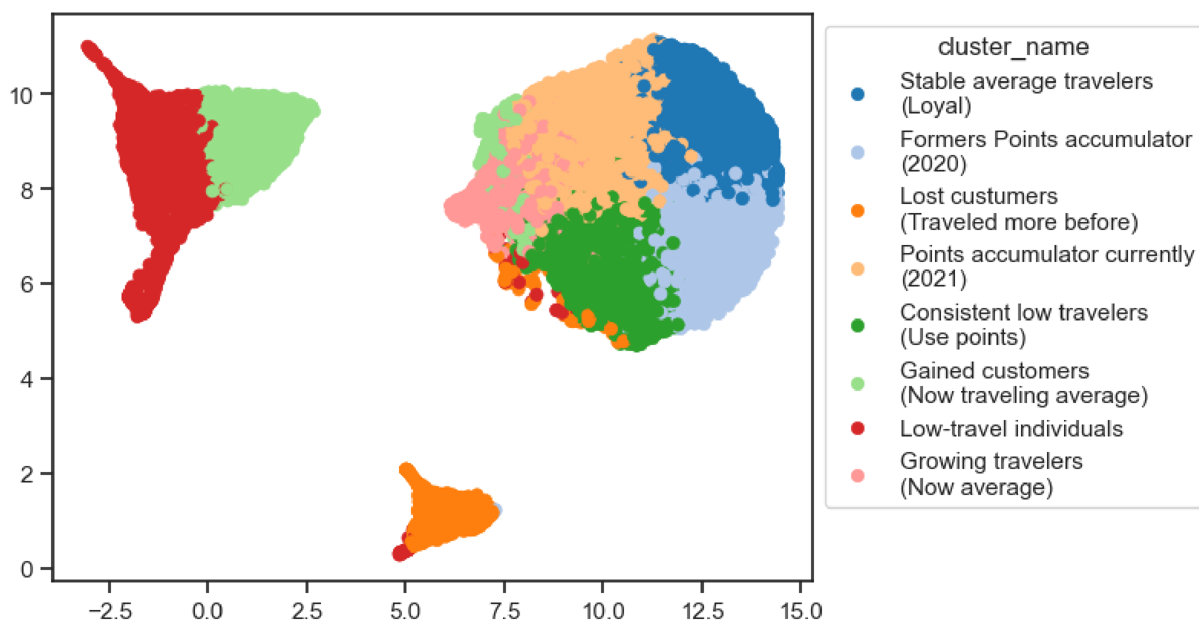


Cluster Profiling:
KMeans Clustering with 5 Customer Segments
Total_NumFlights_2020
Total_NumFlights_2021
Total_DistanceKM_2020
Total_DistanceKM_2021
Perc_Flights_With_Companions_2020
Perc_Flights_With_Companions_2021
Dollar_Cost_Points_Remaining_2020
Dollar_Cost_Points_Remaining_2021
-0.57
0.08
-0.97
-0.38
0.06
-0.92
0.11
-0.39
-0.04
-0.26
-0.61
0.01
-0.98
-0.44
0.01
-0.93
0.08
-0.43
-0.07
-0.31
-0.54
-0.52
-0.98
-0.35
-0.51
-0.94
-0.54
-0.42
-0.50
-0.43
-0.10
0.24
-0.26
-0.07
0.23
-0.26
0.28
-0.03
0.15
-0.03
0.2
0.0
−0.2
−0.4
−0.6
−0.8
Lost customers
(Traveling much less now)
Points accumulators
(Loyal travelers)
Gained customers
(New travelers)
Moderate travelers
(Current average)
Previously average travelers
(Traveling less now)
Customer Segments

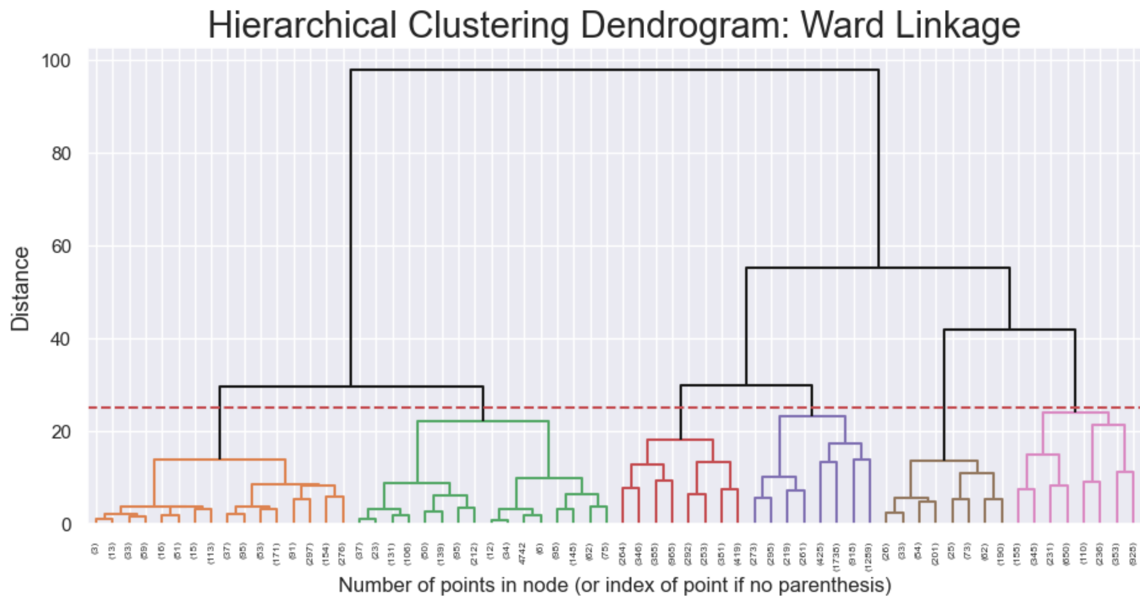


10
8
6
4
2
0
−2.5
0.0
2.5
5.0
7.5
10.0
12.5
15.0
cluster_name
Points accumulators
(Loyal travelers)
Lost customers
(Traveling much less now)
Previously average travelers
(Traveling less now)
Moderate travelers
(Current average)
Gained customers
(New travelers)

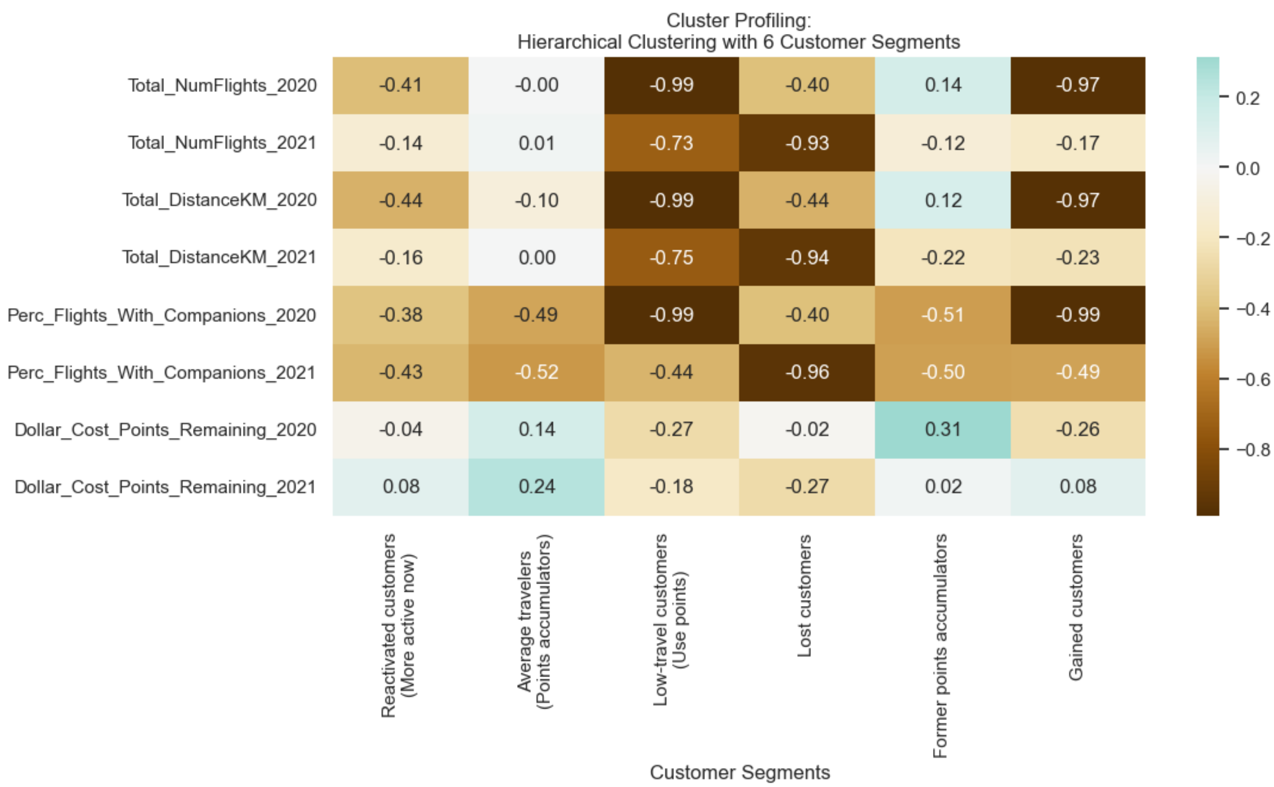

Cluster Profiling:
KMeans Clustering with 8 Customer Segments
Total_NumFlights_2020 -0.41 0.13 -0.93 -0.14 -0.59 0.17 -0.98 -0.18
Total_NumFlights_2021 -0.94 0.14 -0.11 -0.29 -0.10 -0.18 -0.68 0.06
Total_DistanceKM_2020 -0.46 0.08 -0.94 -0.21 -0.58 0.13 -0.98 -0.28
Total_DistanceKM_2021 -0.95 0.13 -0.15 -0.33 -0.13 -0.26 -0.71 0.02
Perc_Flights_With_Companions_2020 -0.40 -0.52 -0.95 -0.48 0.02 -0.52 -0.99 -0.49
Perc_Flights_With_Companions_2021 -0.96 -0.52 -0.51 -0.35 -0.49 -0.51 -0.43 -0.56
Dollar_Cost_Points_Remaining_2020 -0.03 0.29 -0.24 0.07 -0.15 0.32 -0.27 0.03
Dollar_Cost_Points_Remaining_2021 -0.27 0.31 0.13 -0.06 0.11 0.02 -0.18 0.25
Lost custumers (Traveled more before)
Stable average travelers (Loyal)
Gained customers (Now traveling average)
Consistent low travelers (Use points)
Growing travelers (Now average)
Formers Points accumulator (2020)
Low-travel individuals
Points accumulator currently (2021)
Customer Segments
0.2
0.0
−0.2
−0.4
−0.6
−0.8

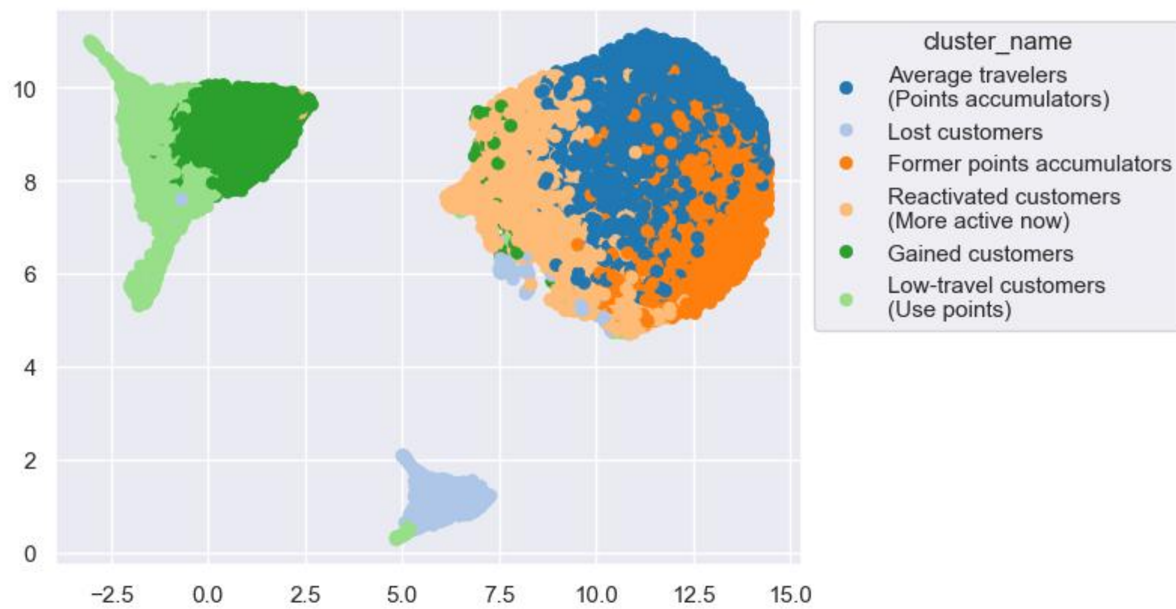

cluster_name
Stable average travelers (Loyal)
Formers Points accumulator (2020)
Lost custumers (Traveled more before)
Points accumulator currently (2021)
Consistent low travelers (Use points)
Gained customers (Now traveling average)
Low-travel individuals
Growing travelers (Now average)
10
8
6
4
2
0
−2.5
0.0
2.5
5.0
7.5
10.0
12.5
15.0

Annex17



Annex18





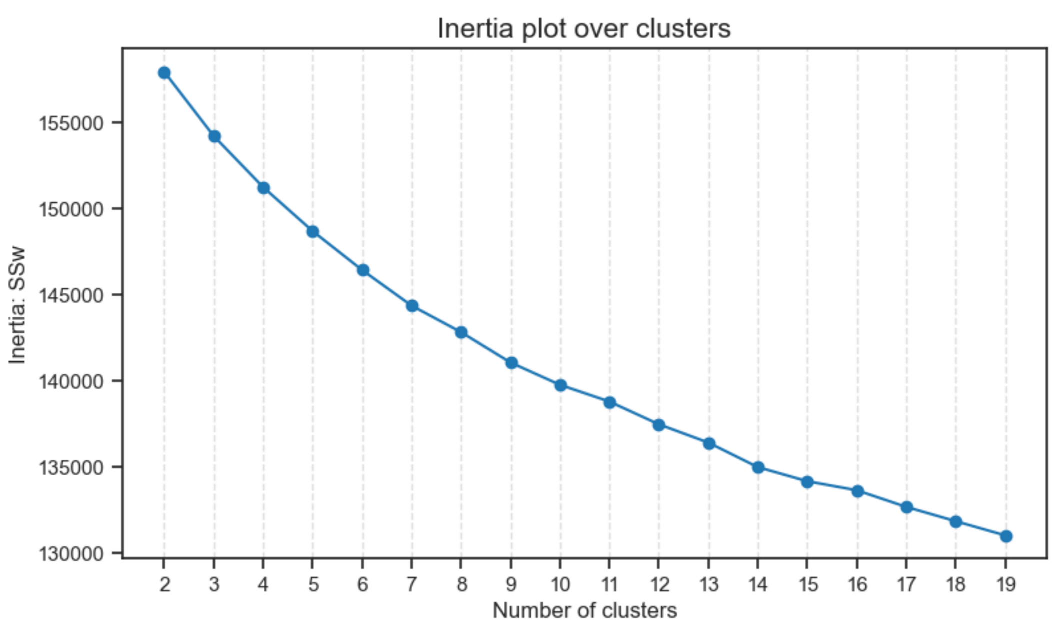
Annex19

Method	Num Clusters	R^2 (Higher is better)	Silhouette Score (Higher is better)	SSW (Lower is better)	SSB (Higher is better)
KMeans	5	0.572619	0.192725	6170.571521	8267.509176
KMeans	**8**	**0.646734**	**0.165211**	**5100.491436**	**9337.586982**
Hierarchical	6	0.559739	0.125167	6356.524338	8081.542355

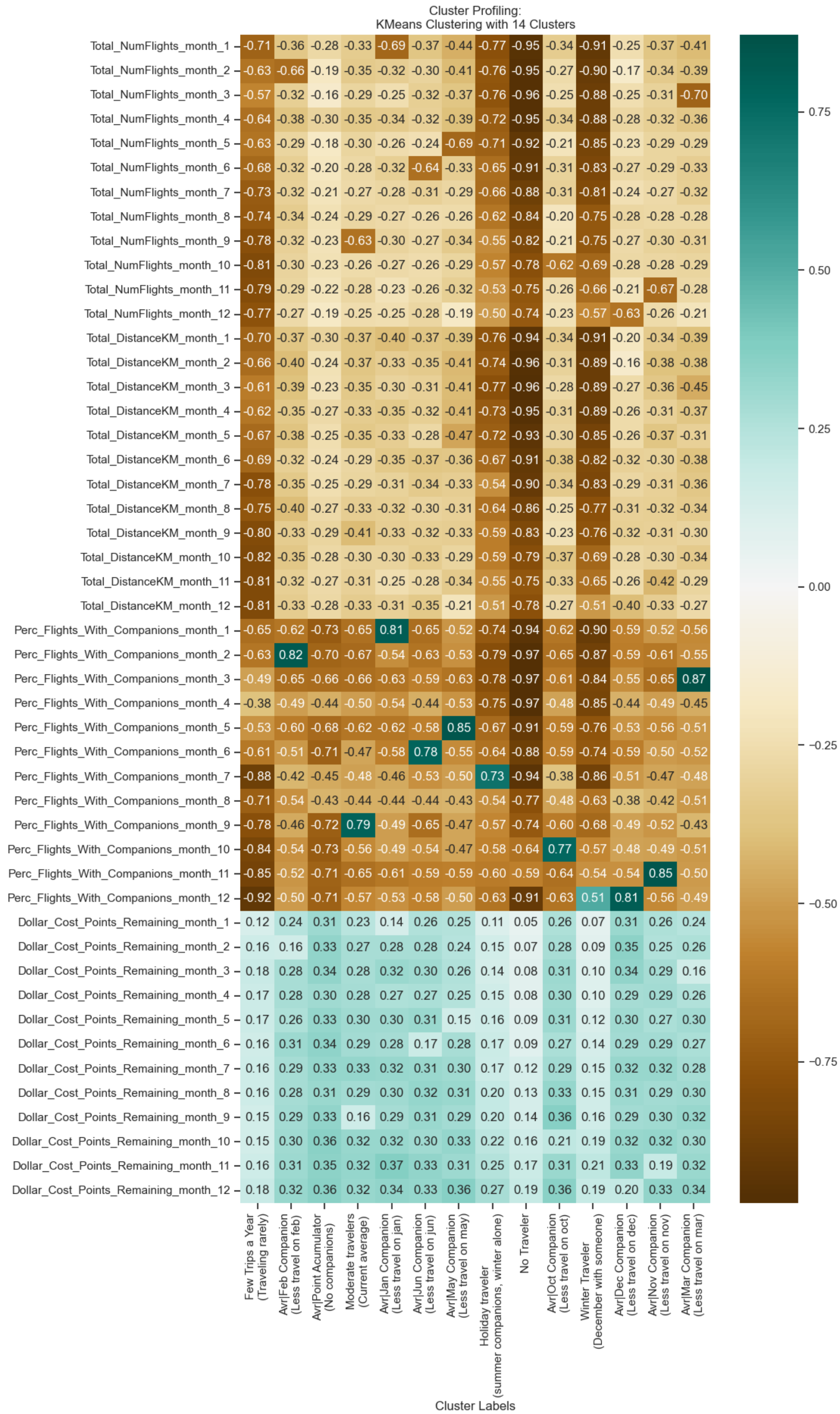
Annex20

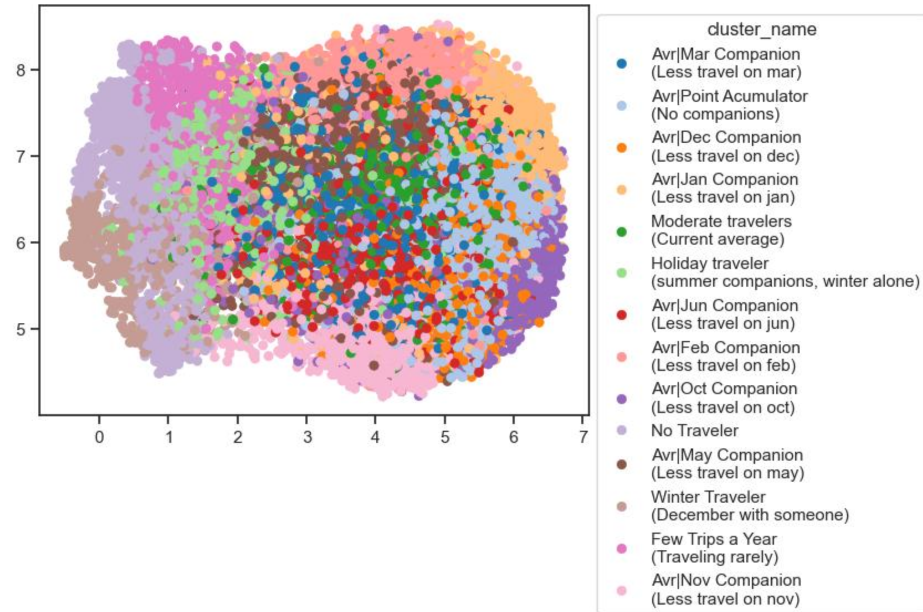
Number Clusters	SSW (Lower is better)	SSB (Higher is better)	Silhouette Score	R^2
2	157,954.4165	20,298.8911	0.098467	0.113877
3	154,226.0334	24,027.3687	0.072249	0.134793
4	151,252.3064	27,001.0000	0.074001	0.151475
5	148,698.4926	29,554.8283	0.075853	0.165802

6	146,438.8141	31,814.5223	0.075887	0.178479
7	144,368.9939	33,884.2830	0.076508	0.190091
8	142,828.2628	35,425.0458	0.057496	0.198734
9	141,065.0892	37,188.1878	0.049909	0.208626
10	139,768.3182	38,484.9633	0.046648	0.215900
11	138,794.2916	39,458.9854	0.034431	0.221365
12	137,471.6882	40,781.6193	0.034310	0.228785
13	136,411.0848	41,842.1922	0.032825	0.234734
14	**134,987.2570**	**43,266.0722**	**0.033037**	**0.242722**
15	134,166.0999	44,087.1771	0.031549	0.247329
16	133,643.1760	44,610.1010	0.028941	0.250262
17	132,675.4491	45,577.8279	0.029834	0.255691
18	131,844.8350	46,408.4419	0.026687	0.260351
19	131,020.6638	47,232.6132	0.026088	0.264975

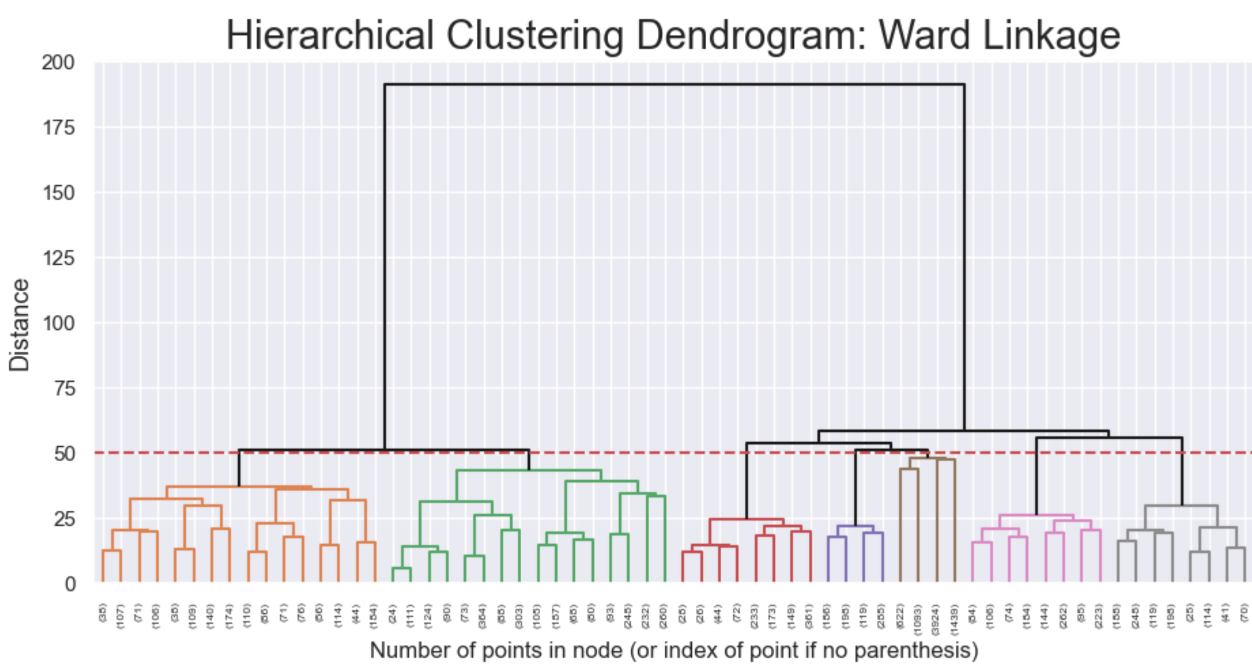


Annex21

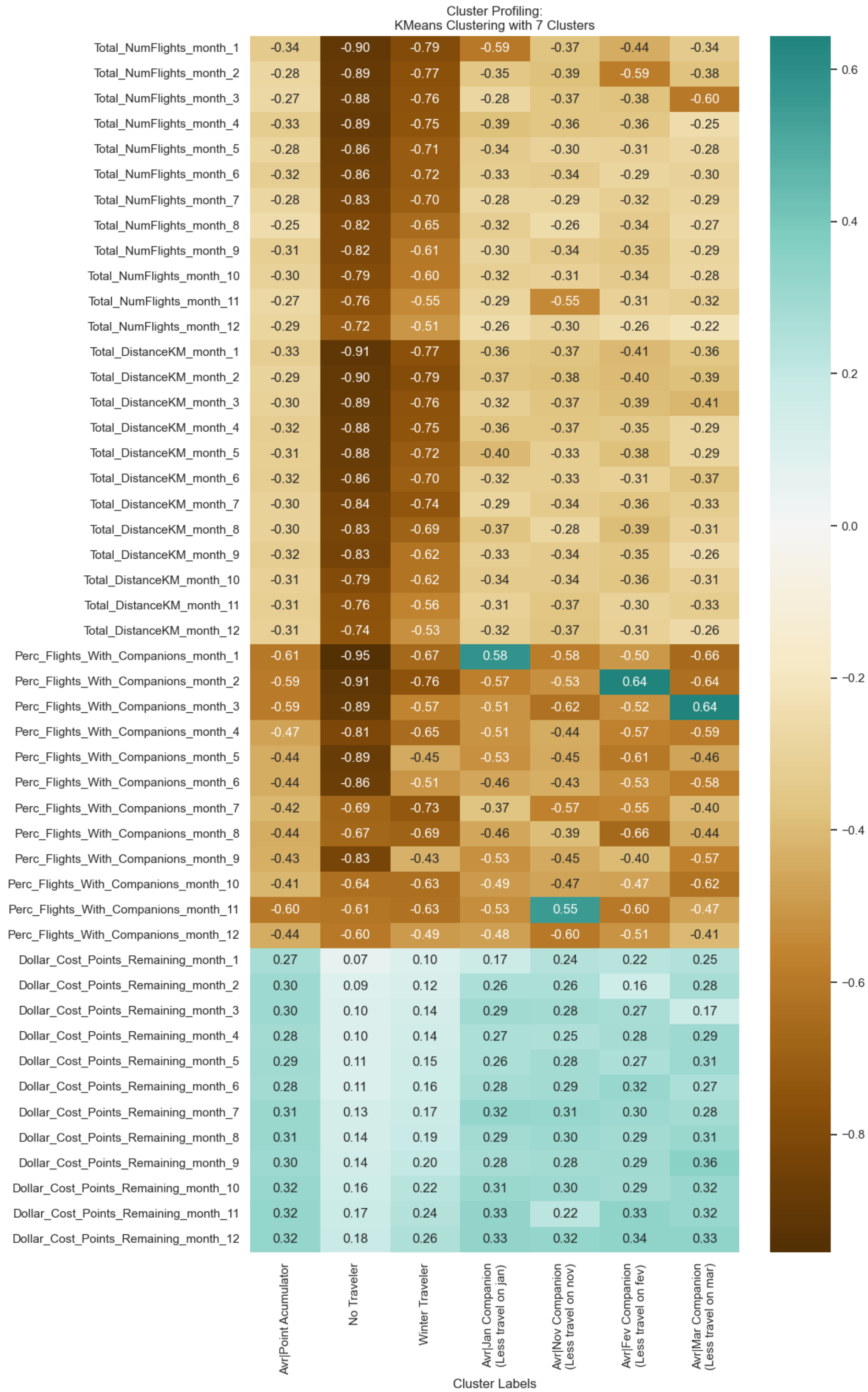

Cluster Profiling:
KMeans Clustering with 14 Clusters
Total_NumFlights_month_1
Total_NumFlights_month_2
Total_NumFlights_month_3
Total_NumFlights_month_4
Total_NumFlights_month_5
Total_NumFlights_month_6
Total_NumFlights_month_7
Total_NumFlights_month_8
Total_NumFlights_month_9
Total_NumFlights_month_10
Total_NumFlights_month_11
Total_NumFlights_month_12
Total_DistanceKM_month_1
Total_DistanceKM_month_2
Total_DistanceKM_month_3
Total_DistanceKM_month_4
Total_DistanceKM_month_5
Total_DistanceKM_month_6
Total_DistanceKM_month_7
Total_DistanceKM_month_8
Total_DistanceKM_month_9
Total_DistanceKM_month_10
Total_DistanceKM_month_11
Total_DistanceKM_month_12
Perc_Flights_With_Companions_month_1
Perc_Flights_With_Companions_month_2
Perc_Flights_With_Companions_month_3
Perc_Flights_With_Companions_month_4
Perc_Flights_With_Companions_month_5
Perc_Flights_With_Companions_month_6
Perc_Flights_With_Companions_month_7
Perc_Flights_With_Companions_month_8
Perc_Flights_With_Companions_month_9
Perc_Flights_With_Companions_month_10
Perc_Flights_With_Companions_month_11
Perc_Flights_With_Companions_month_12
Dollar_Cost_Points_Remaining_month_1
Dollar_Cost_Points_Remaining_month_2
Dollar_Cost_Points_Remaining_month_3
Dollar_Cost_Points_Remaining_month_4
Dollar_Cost_Points_Remaining_month_5
Dollar_Cost_Points_Remaining_month_6
Dollar_Cost_Points_Remaining_month_7
Dollar_Cost_Points_Remaining_month_8
Dollar_Cost_Points_Remaining_month_9
Dollar_Cost_Points_Remaining_month_10
Dollar_Cost_Points_Remaining_month_11
Dollar_Cost_Points_Remaining_month_12
Few Trips a Year (Traveling rarely)
Avr|Feb Companion (Less travel on feb)
Avr|Point Acumulator (No companions)
Moderate travelers (Current average)
Avr|Jan Companion (Less travel on jan)
Avr|Jun Companion (Less travel on jun)
Avr|May Companion (Less travel on may)
Holiday traveler (summer companions, winter alone)
No Traveler
Avr|Oct Companion (Less travel on oct)
Winter Traveler (December with someone)
Avr|Dec Companion (Less travel on dec)
Avr|Nov Companion (Less travel on nov)
Avr|Mar Companion (Less travel on mar)
Cluster Labels
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75

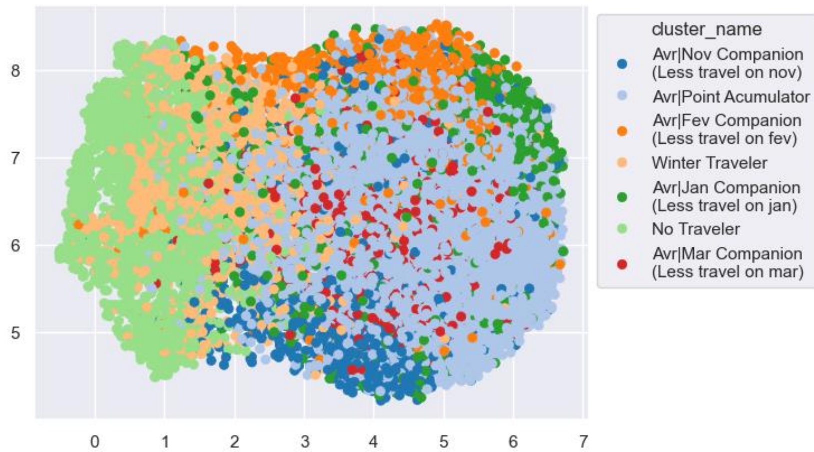

cluster_name
Avr|Mar Companion (Less travel on mar)
Avr|Point Acumulator (No companions)
Avr|Dec Companion (Less travel on dec)
Avr|Jan Companion (Less travel on jan)
Moderate travelers (Current average)
Holiday traveler (summer companions, winter alone)
Avr|Jun Companion (Less travel on jun)
Avr|Feb Companion (Less travel on feb)
Avr|Oct Companion (Less travel on oct)
No Traveler
Avr|May Companion (Less travel on may)
Winter Traveler (December with someone)
Few Trips a Year (Traveling rarely)
Avr|Nov Companion (Less travel on nov)

Annex22

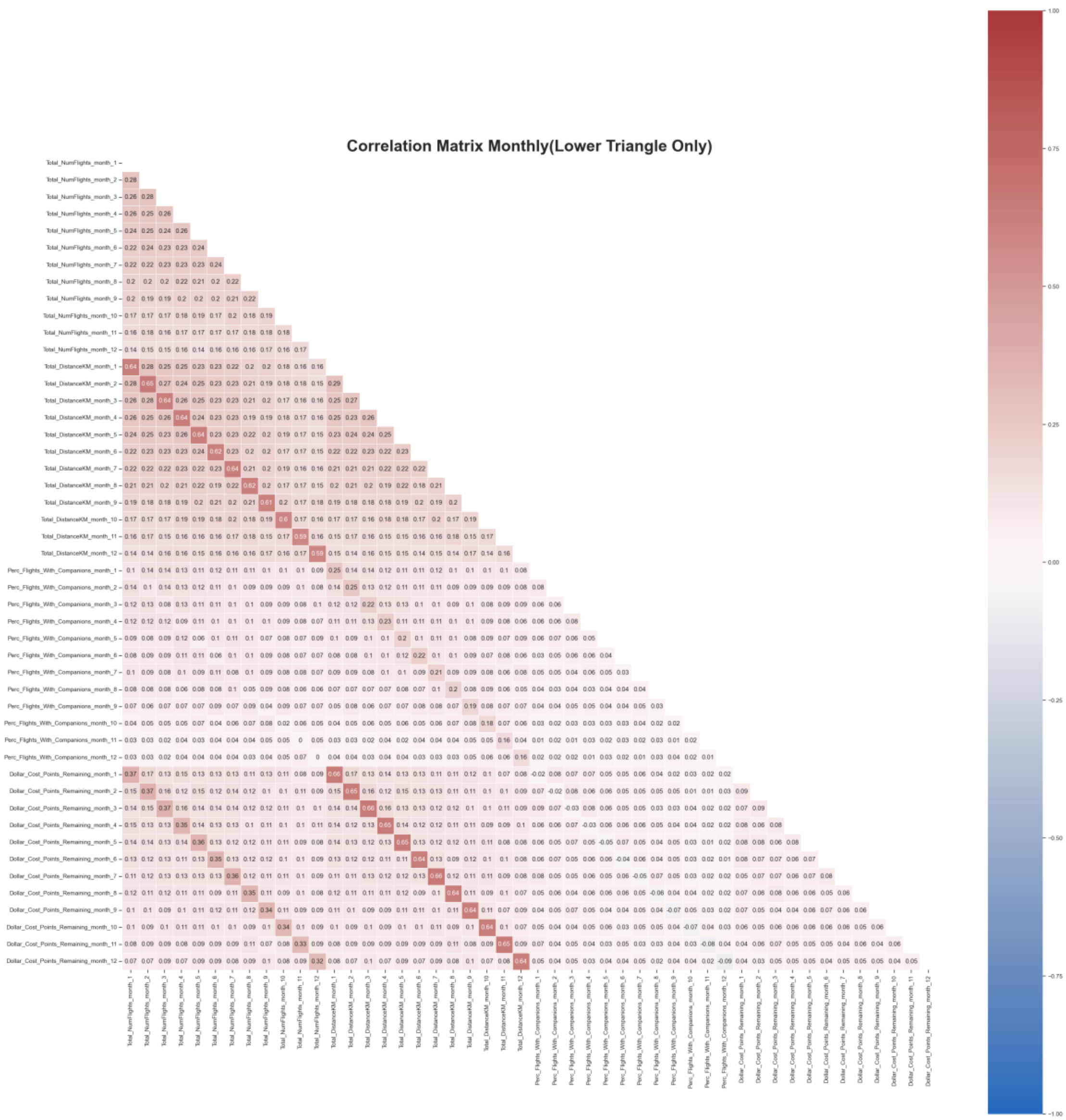

Hierarchical Clustering Dendrogram: Ward Linkage
Distance
Number of points in node (or index of point if no parenthesis)

Annex23

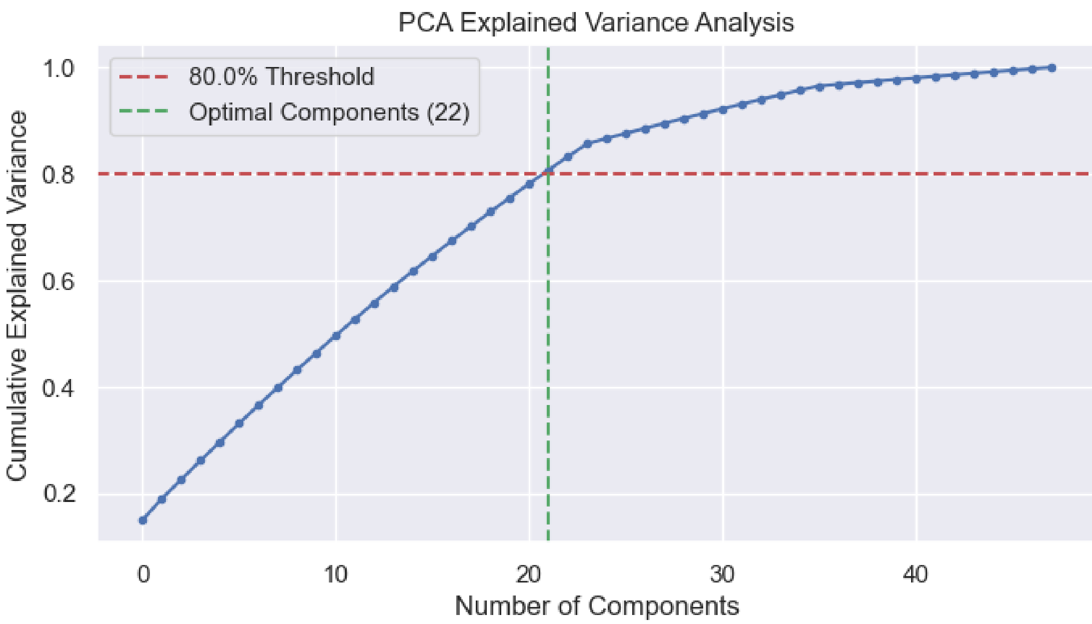

Cluster Profiling:
KMeans Clustering with 7 Clusters
Total_NumFlights_month_1 -0.34 -0.90 -0.79 -0.59 -0.37 -0.44 -0.34
Total_NumFlights_month_2 -0.28 -0.89 -0.77 -0.35 -0.39 -0.59 -0.38
Total_NumFlights_month_3 -0.27 -0.88 -0.76 -0.28 -0.37 -0.38 -0.60
Total_NumFlights_month_4 -0.33 -0.89 -0.75 -0.39 -0.36 -0.36 -0.25
Total_NumFlights_month_5 -0.28 -0.86 -0.71 -0.34 -0.30 -0.31 -0.28
Total_NumFlights_month_6 -0.32 -0.86 -0.72 -0.33 -0.34 -0.29 -0.30
Total_NumFlights_month_7 -0.28 -0.83 -0.70 -0.28 -0.29 -0.32 -0.29
Total_NumFlights_month_8 -0.25 -0.82 -0.65 -0.32 -0.26 -0.34 -0.27
Total_NumFlights_month_9 -0.31 -0.82 -0.61 -0.30 -0.34 -0.35 -0.29
Total_NumFlights_month_10 -0.30 -0.79 -0.60 -0.32 -0.31 -0.34 -0.28
Total_NumFlights_month_11 -0.27 -0.76 -0.55 -0.29 -0.55 -0.31 -0.32
Total_NumFlights_month_12 -0.29 -0.72 -0.51 -0.26 -0.30 -0.26 -0.22
Total_DistanceKM_month_1 -0.33 -0.91 -0.77 -0.36 -0.37 -0.41 -0.36
Total_DistanceKM_month_2 -0.29 -0.90 -0.79 -0.37 -0.38 -0.40 -0.39
Total_DistanceKM_month_3 -0.30 -0.89 -0.76 -0.32 -0.37 -0.39 -0.41
Total_DistanceKM_month_4 -0.32 -0.88 -0.75 -0.36 -0.37 -0.35 -0.29
Total_DistanceKM_month_5 -0.31 -0.88 -0.72 -0.40 -0.33 -0.38 -0.29
Total_DistanceKM_month_6 -0.32 -0.86 -0.70 -0.32 -0.33 -0.31 -0.37
Total_DistanceKM_month_7 -0.30 -0.84 -0.74 -0.29 -0.34 -0.36 -0.33
Total_DistanceKM_month_8 -0.30 -0.83 -0.69 -0.37 -0.28 -0.39 -0.31
Total_DistanceKM_month_9 -0.32 -0.83 -0.62 -0.33 -0.34 -0.35 -0.26
Total_DistanceKM_month_10 -0.31 -0.79 -0.62 -0.34 -0.34 -0.36 -0.31
Total_DistanceKM_month_11 -0.31 -0.76 -0.56 -0.31 -0.37 -0.30 -0.33
Total_DistanceKM_month_12 -0.31 -0.74 -0.53 -0.32 -0.37 -0.31 -0.26
Perc_Flights_With_Companions_month_1 -0.61 -0.95 -0.67 0.58 -0.58 -0.50 -0.66
Perc_Flights_With_Companions_month_2 -0.59 -0.91 -0.76 -0.57 -0.53 0.64 -0.64
Perc_Flights_With_Companions_month_3 -0.59 -0.89 -0.57 -0.51 -0.62 -0.52 0.64
Perc_Flights_With_Companions_month_4 -0.47 -0.81 -0.65 -0.51 -0.44 -0.57 -0.59
Perc_Flights_With_Companions_month_5 -0.44 -0.89 -0.45 -0.53 -0.45 -0.61 -0.46
Perc_Flights_With_Companions_month_6 -0.44 -0.86 -0.51 -0.46 -0.43 -0.53 -0.58
Perc_Flights_With_Companions_month_7 -0.42 -0.69 -0.73 -0.37 -0.57 -0.55 -0.40
Perc_Flights_With_Companions_month_8 -0.44 -0.67 -0.69 -0.46 -0.39 -0.66 -0.44
Perc_Flights_With_Companions_month_9 -0.43 -0.83 -0.43 -0.53 -0.45 -0.40 -0.57
Perc_Flights_With_Companions_month_10 -0.41 -0.64 -0.63 -0.49 -0.47 -0.47 -0.62
Perc_Flights_With_Companions_month_11 -0.60 -0.61 -0.63 -0.53 0.55 -0.60 -0.47
Perc_Flights_With_Companions_month_12 -0.44 -0.60 -0.49 -0.48 -0.60 -0.51 -0.41
Dollar_Cost_Points_Remaining_month_1 0.27 0.07 0.10 0.17 0.24 0.22 0.25
Dollar_Cost_Points_Remaining_month_2 0.30 0.09 0.12 0.26 0.26 0.16 0.28
Dollar_Cost_Points_Remaining_month_3 0.30 0.10 0.14 0.29 0.28 0.27 0.17
Dollar_Cost_Points_Remaining_month_4 0.28 0.10 0.14 0.27 0.25 0.28 0.29
Dollar_Cost_Points_Remaining_month_5 0.29 0.11 0.15 0.26 0.28 0.27 0.31
Dollar_Cost_Points_Remaining_month_6 0.28 0.11 0.16 0.28 0.29 0.32 0.27
Dollar_Cost_Points_Remaining_month_7 0.31 0.13 0.17 0.32 0.31 0.30 0.28
Dollar_Cost_Points_Remaining_month_8 0.31 0.14 0.19 0.29 0.30 0.29 0.31
Dollar_Cost_Points_Remaining_month_9 0.30 0.14 0.20 0.28 0.28 0.29 0.36
Dollar_Cost_Points_Remaining_month_10 0.32 0.16 0.22 0.31 0.30 0.29 0.32
Dollar_Cost_Points_Remaining_month_11 0.32 0.17 0.24 0.33 0.22 0.33 0.32
Dollar_Cost_Points_Remaining_month_12 0.32 0.18 0.26 0.33 0.32 0.34 0.33
Avr|Point Acumulator
No Traveler
Winter Traveler
Avr|Jan Companion (Less travel on jan)
Avr|Nov Companion (Less travel on nov)
Avr|Fev Companion (Less travel on fev)
Avr|Mar Companion (Less travel on mar)
Cluster Labels
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8


cluster_name
Avr|Nov Companion (Less travel on nov)
Avr|Point Acumulator
Avr|Fev Companion (Less travel on fev)
Winter Traveler
Avr|Jan Companion (Less travel on jan)
No Traveler
Avr|Mar Companion (Less travel on mar)
8
7
6
5
0
1
2
3
4
5
6
7

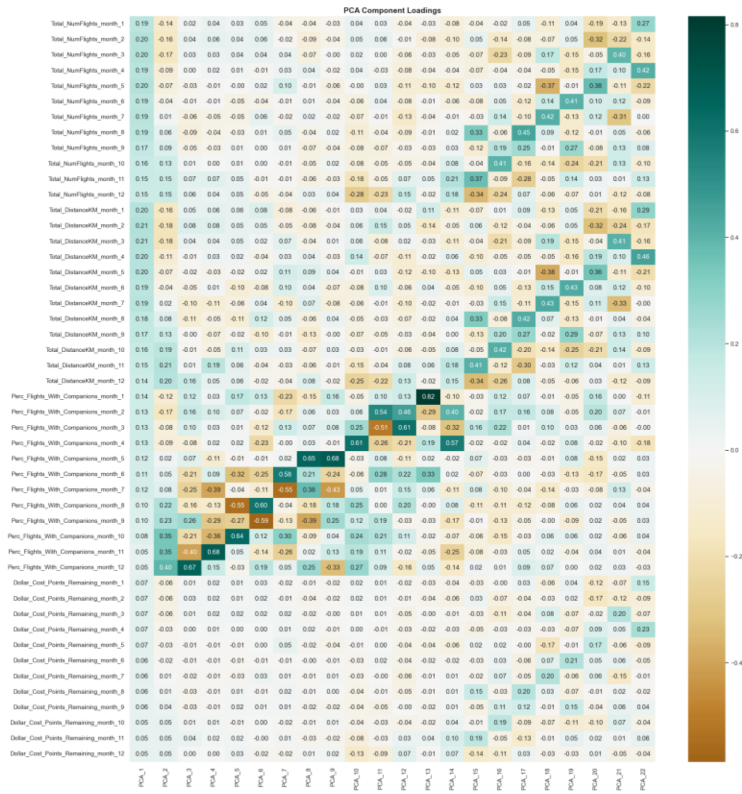
Annex24

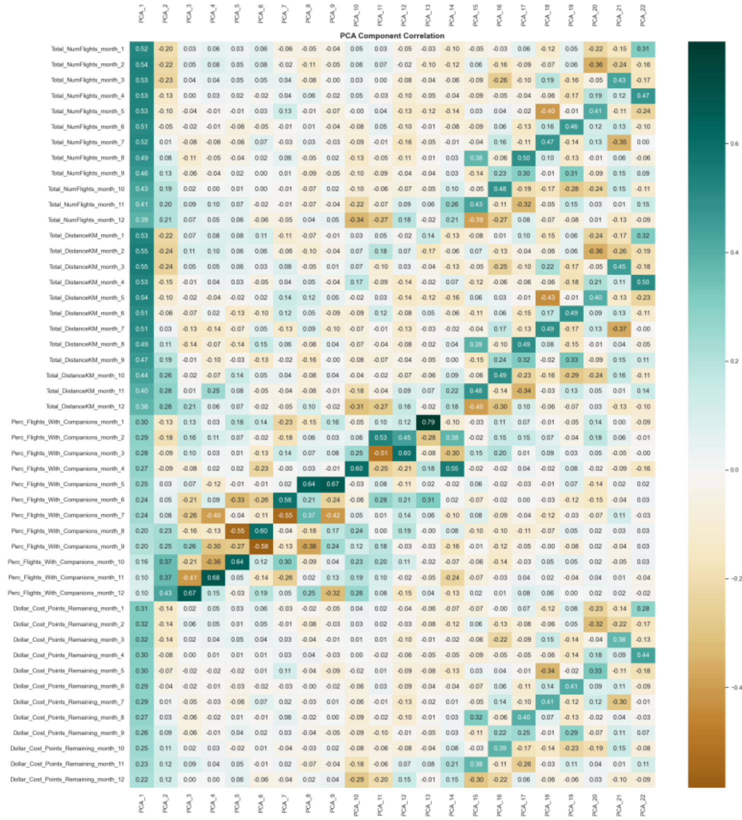

Correlation Matrix Monthly(Lower Triangle Only)
1.00
0.75
0.50
0.25
0.00
-0.25
-0.50
-0.75
-1.00

Annex25


PCA Explained Variance Analysis
80.0% Threshold
Optimal Components (22)
Cumulative Explained Variance
1.0
0.8
0.6
0.4
0.2
0
10
20
30
40
Number of Components

Annex26





Component	Suggested Name	Interpretation
PCA 1	**The Frequent Flyer**	**Global Volume.** Represents overall usage intensity. High positive loadings across all NumFlights and DistanceKM months. Measures *how much* they fly year-round.
PCA 2	**The Winter Socializer**	**Q4 Social vs. Summer Solo.** Strongly positive for Companion travel in Oct-Dec; negative for May-Aug. Indicates people who travel with family/friends for holidays but fly solo in summer.
PCA 3	**The Holiday Homecomer**	**December Social Spike.** Distinct positive spike for Companion travel in Month 12 (Dec) vs. negative in Month 8/9. Captures Christmas/Hanukkah family travel.
PCA 4	**The Valentine's Traveler**	**February Social Spike.** High positive loading for Companion travel in Month 2 (Feb) contrasting with Month 10/11. Captures early-year social trips.
PCA 5	**The Mid-Year Companion**	**June/July Social vs. Aug/Sept Solo.** Captures travelers who are social in early summer but switch to solo travel as autumn approaches.
PCA 6	**The Pre-Summer Socializer**	**May Social Spike.** Positive loading for Companion travel in Month 5 (May) contrasting with Month 6/7. Captures late-spring social trips before the summer peak.
PCA 7	**The July Soloist**	**June/Aug Social vs. July Solo.** Captures a "sandwich" pattern: social travel in flank summer months (6 & 8) but solo travel during the peak of July (7).
PCA 8	**The Late Spring Pivot**	**May Social vs. June Solo.** Distinguishes customers who travel socially in May but switch to solo immediately in June.
PCA 9	**The Spring/Summer Contrast**	**May Social vs. July Solo.** Similar to PCA 8 but contrasts May specifically against July solo travel.
PCA 10	**The Spring Break Socializer**	**April Social Spike.** Dominated by a strong positive loading on Companion travel in Month 4 (April).
PCA 11	**The March Social / Q4 Dormant**	**March Social vs. Low Q4 Volume.** High companion travel in March combined with notably *low* flight volume in Oct-Dec.
PCA 12	**The March Social / Nov Active**	**March Social vs. High Nov Volume.** High companion travel in March combined with *high* flight volume in November (contrasting with PCA 11).
PCA 13	**The New Year's Companion**	**January Social Spike.** Driven almost entirely by Companion travel in Month 1 (Jan).

PCA 14	**The Thanksgiving Socializer**	**November Social Spike.** Positive loading for Companion travel in Month 11 vs. negative in Feb.
PCA 15	**The Late Autumn Surge**	**November Volume Peak.** Driven by high Flight/Distance numbers in Month 11 and 12, independent of social status.
PCA 16	**The October Peak**	**October Volume Spike.** High Flight/Distance volume specifically in Month 10 vs low volume in Month 11.
PCA 17	**The Late Summer Flyer**	**August Volume Spike.** High Flight/Distance volume specifically in Month 8 (Aug).
PCA 18	**The Mid-Summer Flyer**	**July Volume Spike.** High Flight/Distance volume specifically in Month 7 (July).
PCA 19	**The Early Summer Flyer**	**June Volume Spike.** High Flight/Distance volume specifically in Month 6 (June).
PCA 20	**The May Flyer**	**May Volume Spike.** High Flight/Distance volume specifically in Month 5 (May) vs April.
PCA 21	**The April Flyer**	**April Volume Spike.** High Flight/Distance volume specifically in Month 4 (April) vs March.
PCA 22	**The March Flyer**	**March Volume Spike.** High Flight/Distance volume specifically in Month 3 (March).

PC	Focus	Description / Pattern
PCA 1–14	**Social Travel Patterns**	Capture behavior related to *travel companions vs. solo travel*, highlighting seasonal or holiday trends. Examples: Winter socializers (PCA 2), December family spikes (PCA 3), Valentine's trips (PCA 4).
PCA 15–22	**Monthly Volume Spikes**	Capture *high flight/distance activity in specific months*, independent of social context. Examples: November surge (PCA 15), October peak (PCA 16), March flyer (PCA 22).

Annex27

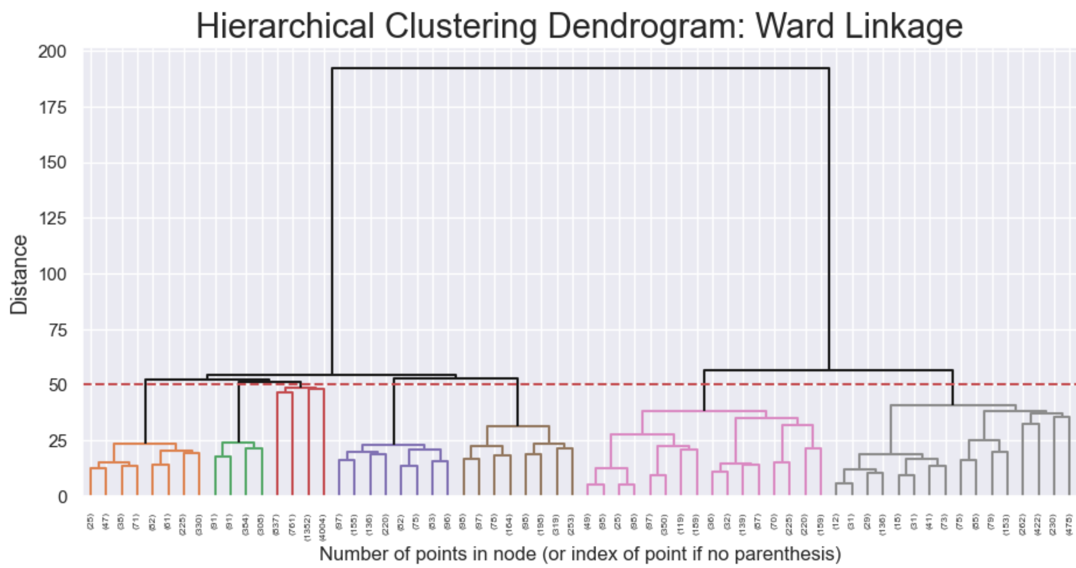
Num clusters	SSW (Lower better)	SSB (Higher better)	Silhouette Score (Higher better)	R^2 (Higher better)
2	123,625.63	20,296.53	0.1365	0.1410
3	120,259.00	23,663.46	0.0906	0.1644
4	116,936.96	26,985.36	0.0864	0.1875
5	114,316.59	29,605.65	0.0853	0.2057
6	112,057.12	31,865.13	0.0848	0.2214
7	110,046.41	33,875.80	0.0850	0.2354
8	108,508.89	35,413.27	0.0766	0.2461
9	106,987.58	36,934.63	0.0750	0.2566
10	105,492.81	38,429.34	0.0639	0.2670
11	104,392.62	39,529.54	0.0574	0.2747
12	**103,421.87**	**40,500.30**	**0.0531**	**0.2814**
13	102,086.45	41,835.73	0.0534	0.2907
14	101,178.08	42,744.07	0.0523	0.2970
15	100,209.53	43,712.64	0.0512	0.3037
16	99,868.01	44,054.15	0.0494	0.3061
17	98,914.87	45,007.28	0.0479	0.3127
18	97,986.17	45,936.03	0.0462	0.3192
19	96,899.16	47,022.99	0.0472	0.3267

Inertia plot over clusters
Inertia: SSw
120000
115000
110000
105000
100000
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
Number of clusters

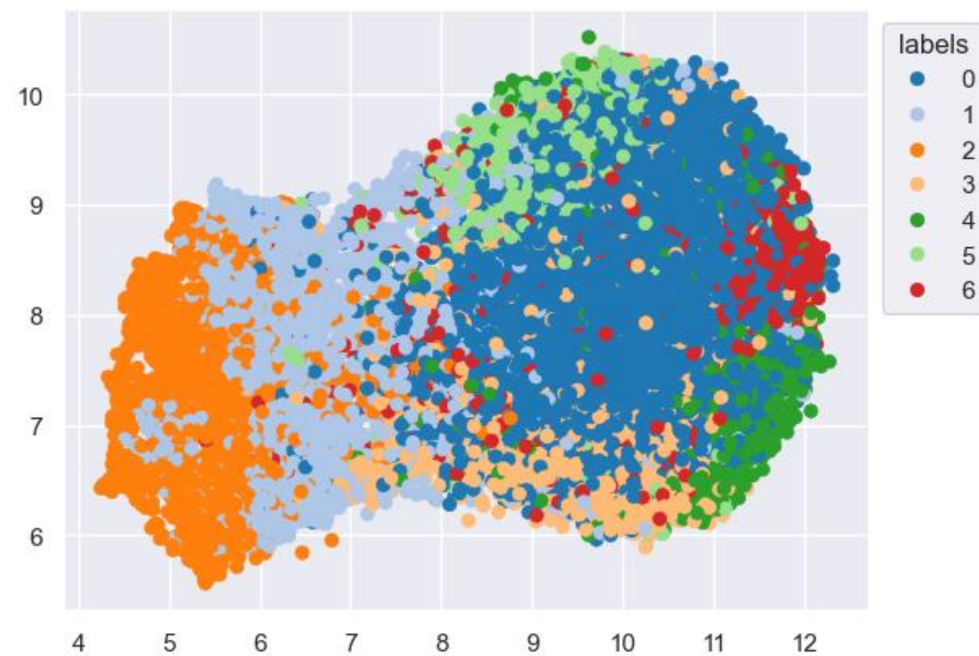
Annex28

labels
0
1
2
3
4
5
6
7
8
9
10
11

Annex29



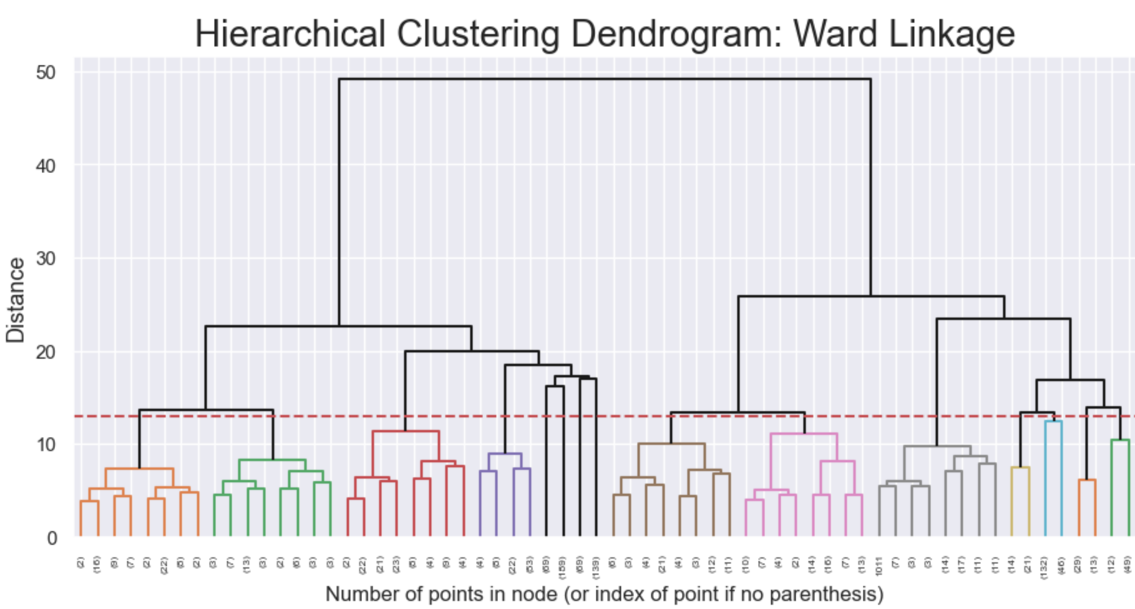
Annex30



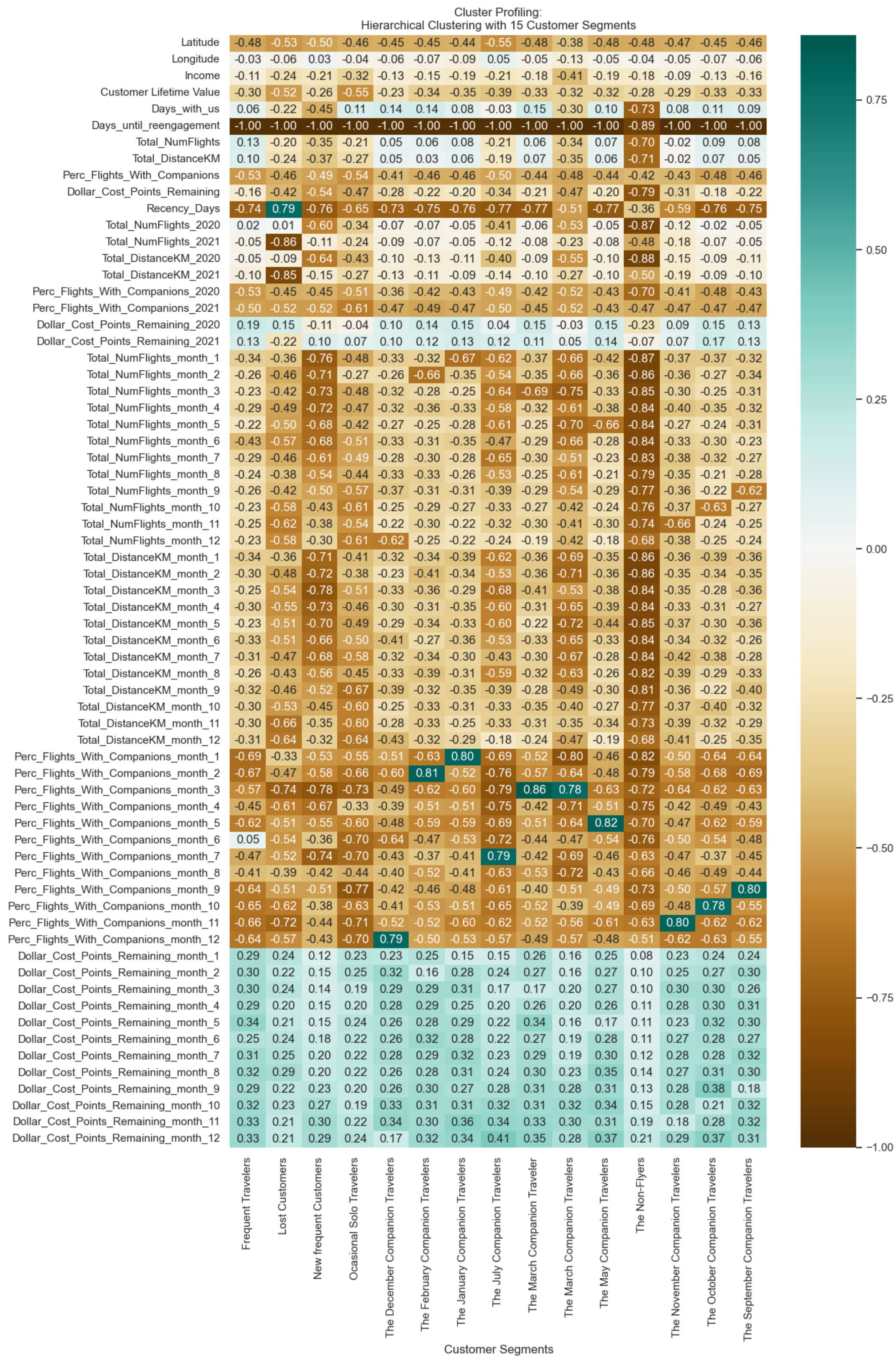
Annex31

Method	Num Clusters	R^2 (Higher better)	Silhouette Score (Higher better)	SSW (Lower better)	SSB (Higher better)
(PCA) KMeans	12	0.2814	0.0531	103,421.87	40,500.30
(PCA) Hierarchical	7	0.17796	0.01918	118,309.90	25,612.24
KMeans	**14**	**0.24272**	**0.03304**	**134,987.26**	**43,266.07**
Hierarchical	7	0.14340	0.01620	152,692.09	25,561.19

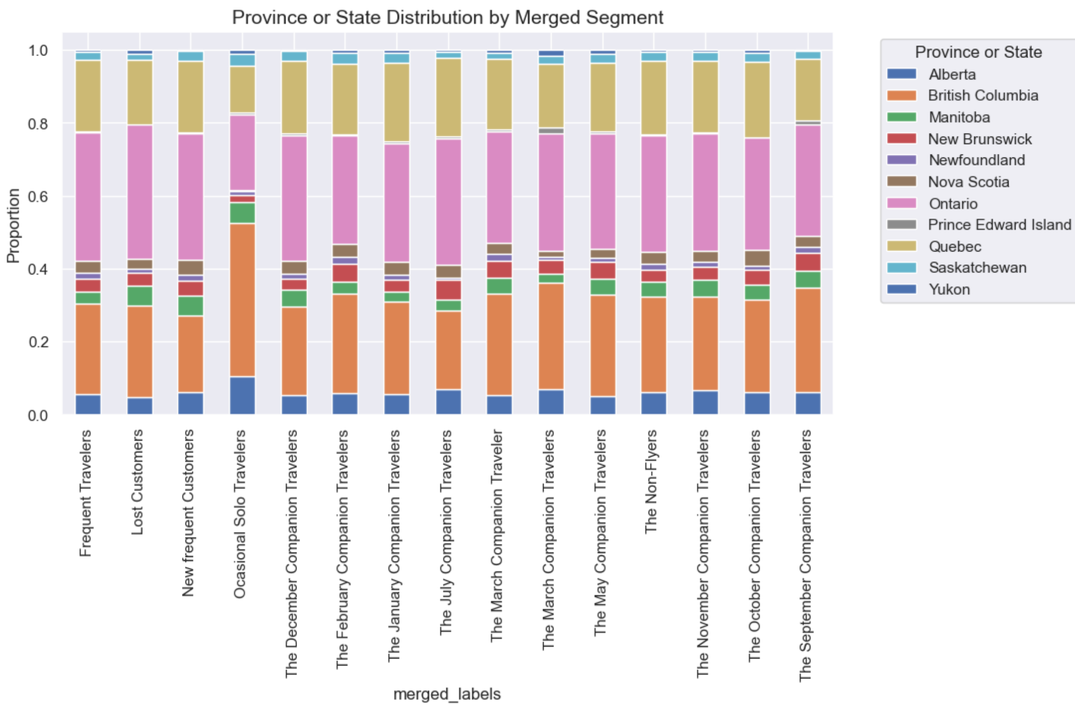
Annex32

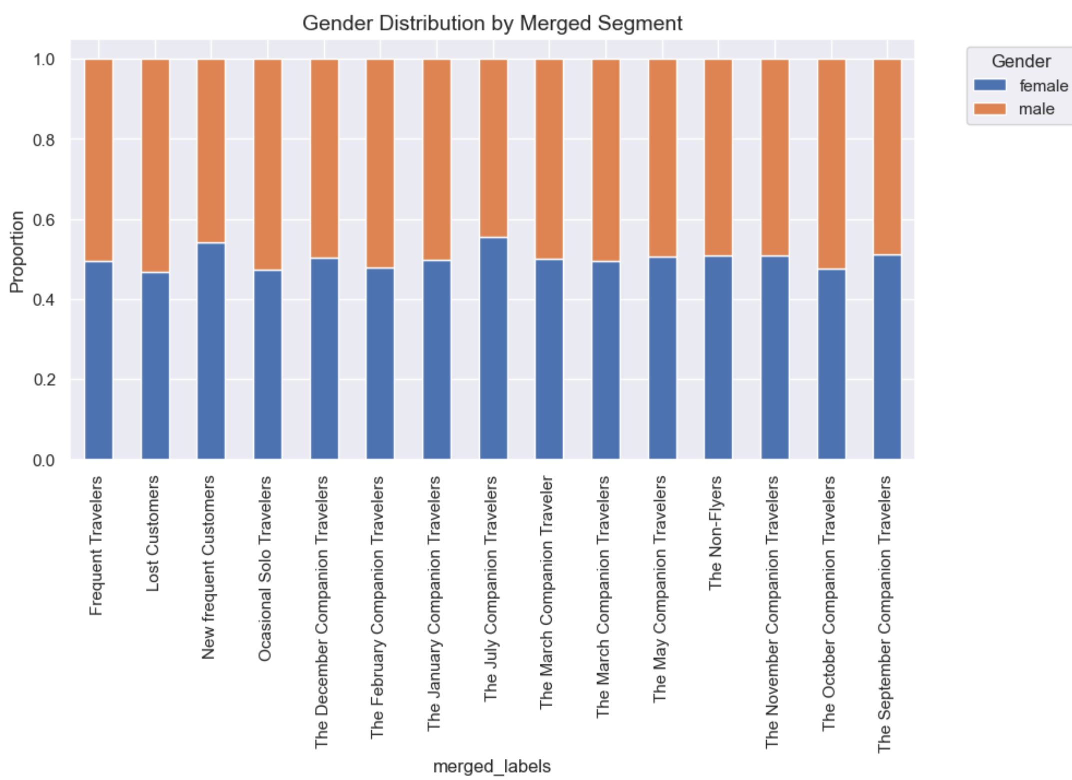


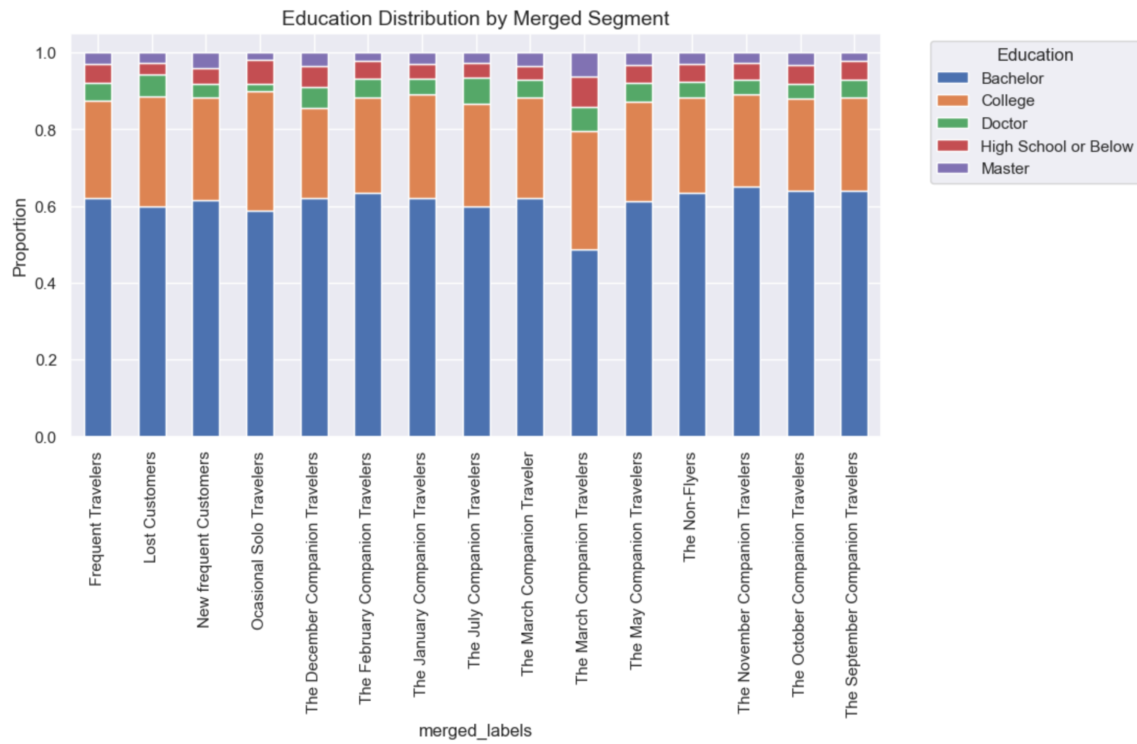
Annex33

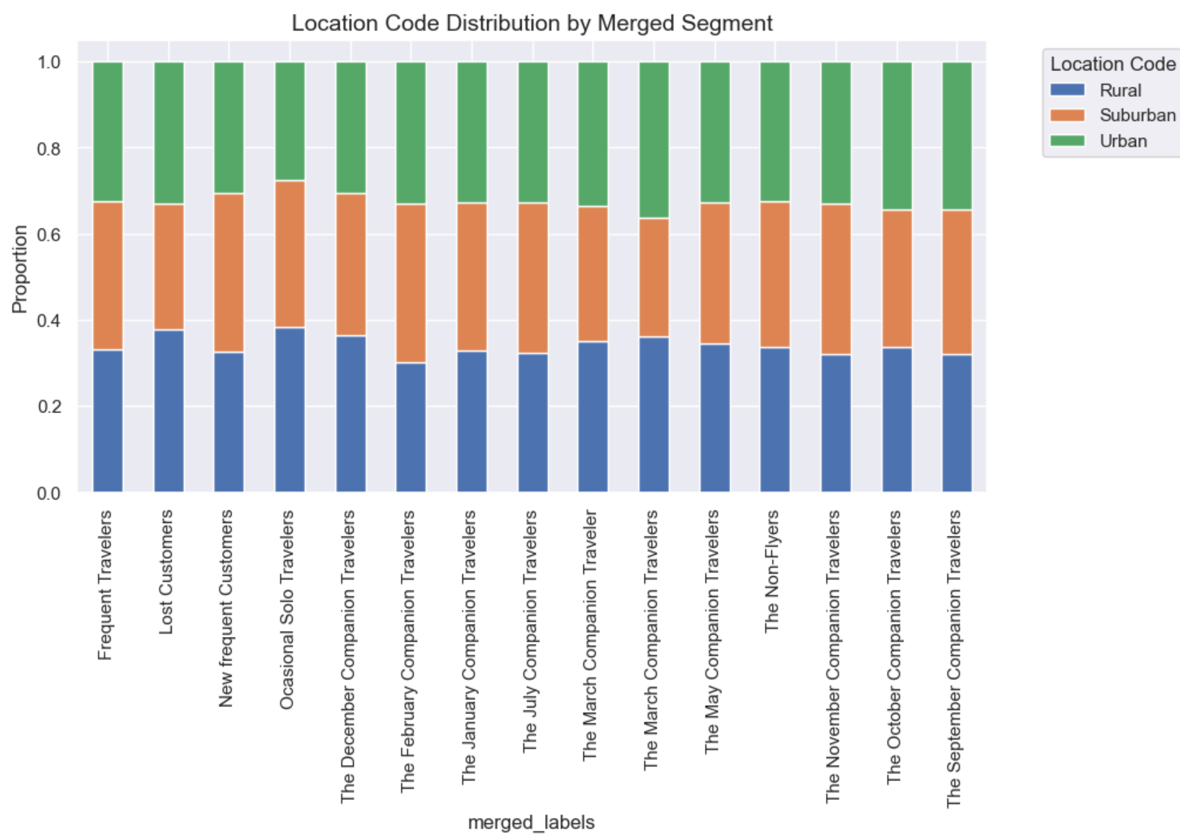


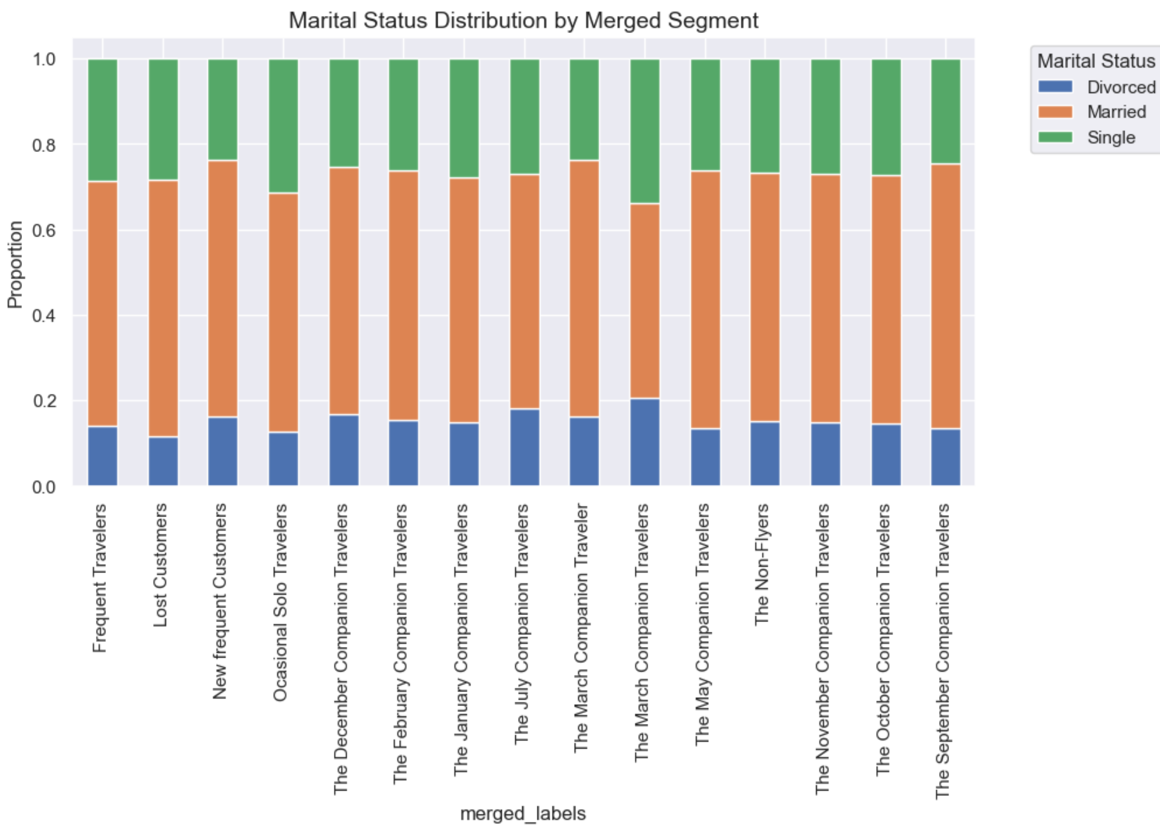
Cluster Profiling:
Hierarchical Clustering with 15 Customer Segments
Latitude
Longitude
Income
Customer Lifetime Value
Days_with_us
Days_until_reengagement
Total_NumFlights
Total_DistanceKM
Perc_Flights_With_Companions
Dollar_Cost_Points_Remaining
Recency_Days
Total_NumFlights_2020
Total_NumFlights_2021
Total_DistanceKM_2020
Total_DistanceKM_2021
Perc_Flights_With_Companions_2020
Perc_Flights_With_Companions_2021
Dollar_Cost_Points_Remaining_2020
Dollar_Cost_Points_Remaining_2021
Total_NumFlights_month_1
Total_NumFlights_month_2
Total_NumFlights_month_3
Total_NumFlights_month_4
Total_NumFlights_month_5
Total_NumFlights_month_6
Total_NumFlights_month_7
Total_NumFlights_month_8
Total_NumFlights_month_9
Total_NumFlights_month_10
Total_NumFlights_month_11
Total_NumFlights_month_12
Total_DistanceKM_month_1
Total_DistanceKM_month_2
Total_DistanceKM_month_3
Total_DistanceKM_month_4
Total_DistanceKM_month_5
Total_DistanceKM_month_6
Total_DistanceKM_month_7
Total_DistanceKM_month_8
Total_DistanceKM_month_9
Total_DistanceKM_month_10
Total_DistanceKM_month_11
Total_DistanceKM_month_12
Perc_Flights_With_Companions_month_1
Perc_Flights_With_Companions_month_2
Perc_Flights_With_Companions_month_3
Perc_Flights_With_Companions_month_4
Perc_Flights_With_Companions_month_5
Perc_Flights_With_Companions_month_6
Perc_Flights_With_Companions_month_7
Perc_Flights_With_Companions_month_8
Perc_Flights_With_Companions_month_9
Perc_Flights_With_Companions_month_10
Perc_Flights_With_Companions_month_11
Perc_Flights_With_Companions_month_12
Dollar_Cost_Points_Remaining_month_1
Dollar_Cost_Points_Remaining_month_2
Dollar_Cost_Points_Remaining_month_3
Dollar_Cost_Points_Remaining_month_4
Dollar_Cost_Points_Remaining_month_5
Dollar_Cost_Points_Remaining_month_6
Dollar_Cost_Points_Remaining_month_7
Dollar_Cost_Points_Remaining_month_8
Dollar_Cost_Points_Remaining_month_9
Dollar_Cost_Points_Remaining_month_10
Dollar_Cost_Points_Remaining_month_11
Dollar_Cost_Points_Remaining_month_12
Frequent Travelers
Lost Customers
New frequent Customers
Ocasional Solo Travelers
The December Companion Travelers
The February Companion Travelers
The January Companion Travelers
The July Companion Travelers
The March Companion Traveler
The March Companion Travelers
The May Companion Travelers
The Non-Flyers
The November Companion Travelers
The October Companion Travelers
The September Companion Travelers
Customer Segments
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

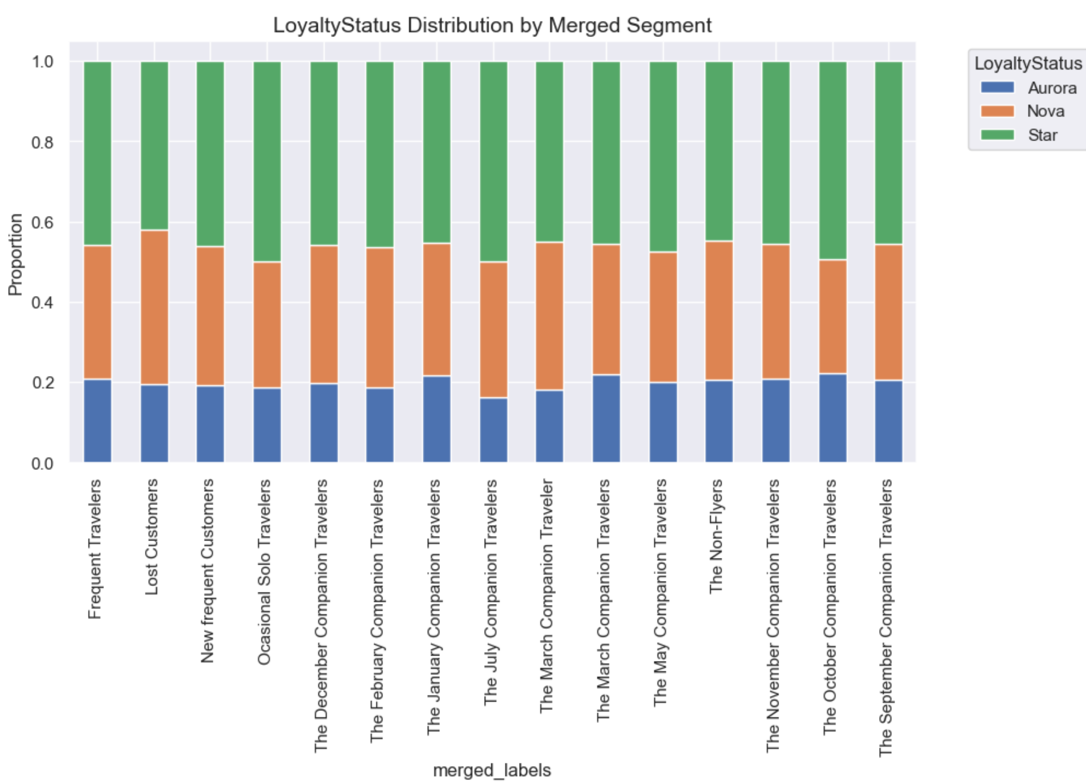

Province or State Distribution by Merged Segment
Proportion
merged_labels
Province or State
Alberta
British Columbia
Manitoba
New Brunswick
Newfoundland
Nova Scotia
Ontario
Prince Edward Island
Quebec
Saskatchewan
Yukon
Frequent Travelers
Lost Customers
New frequent Customers
Ocasional Solo Travelers
The December Companion Travelers
The February Companion Travelers
The January Companion Travelers
The July Companion Travelers
The March Companion Traveler
The March Companion Travelers
The May Companion Travelers
The Non-Flyers
The November Companion Travelers
The October Companion Travelers
The September Companion Travelers

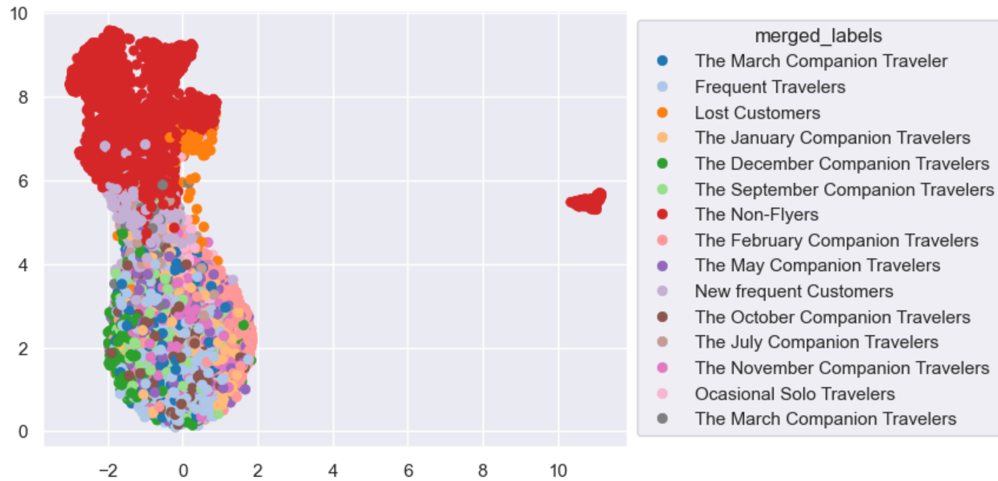

Gender Distribution by Merged Segment
Proportion
merged_labels
Gender
female
male
Frequent Travelers
Lost Customers
New frequent Customers
Ocasional Solo Travelers
The December Companion Travelers
The February Companion Travelers
The January Companion Travelers
The July Companion Travelers
The March Companion Traveler
The March Companion Travelers
The May Companion Travelers
The Non-Flyers
The November Companion Travelers
The October Companion Travelers
The September Companion Travelers


Education Distribution by Merged Segment
Proportion
0.0
0.2
0.4
0.6
0.8
1.0
Education
Bachelor
College
Doctor
High School or Below
Master
Frequent Travelers
Lost Customers
New frequent Customers
Ocasional Solo Travelers
The December Companion Travelers
The February Companion Travelers
The January Companion Travelers
The July Companion Travelers
The March Companion Traveler
The March Companion Travelers
The May Companion Travelers
The Non-Flyers
The November Companion Travelers
The October Companion Travelers
The September Companion Travelers
merged_labels


Location Code Distribution by Merged Segment
Proportion
0.0
0.2
0.4
0.6
0.8
1.0
Location Code
Rural
Suburban
Urban
Frequent Travelers
Lost Customers
New frequent Customers
Ocasional Solo Travelers
The December Companion Travelers
The February Companion Travelers
The January Companion Travelers
The July Companion Travelers
The March Companion Traveler
The March Companion Travelers
The May Companion Travelers
The Non-Flyers
The November Companion Travelers
The October Companion Travelers
The September Companion Travelers
merged_labels


Marital Status Distribution by Merged Segment
Proportion
1.0
0.8
0.6
0.4
0.2
0.0
Marital Status
Divorced
Married
Single
Frequent Travelers
Lost Customers
New frequent Customers
Ocasional Solo Travelers
The December Companion Travelers
The February Companion Travelers
The January Companion Travelers
The July Companion Travelers
The March Companion Traveler
The March Companion Travelers
The May Companion Travelers
The Non-Flyers
The November Companion Travelers
The October Companion Travelers
The September Companion Travelers
merged_labels

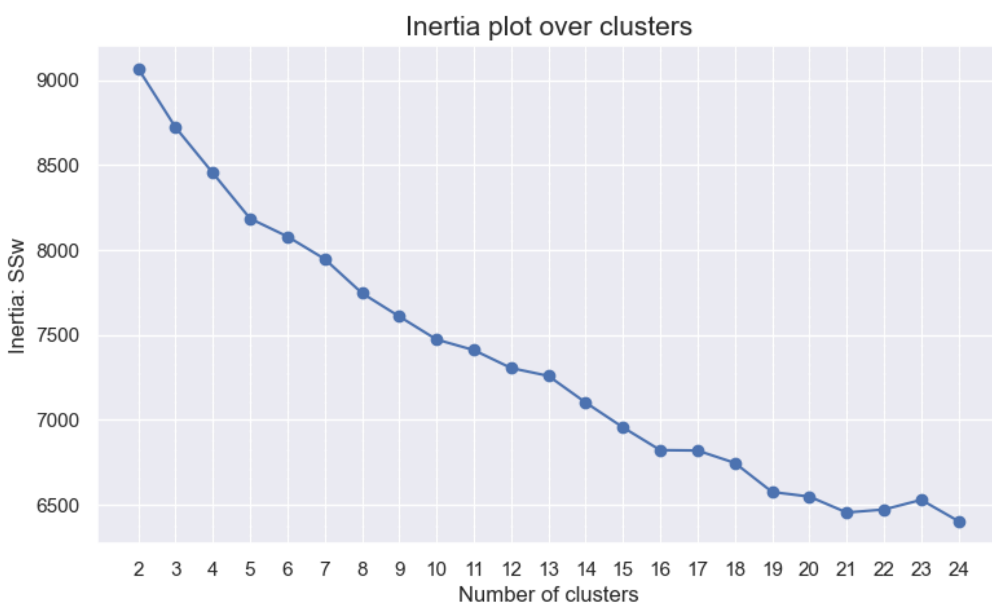

LoyaltyStatus Distribution by Merged Segment
Proportion
1.0
0.8
0.6
0.4
0.2
0.0
LoyaltyStatus
Aurora
Nova
Star
Frequent Travelers
Lost Customers
New frequent Customers
Ocasional Solo Travelers
The December Companion Travelers
The February Companion Travelers
The January Companion Travelers
The July Companion Travelers
The March Companion Traveler
The March Companion Travelers
The May Companion Travelers
The Non-Flyers
The November Companion Travelers
The October Companion Travelers
The September Companion Travelers
merged_labels



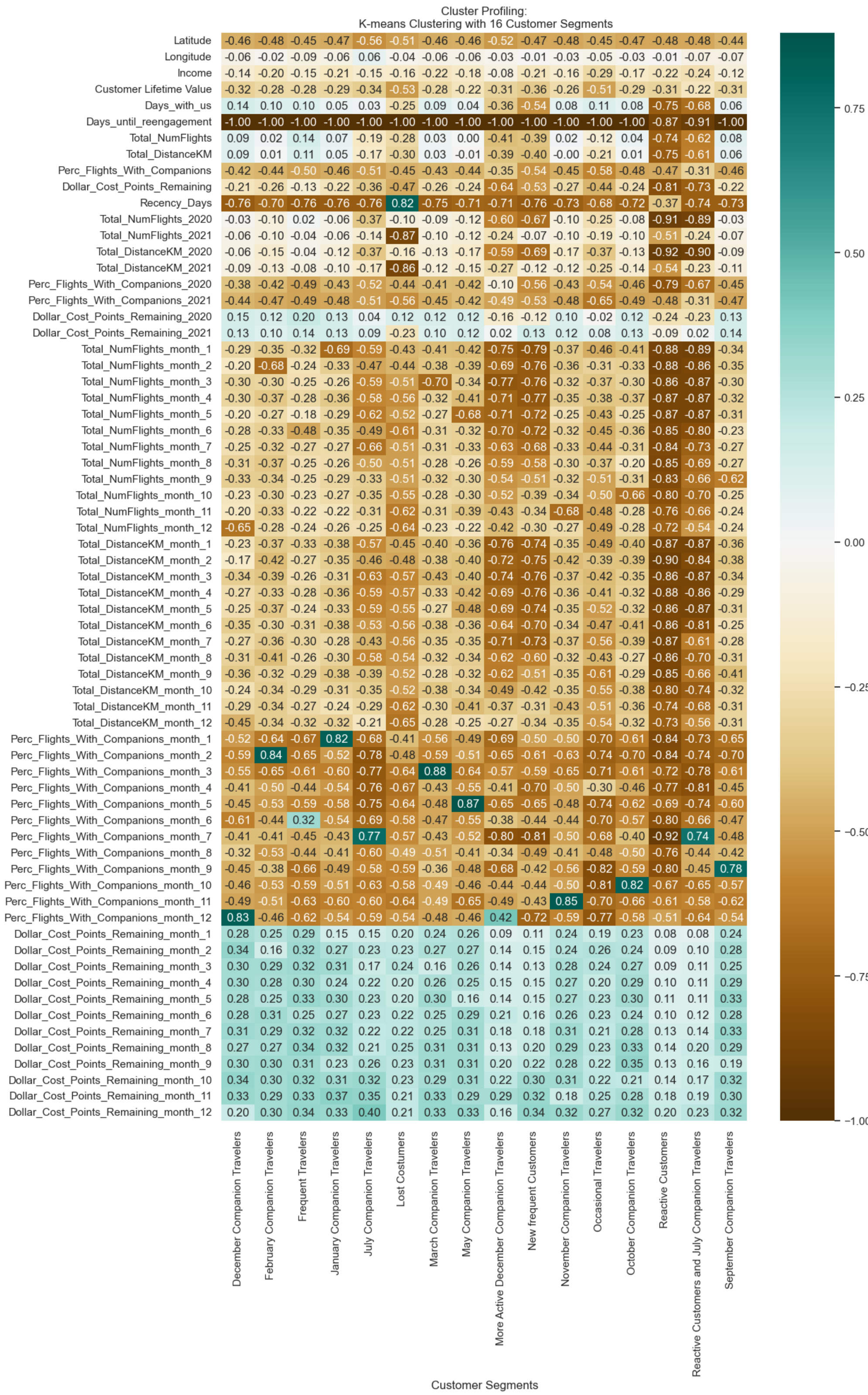
Annex34

Number of Clusters	SSW (↓ Better)	SSB (↑ Better)	Silhouette Score (↑ Better)	R^2 (↑ Better)
2	9,067.58	1,490.20	0.158	0.141
3	8,721.59	1,836.19	0.078	0.174
4	8,452.94	2,104.84	0.070	0.199
5	8,184.72	2,373.06	0.052	0.225
6	8,079.18	2,478.60	0.050	0.235
7	7,946.84	2,610.94	0.037	0.247
8	7,745.75	2,812.03	0.060	0.266

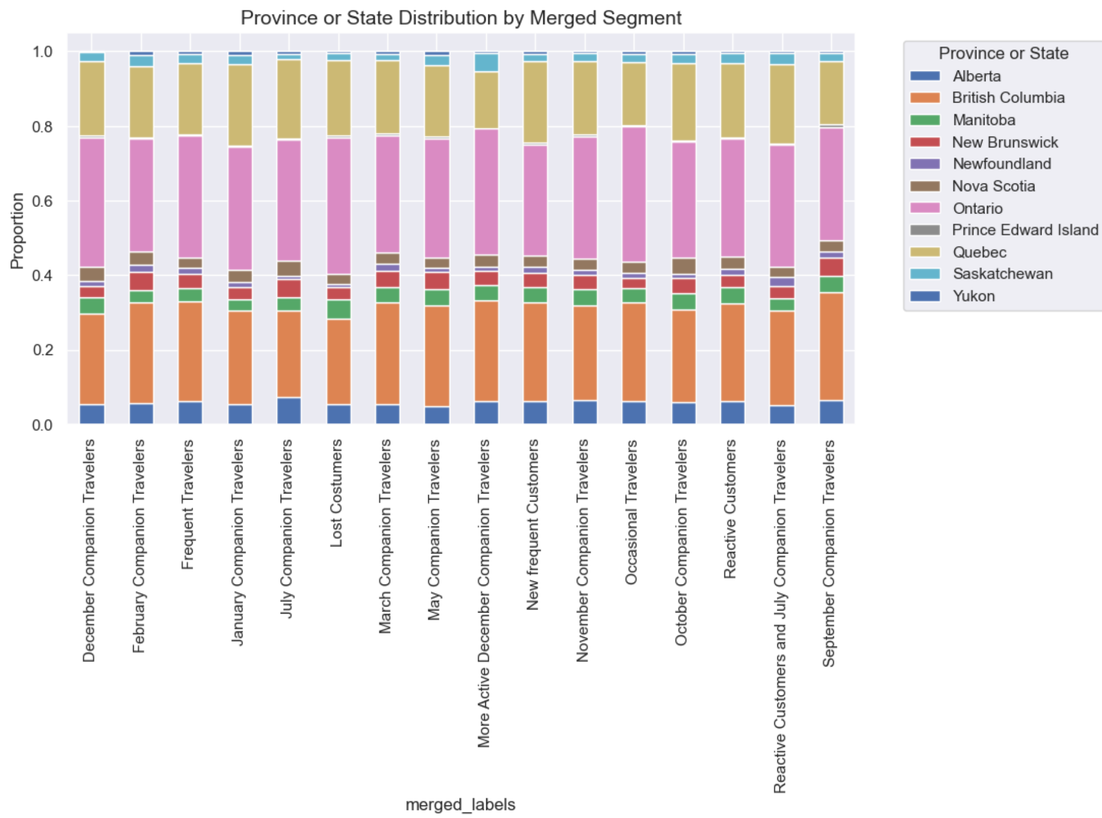
9	7,608.40	2,949.38	0.061	0.279
10	7,472.63	3,085.15	0.066	0.292
11	7,410.26	3,147.52	0.062	0.298
12	7,304.56	3,253.22	0.063	0.308
13	7,258.39	3,299.39	0.053	0.313
14	7,101.30	3,456.47	0.069	0.327
15	6,955.36	3,602.42	0.076	0.341
16	**6,821.04**	**3,736.74**	**0.084**	**0.354**
17	6,818.91	3,738.87	0.057	0.354
18	6,745.93	3,811.85	0.063	0.361
19	6,574.92	3,982.86	0.062	0.377
20	6,547.71	4,010.07	0.072	0.380
21	6,453.97	4,103.81	0.076	0.389
22	6,471.58	4,086.20	0.092	0.387
23	6,528.38	4,029.40	0.078	0.382
24	6,402.77	4,155.01	0.083	0.394



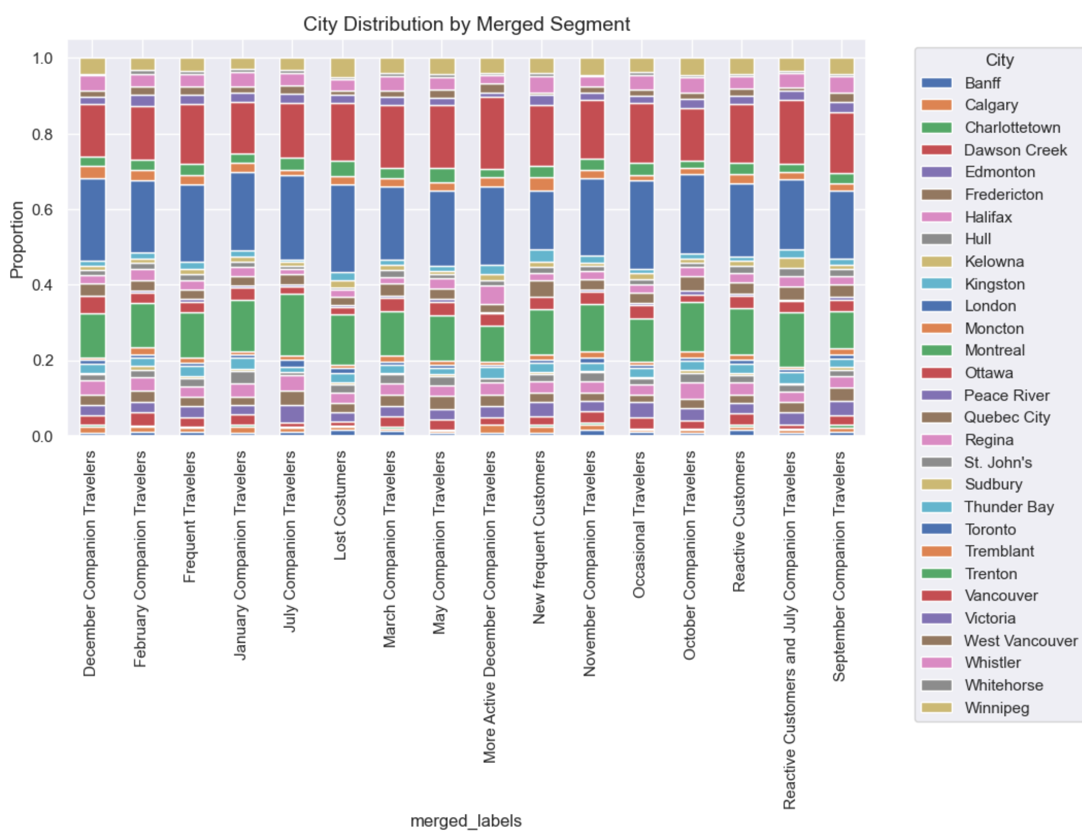
Annex35



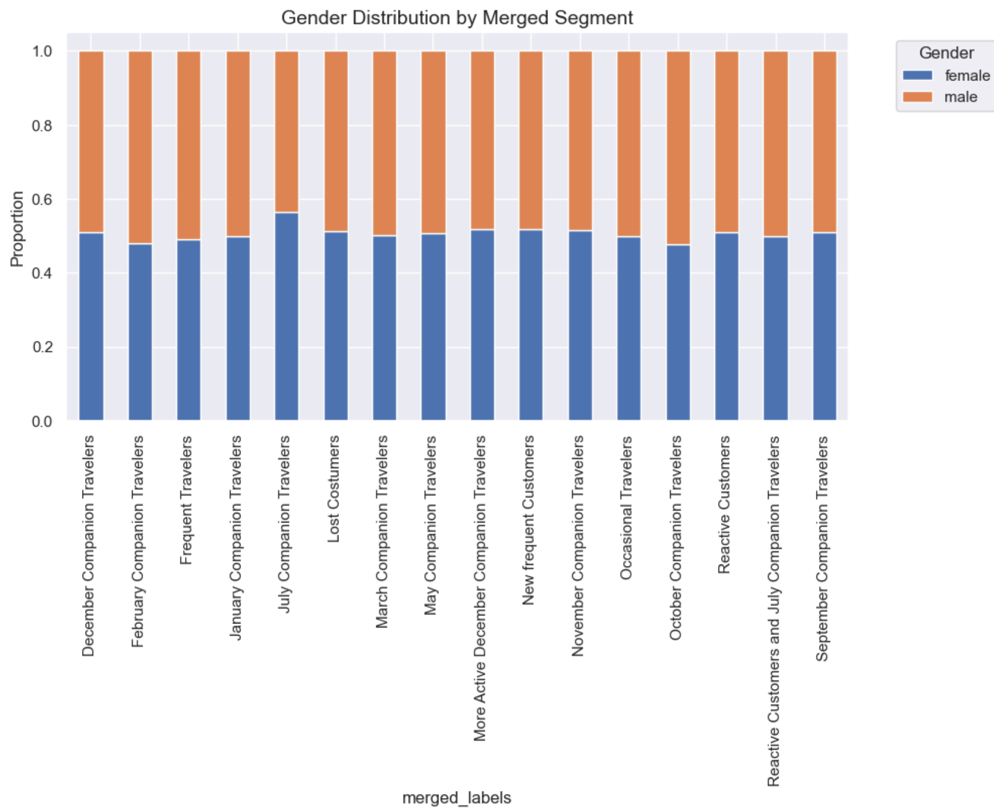
Cluster Profiling:
K-means Clustering with 16 Customer Segments
Latitude
Longitude
Income
Customer Lifetime Value
Days_with_us
Days_until_reengagement
Total_NumFlights
Total_DistanceKM
Perc_Flights_With_Companions
Dollar_Cost_Points_Remaining
Recency_Days
Total_NumFlights_2020
Total_NumFlights_2021
Total_DistanceKM_2020
Total_DistanceKM_2021
Perc_Flights_With_Companions_2020
Perc_Flights_With_Companions_2021
Dollar_Cost_Points_Remaining_2020
Dollar_Cost_Points_Remaining_2021
Total_NumFlights_month_1
Total_NumFlights_month_2
Total_NumFlights_month_3
Total_NumFlights_month_4
Total_NumFlights_month_5
Total_NumFlights_month_6
Total_NumFlights_month_7
Total_NumFlights_month_8
Total_NumFlights_month_9
Total_NumFlights_month_10
Total_NumFlights_month_11
Total_NumFlights_month_12
Total_DistanceKM_month_1
Total_DistanceKM_month_2
Total_DistanceKM_month_3
Total_DistanceKM_month_4
Total_DistanceKM_month_5
Total_DistanceKM_month_6
Total_DistanceKM_month_7
Total_DistanceKM_month_8
Total_DistanceKM_month_9
Total_DistanceKM_month_10
Total_DistanceKM_month_11
Total_DistanceKM_month_12
Perc_Flights_With_Companions_month_1
Perc_Flights_With_Companions_month_2
Perc_Flights_With_Companions_month_3
Perc_Flights_With_Companions_month_4
Perc_Flights_With_Companions_month_5
Perc_Flights_With_Companions_month_6
Perc_Flights_With_Companions_month_7
Perc_Flights_With_Companions_month_8
Perc_Flights_With_Companions_month_9
Perc_Flights_With_Companions_month_10
Perc_Flights_With_Companions_month_11
Perc_Flights_With_Companions_month_12
Dollar_Cost_Points_Remaining_month_1
Dollar_Cost_Points_Remaining_month_2
Dollar_Cost_Points_Remaining_month_3
Dollar_Cost_Points_Remaining_month_4
Dollar_Cost_Points_Remaining_month_5
Dollar_Cost_Points_Remaining_month_6
Dollar_Cost_Points_Remaining_month_7
Dollar_Cost_Points_Remaining_month_8
Dollar_Cost_Points_Remaining_month_9
Dollar_Cost_Points_Remaining_month_10
Dollar_Cost_Points_Remaining_month_11
Dollar_Cost_Points_Remaining_month_12
December Companion Travelers
February Companion Travelers
Frequent Travelers
January Companion Travelers
July Companion Travelers
Lost Costumers
March Companion Travelers
May Companion Travelers
More Active December Companion Travelers
New frequent Customers
November Companion Travelers
Occasional Travelers
October Companion Travelers
Reactive Customers
Reactive Customers and July Companion Travelers
September Companion Travelers
Customer Segments
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

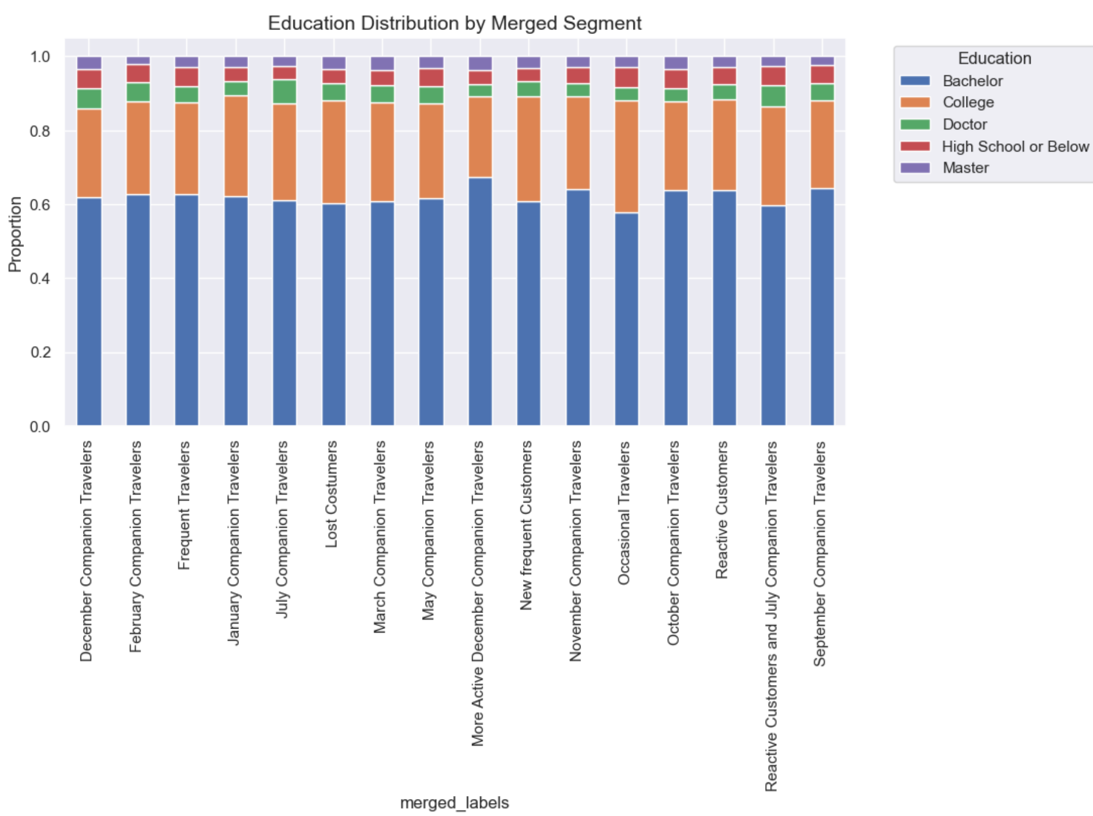


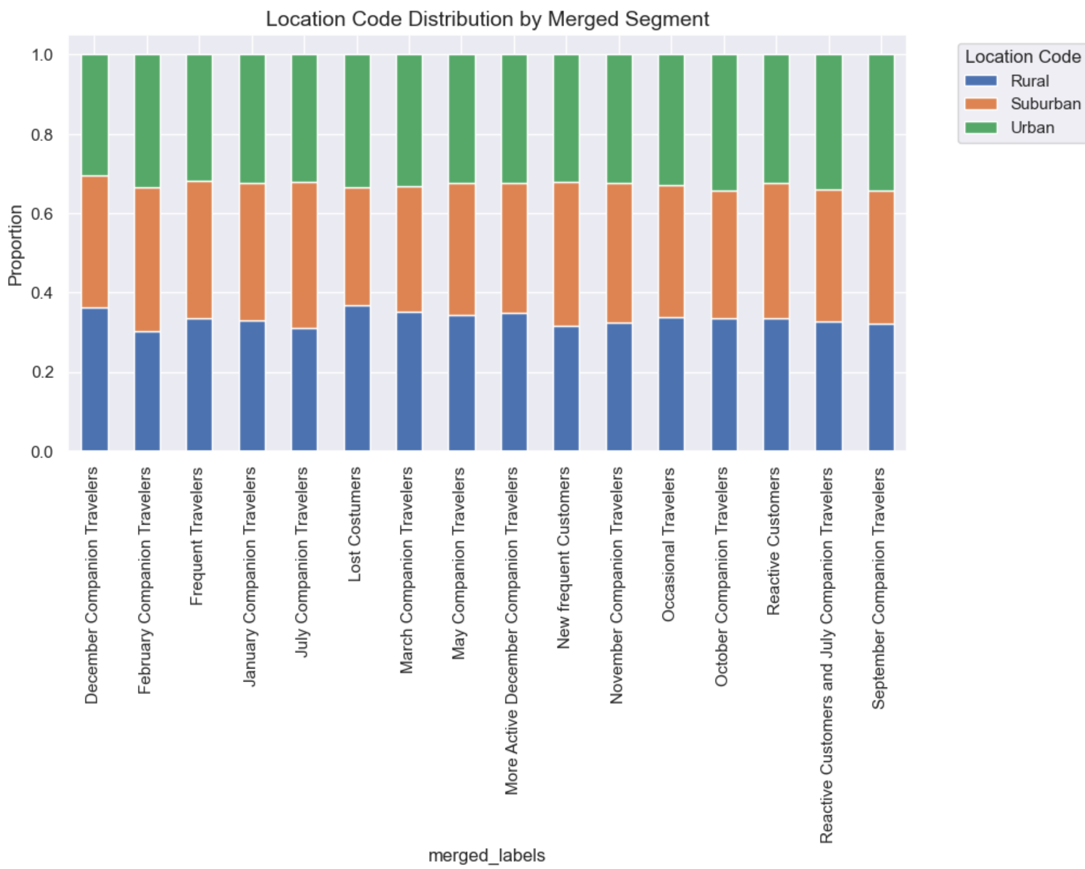
Province or State Distribution by Merged Segment
Proportion
1.0
0.8
0.6
0.4
0.2
0.0
December Companion Travelers
February Companion Travelers
Frequent Travelers
January Companion Travelers
July Companion Travelers
Lost Costumers
March Companion Travelers
May Companion Travelers
More Active December Companion Travelers
New frequent Customers
November Companion Travelers
Occasional Travelers
October Companion Travelers
Reactive Customers
Reactive Customers and July Companion Travelers
September Companion Travelers
merged_labels
Province or State
Alberta
British Columbia
Manitoba
New Brunswick
Newfoundland
Nova Scotia
Ontario
Prince Edward Island
Quebec
Saskatchewan
Yukon

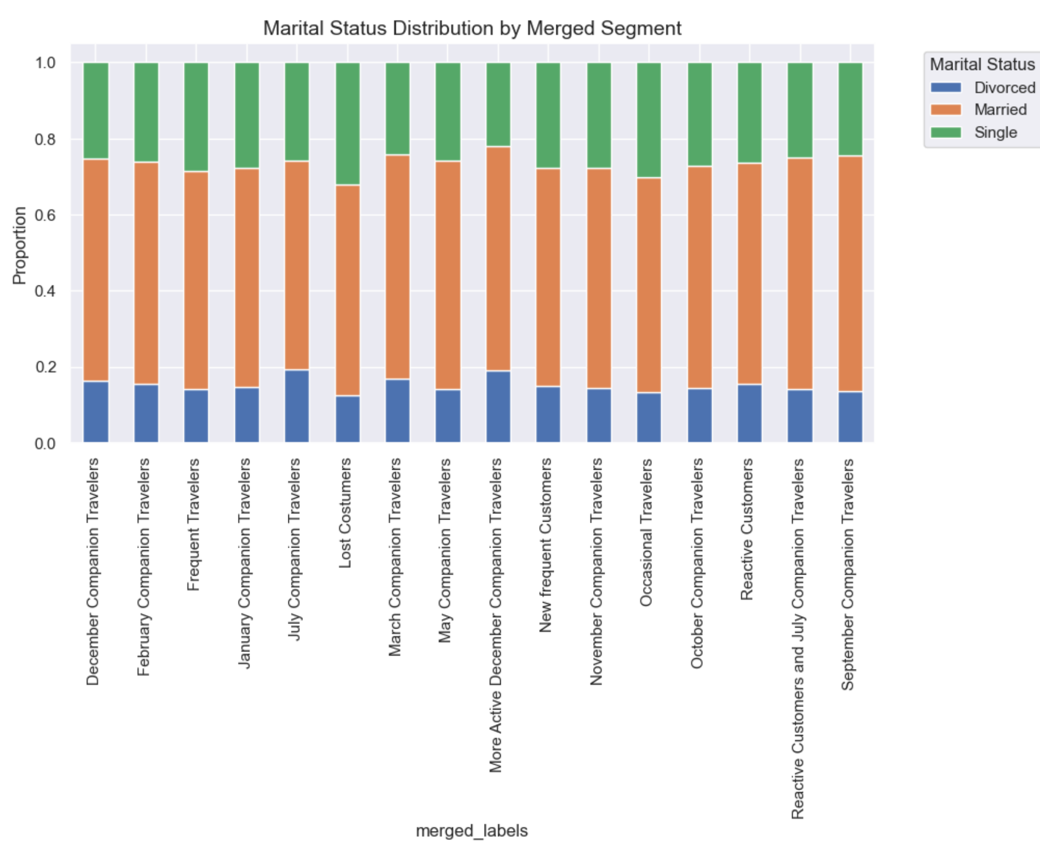


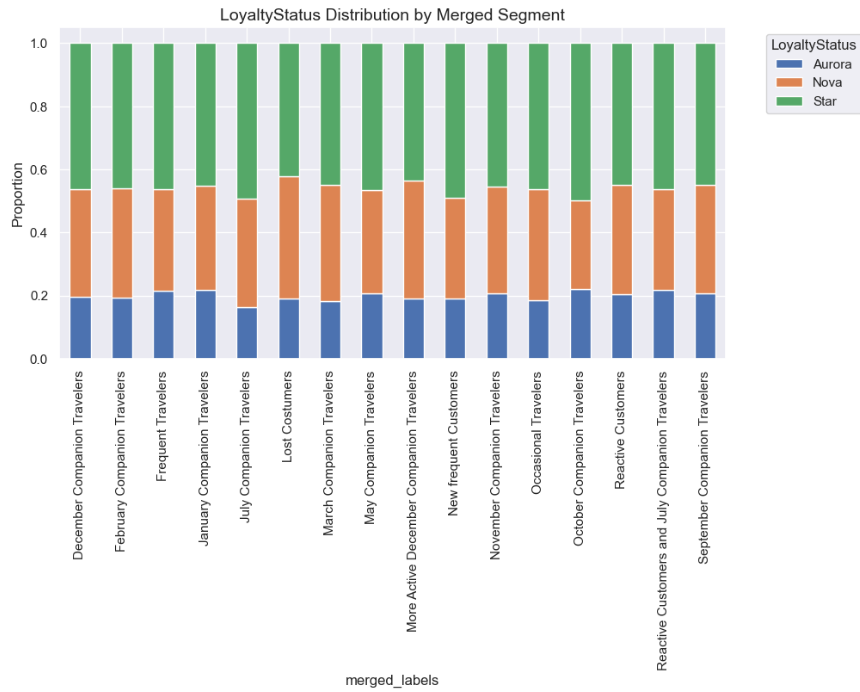
City Distribution by Merged Segment
Proportion
1.0
0.8
0.6
0.4
0.2
0.0
December Companion Travelers
February Companion Travelers
Frequent Travelers
January Companion Travelers
July Companion Travelers
Lost Costumers
March Companion Travelers
May Companion Travelers
More Active December Companion Travelers
New frequent Customers
November Companion Travelers
Occasional Travelers
October Companion Travelers
Reactive Customers
Reactive Customers and July Companion Travelers
September Companion Travelers
merged_labels
City
Banff
Calgary
Charlottetown
Dawson Creek
Edmonton
Fredericton
Halifax
Hull
Kelowna
Kingston
London
Moncton
Montreal
Ottawa
Peace River
Quebec City
Regina
St. John's
Sudbury
Thunder Bay
Toronto
Tremblant
Trenton
Vancouver
Victoria
West Vancouver
Whistler
Whitehorse
Winnipeg

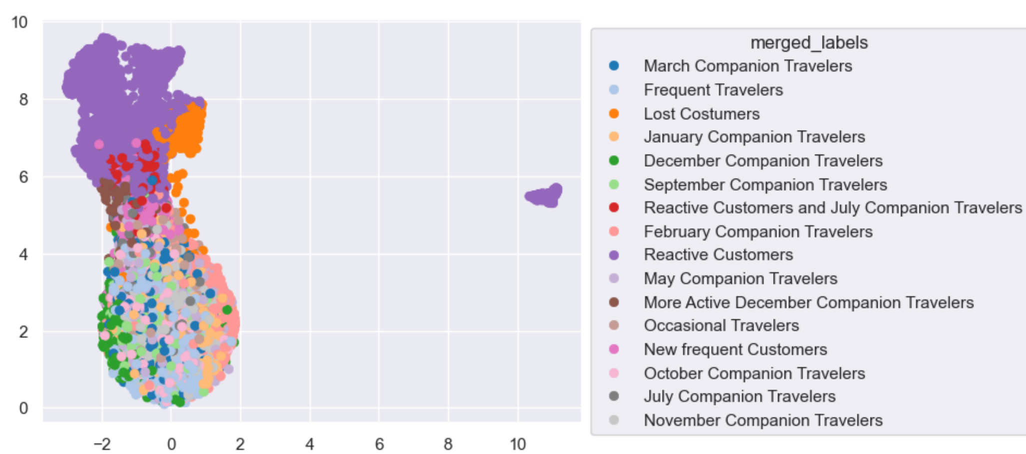

Gender Distribution by Merged Segment
Gender
female
male
Proportion
1.0
0.8
0.6
0.4
0.2
0.0
December Companion Travelers
February Companion Travelers
Frequent Travelers
January Companion Travelers
July Companion Travelers
Lost Costumers
March Companion Travelers
May Companion Travelers
More Active December Companion Travelers
New frequent Customers
November Companion Travelers
Occasional Travelers
October Companion Travelers
Reactive Customers
Reactive Customers and July Companion Travelers
September Companion Travelers
merged_labels

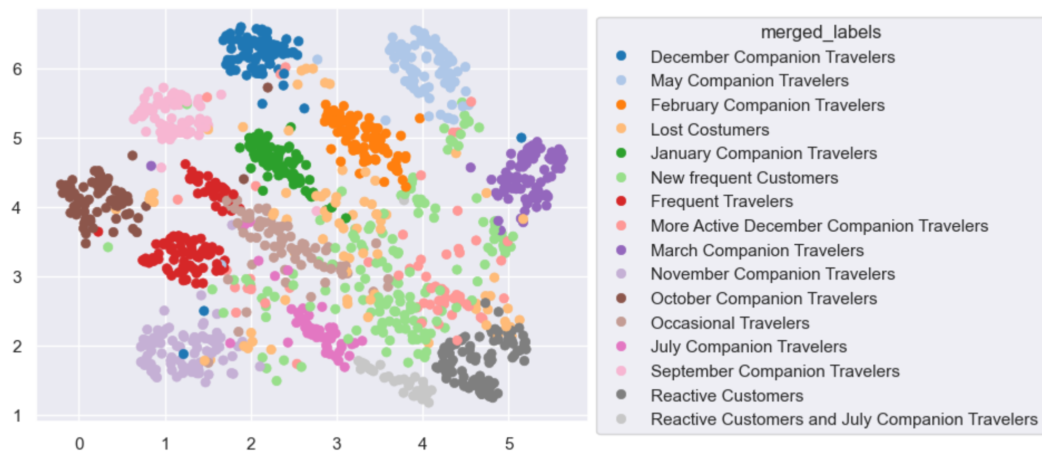

Education Distribution by Merged Segment
Education
Bachelor
College
Doctor
High School or Below
Master
Proportion
1.0
0.8
0.6
0.4
0.2
0.0
December Companion Travelers
February Companion Travelers
Frequent Travelers
January Companion Travelers
July Companion Travelers
Lost Costumers
March Companion Travelers
May Companion Travelers
More Active December Companion Travelers
New frequent Customers
November Companion Travelers
Occasional Travelers
October Companion Travelers
Reactive Customers
Reactive Customers and July Companion Travelers
September Companion Travelers
merged_labels


Location Code Distribution by Merged Segment
Proportion
1.0
0.8
0.6
0.4
0.2
0.0
Location Code
Rural
Suburban
Urban
December Companion Travelers
February Companion Travelers
Frequent Travelers
January Companion Travelers
July Companion Travelers
Lost Costumers
March Companion Travelers
May Companion Travelers
More Active December Companion Travelers
New frequent Customers
November Companion Travelers
Occasional Travelers
October Companion Travelers
Reactive Customers
Reactive Customers and July Companion Travelers
September Companion Travelers
merged_labels

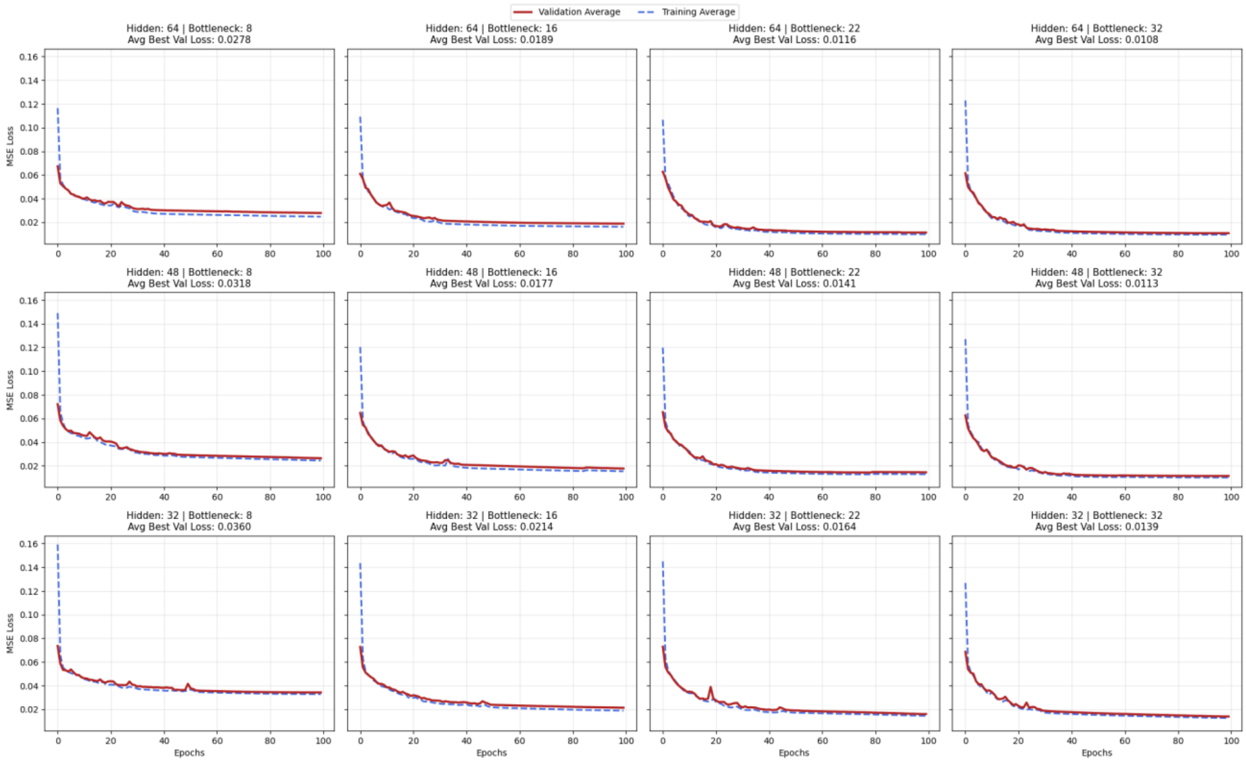

Marital Status Distribution by Merged Segment
Proportion
1.0
0.8
0.6
0.4
0.2
0.0
Marital Status
Divorced
Married
Single
December Companion Travelers
February Companion Travelers
Frequent Travelers
January Companion Travelers
July Companion Travelers
Lost Costumers
March Companion Travelers
May Companion Travelers
More Active December Companion Travelers
New frequent Customers
November Companion Travelers
Occasional Travelers
October Companion Travelers
Reactive Customers
Reactive Customers and July Companion Travelers
September Companion Travelers
merged_labels


LoyaltyStatus Distribution by Merged Segment
Proportion
1.0
0.8
0.6
0.4
0.2
0.0
LoyaltyStatus
Aurora
Nova
Star
December Companion Travelers
February Companion Travelers
Frequent Travelers
January Companion Travelers
July Companion Travelers
Lost Costumers
March Companion Travelers
May Companion Travelers
More Active December Companion Travelers
New frequent Customers
November Companion Travelers
Occasional Travelers
October Companion Travelers
Reactive Customers
Reactive Customers and July Companion Travelers
September Companion Travelers
merged_labels

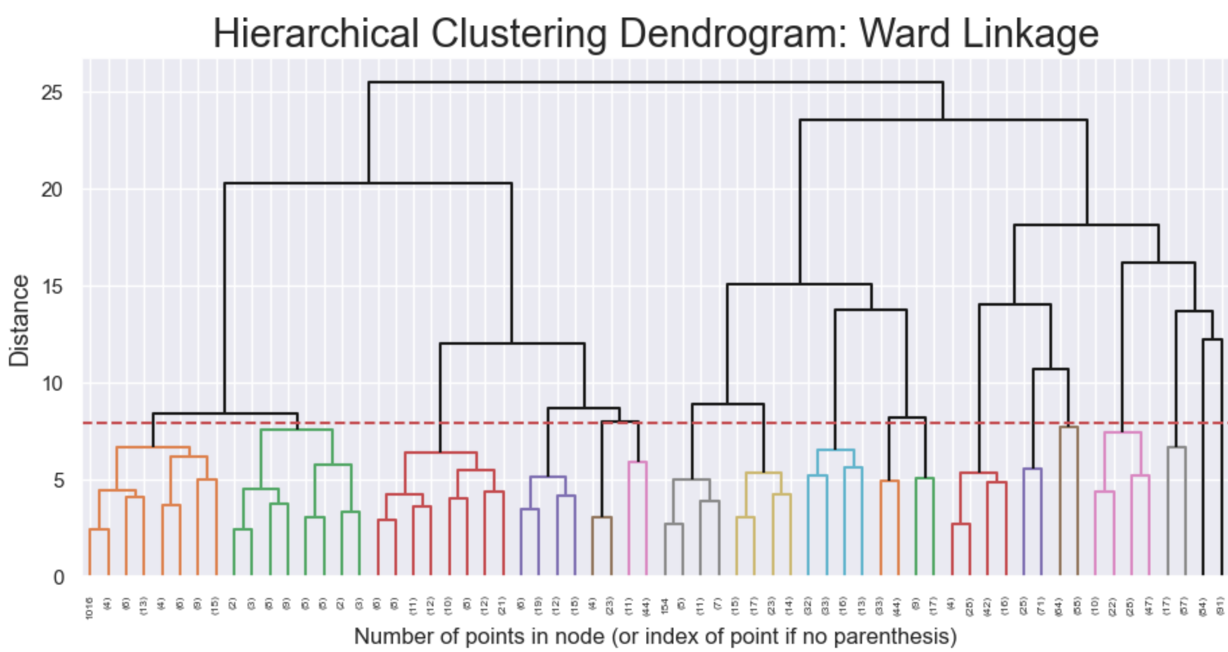

10
8
6
4
2
0
−2
0
2
4
6
8
10
merged_labels
March Companion Travelers
Frequent Travelers
Lost Costumers
January Companion Travelers
December Companion Travelers
September Companion Travelers
Reactive Customers and July Companion Travelers
February Companion Travelers
Reactive Customers
May Companion Travelers
More Active December Companion Travelers
Occasional Travelers
New frequent Customers
October Companion Travelers
July Companion Travelers
November Companion Travelers


6
5
4
3
2
1
0
1
2
3
4
5
merged_labels
December Companion Travelers
May Companion Travelers
February Companion Travelers
Lost Costumers
January Companion Travelers
New frequent Customers
Frequent Travelers
More Active December Companion Travelers
March Companion Travelers
November Companion Travelers
October Companion Travelers
Occasional Travelers
July Companion Travelers
September Companion Travelers
Reactive Customers
Reactive Customers and July Companion Travelers

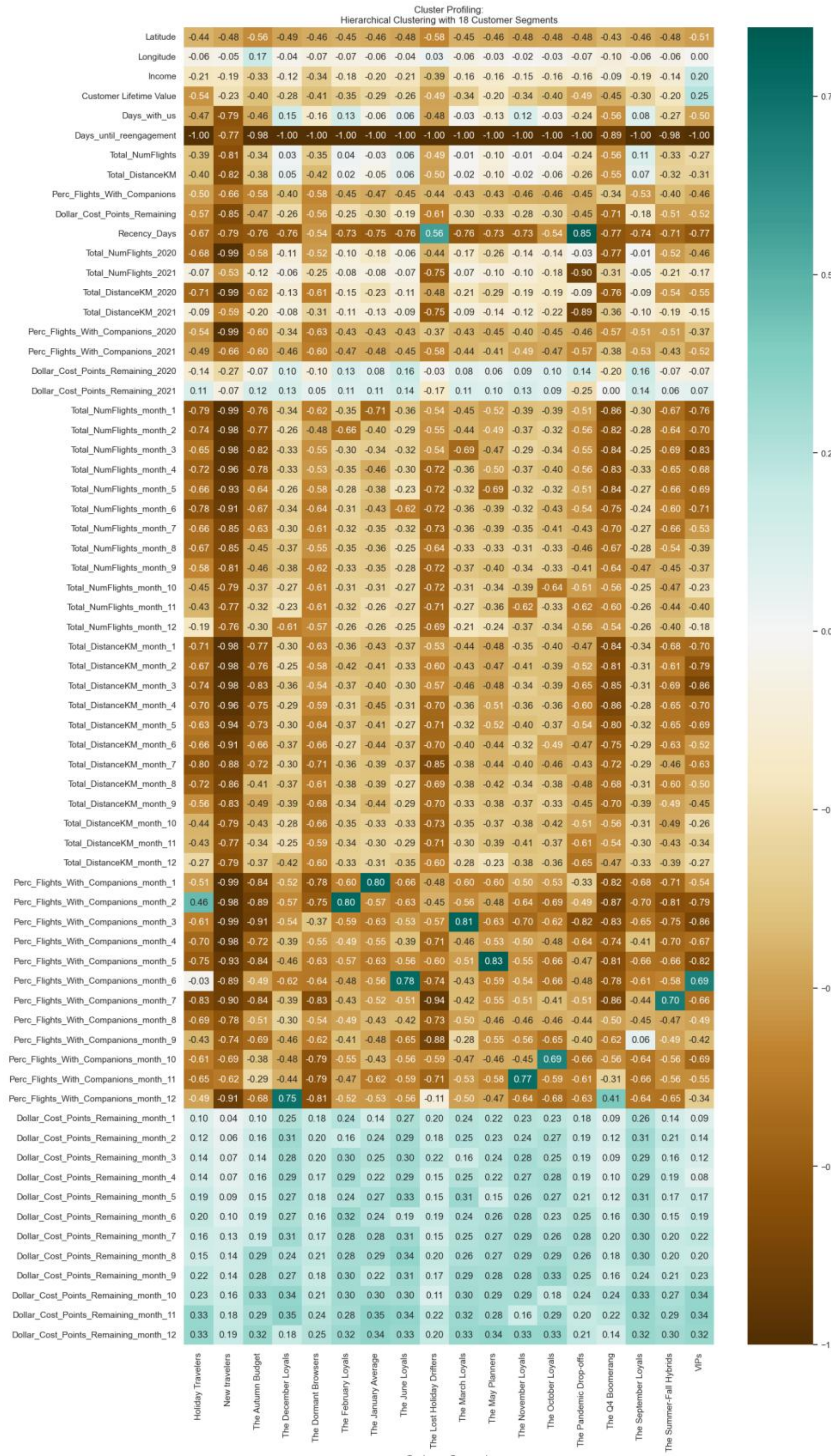
Annex36



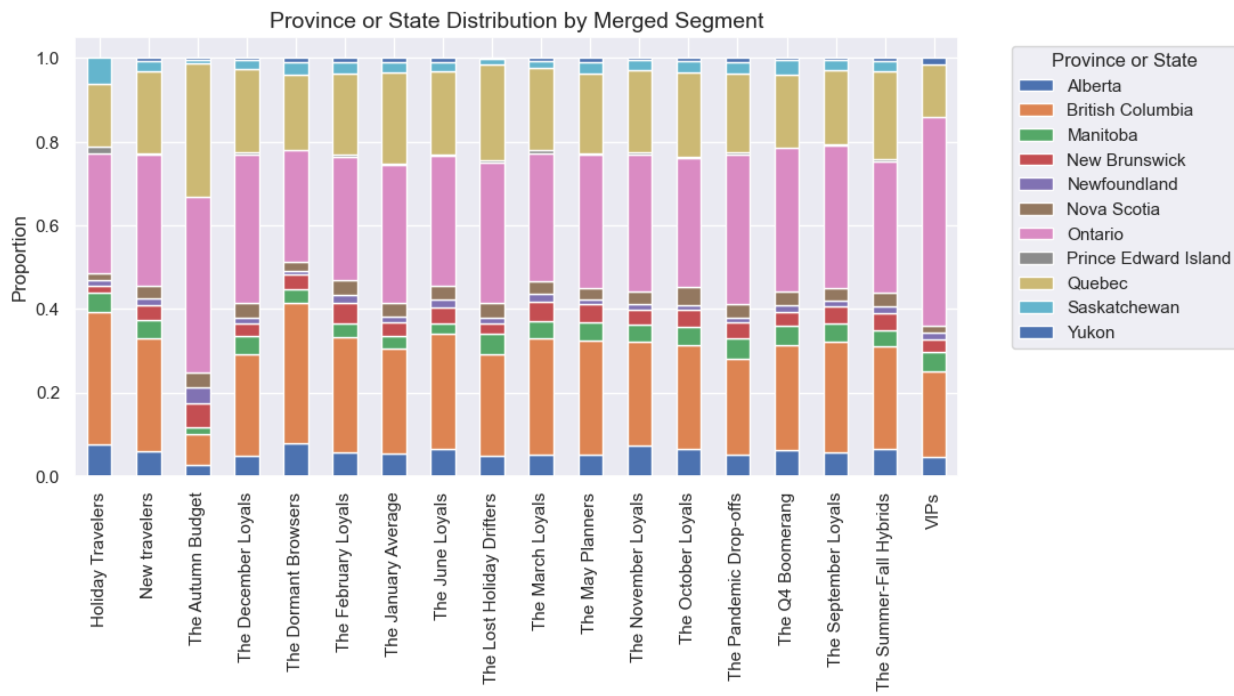
Annex37

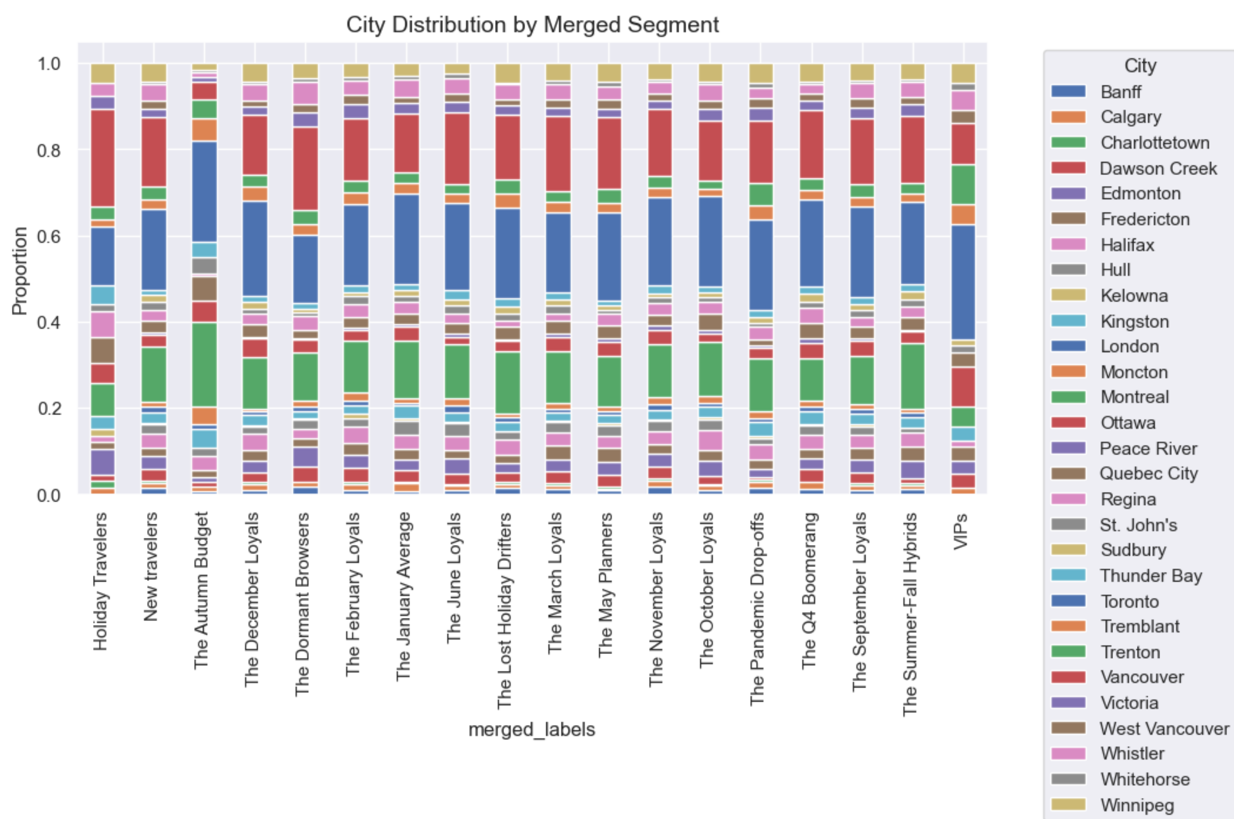


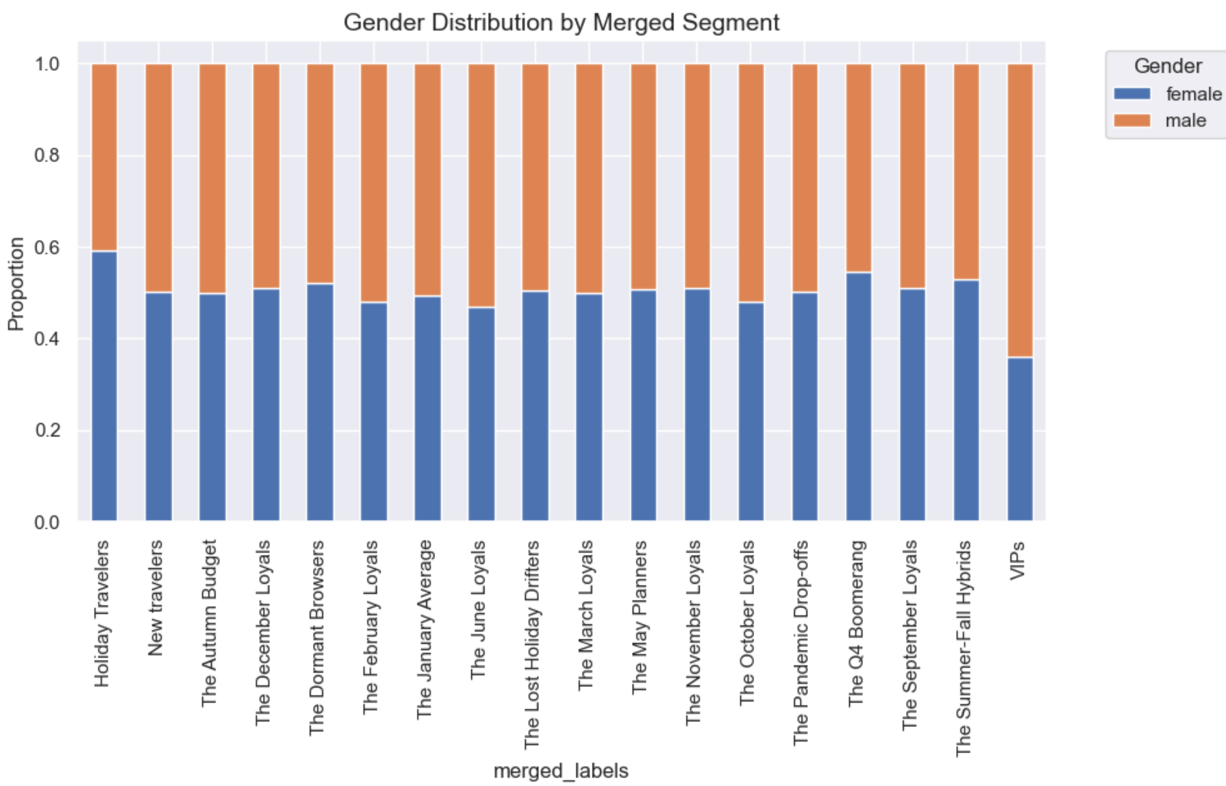
Annex38

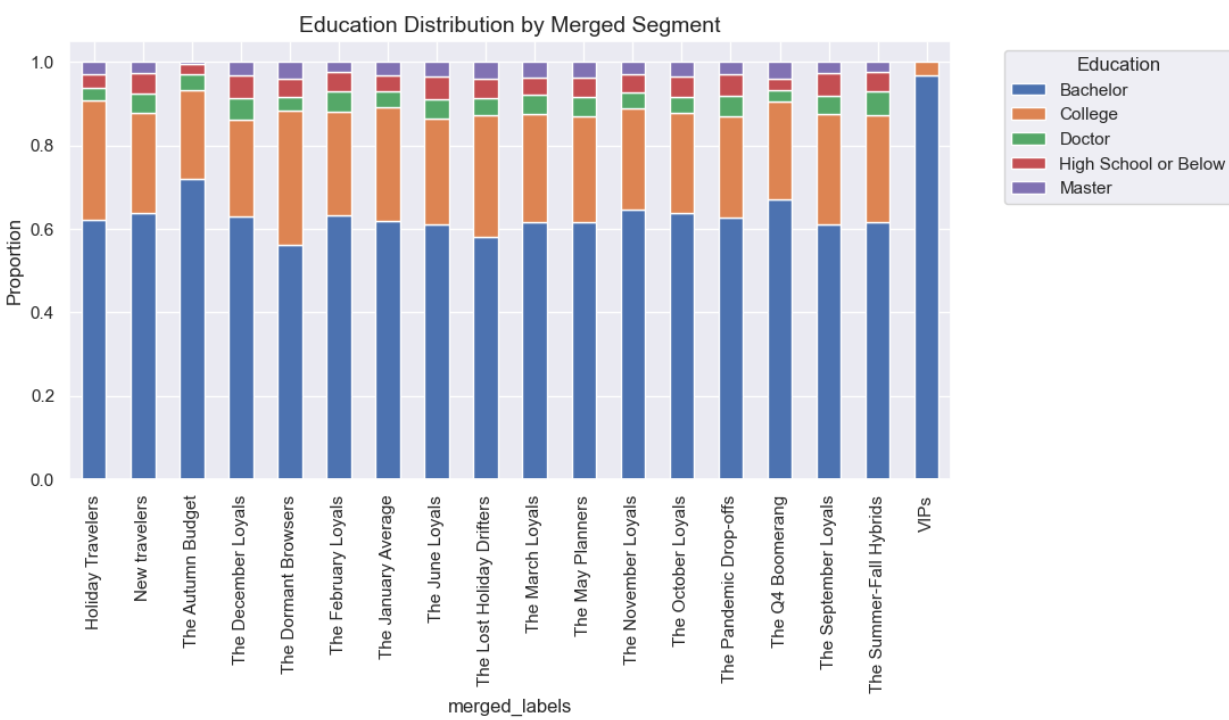


Cluster Profiling:
Hierarchical Clustering with 18 Customer Segments
Latitude
Longitude
Income
Customer Lifetime Value
Days_with_us
Days_until_reengagement
Total_NumFlights
Total_DistanceKM
Perc_Flights_With_Companions
Dollar_Cost_Points_Remaining
Recency_Days
Total_NumFlights_2020
Total_NumFlights_2021
Total_DistanceKM_2020
Total_DistanceKM_2021
Perc_Flights_With_Companions_2020
Perc_Flights_With_Companions_2021
Dollar_Cost_Points_Remaining_2020
Dollar_Cost_Points_Remaining_2021
Total_NumFlights_month_1
Total_NumFlights_month_2
Total_NumFlights_month_3
Total_NumFlights_month_4
Total_NumFlights_month_5
Total_NumFlights_month_6
Total_NumFlights_month_7
Total_NumFlights_month_8
Total_NumFlights_month_9
Total_NumFlights_month_10
Total_NumFlights_month_11
Total_NumFlights_month_12
Total_DistanceKM_month_1
Total_DistanceKM_month_2
Total_DistanceKM_month_3
Total_DistanceKM_month_4
Total_DistanceKM_month_5
Total_DistanceKM_month_6
Total_DistanceKM_month_7
Total_DistanceKM_month_8
Total_DistanceKM_month_9
Total_DistanceKM_month_10
Total_DistanceKM_month_11
Total_DistanceKM_month_12
Perc_Flights_With_Companions_month_1
Perc_Flights_With_Companions_month_2
Perc_Flights_With_Companions_month_3
Perc_Flights_With_Companions_month_4
Perc_Flights_With_Companions_month_5
Perc_Flights_With_Companions_month_6
Perc_Flights_With_Companions_month_7
Perc_Flights_With_Companions_month_8
Perc_Flights_With_Companions_month_9
Perc_Flights_With_Companions_month_10
Perc_Flights_With_Companions_month_11
Perc_Flights_With_Companions_month_12
Dollar_Cost_Points_Remaining_month_1
Dollar_Cost_Points_Remaining_month_2
Dollar_Cost_Points_Remaining_month_3
Dollar_Cost_Points_Remaining_month_4
Dollar_Cost_Points_Remaining_month_5
Dollar_Cost_Points_Remaining_month_6
Dollar_Cost_Points_Remaining_month_7
Dollar_Cost_Points_Remaining_month_8
Dollar_Cost_Points_Remaining_month_9
Dollar_Cost_Points_Remaining_month_10
Dollar_Cost_Points_Remaining_month_11
Dollar_Cost_Points_Remaining_month_12
Holiday Travelers
New travelers
The Autumn Budget
The December Loyals
The Dormant Browsers
The February Loyals
The January Average
The June Loyals
The Lost Holiday Drifters
The March Loyals
The May Planners
The November Loyals
The October Loyals
The Pandemic Drop-offs
The Q4 Boomerang
The September Loyals
The Summer-Fall Hybrids
VIPs
Customer Segments
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

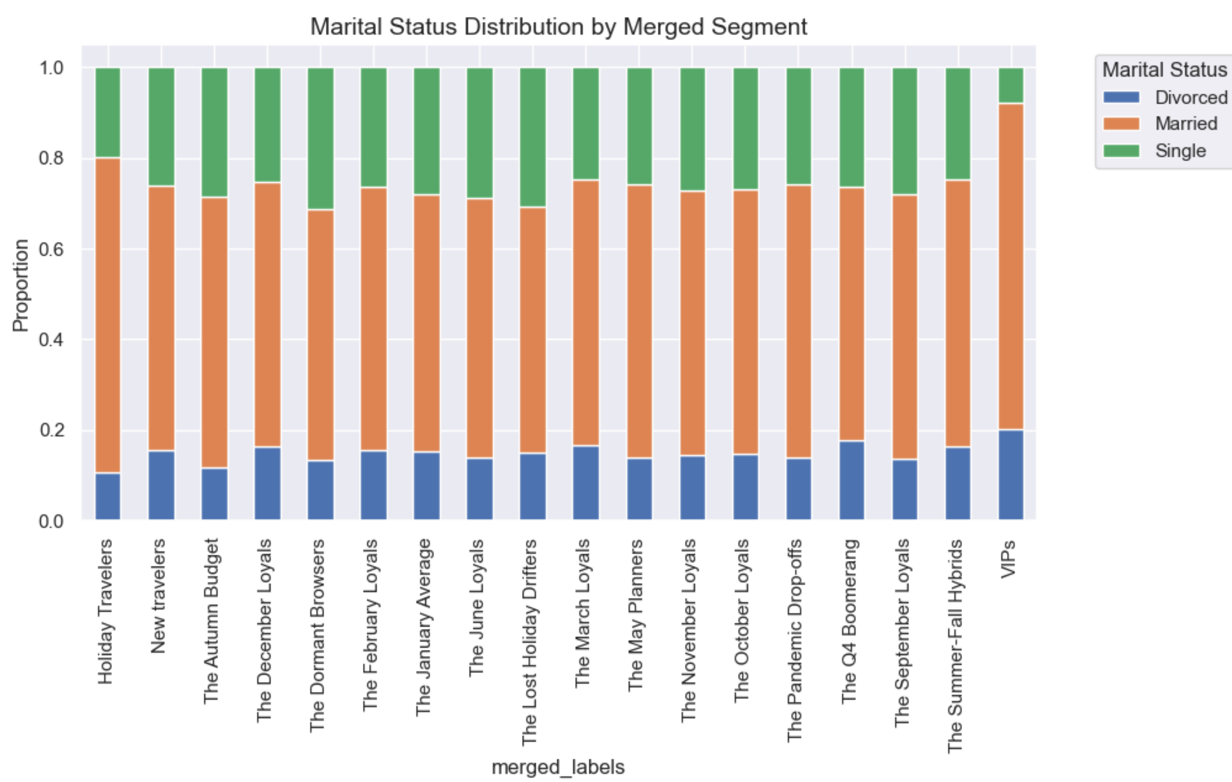

Province or State Distribution by Merged Segment
Proportion
0.0
0.2
0.4
0.6
0.8
1.0
merged_labels
Holiday Travelers
New travelers
The Autumn Budget
The December Loyals
The Dormant Browsers
The February Loyals
The January Average
The June Loyals
The Lost Holiday Drifters
The March Loyals
The May Planners
The November Loyals
The October Loyals
The Pandemic Drop-offs
The Q4 Boomerang
The September Loyals
The Summer-Fall Hybrids
VIPs
Province or State
Alberta
British Columbia
Manitoba
New Brunswick
Newfoundland
Nova Scotia
Ontario
Prince Edward Island
Quebec
Saskatchewan
Yukon

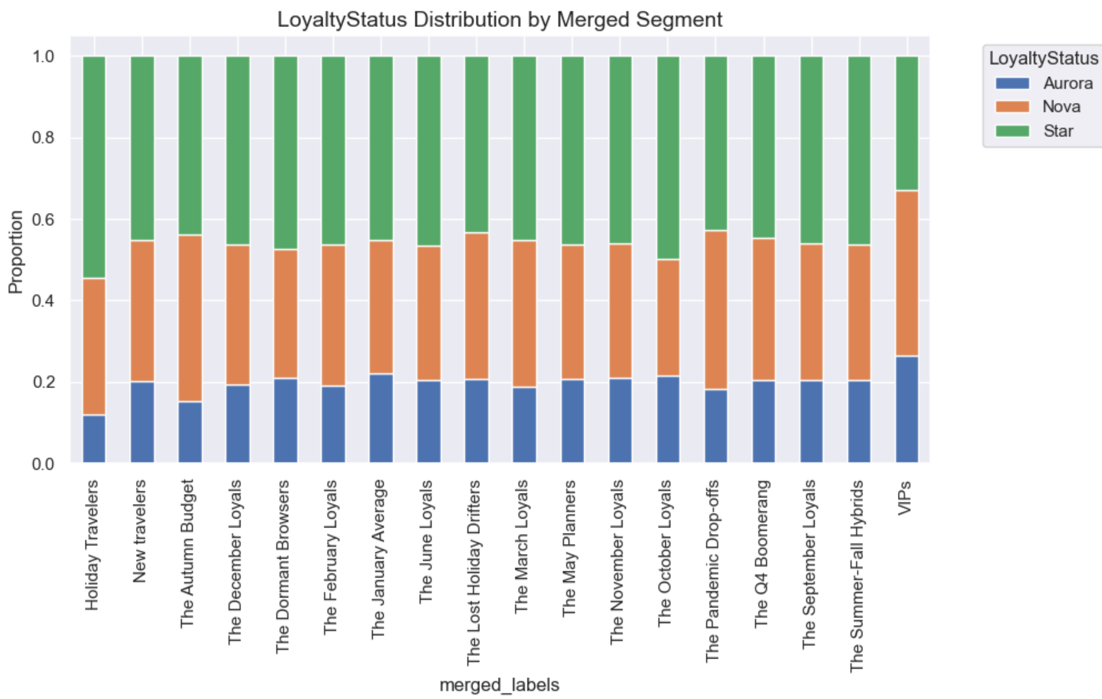

City Distribution by Merged Segment
Proportion
0.0
0.2
0.4
0.6
0.8
1.0
merged_labels
Holiday Travelers
New travelers
The Autumn Budget
The December Loyals
The Dormant Browsers
The February Loyals
The January Average
The June Loyals
The Lost Holiday Drifters
The March Loyals
The May Planners
The November Loyals
The October Loyals
The Pandemic Drop-offs
The Q4 Boomerang
The September Loyals
The Summer-Fall Hybrids
VIPs
City
Banff
Calgary
Charlottetown
Dawson Creek
Edmonton
Fredericton
Halifax
Hull
Kelowna
Kingston
London
Moncton
Montreal
Ottawa
Peace River
Quebec City
Regina
St. John's
Sudbury
Thunder Bay
Toronto
Tremblant
Trenton
Vancouver
Victoria
West Vancouver
Whistler
Whitehorse
Winnipeg

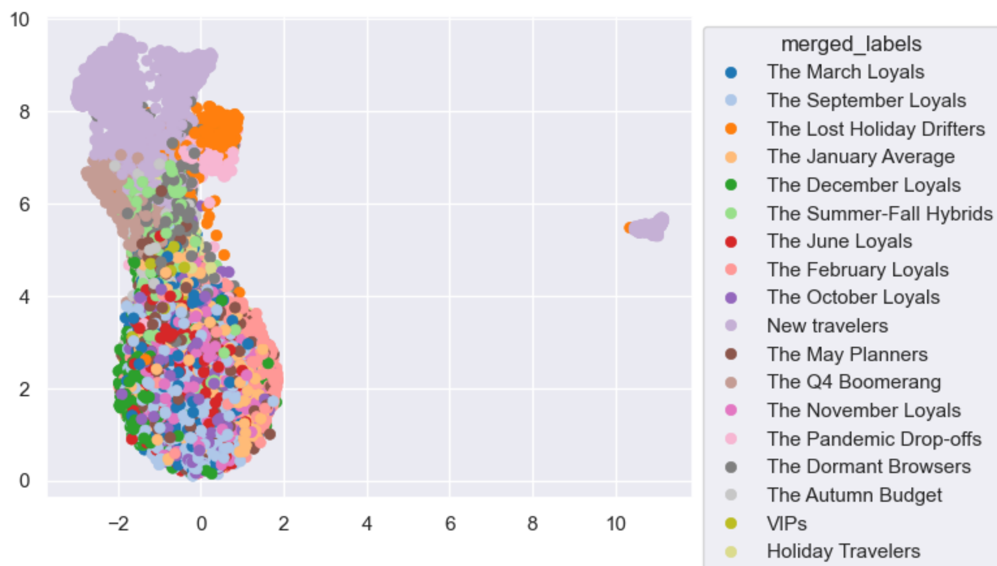

Gender Distribution by Merged Segment
Gender
female
male
Proportion
0.0
0.2
0.4
0.6
0.8
1.0
Holiday Travelers
New travelers
The Autumn Budget
The December Loyals
The Dormant Browsers
The February Loyals
The January Average
The June Loyals
The Lost Holiday Drifters
The March Loyals
The May Planners
The November Loyals
The October Loyals
The Pandemic Drop-offs
The Q4 Boomerang
The September Loyals
The Summer-Fall Hybrids
VIPs
merged_labels

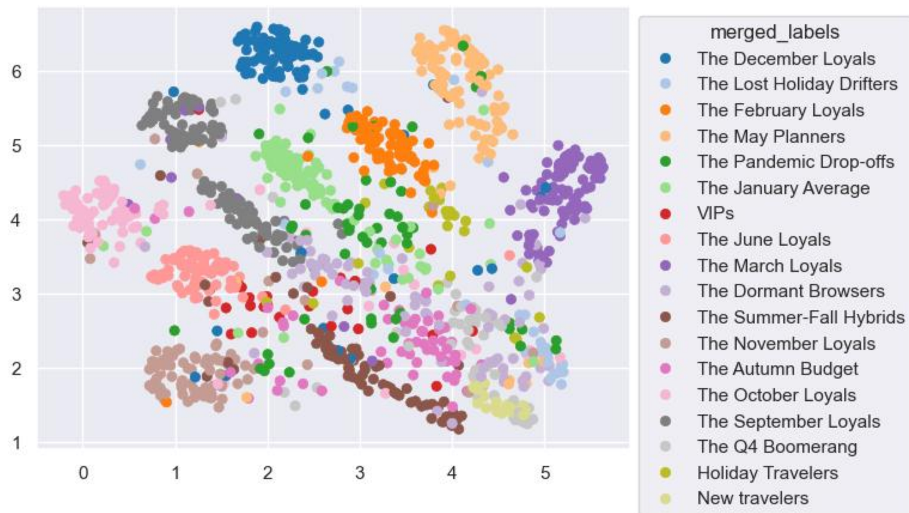

Education Distribution by Merged Segment
Education
Bachelor
College
Doctor
High School or Below
Master
Proportion
0.0
0.2
0.4
0.6
0.8
1.0
Holiday Travelers
New travelers
The Autumn Budget
The December Loyals
The Dormant Browsers
The February Loyals
The January Average
The June Loyals
The Lost Holiday Drifters
The March Loyals
The May Planners
The November Loyals
The October Loyals
The Pandemic Drop-offs
The Q4 Boomerang
The September Loyals
The Summer-Fall Hybrids
VIPs
merged_labels

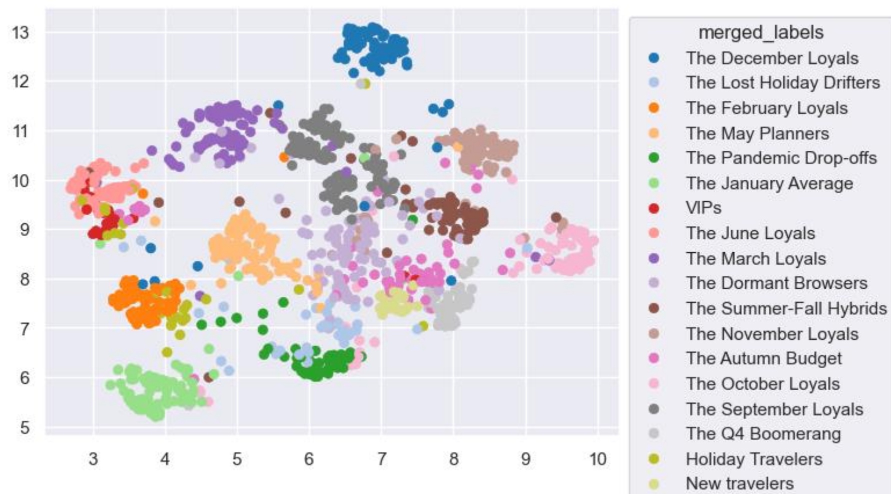

Location Code Distribution by Merged Segment
Proportion
0.0
0.2
0.4
0.6
0.8
1.0
Location Code
Rural
Suburban
Urban
Holiday Travelers
New travelers
The Autumn Budget
The December Loyals
The Dormant Browsers
The February Loyals
The January Average
The June Loyals
The Lost Holiday Drifters
The March Loyals
The May Planners
The November Loyals
The October Loyals
The Pandemic Drop-offs
The Q4 Boomerang
The September Loyals
The Summer-Fall Hybrids
VIPs
merged_labels


Marital Status Distribution by Merged Segment
Proportion
0.0
0.2
0.4
0.6
0.8
1.0
Marital Status
Divorced
Married
Single
Holiday Travelers
New travelers
The Autumn Budget
The December Loyals
The Dormant Browsers
The February Loyals
The January Average
The June Loyals
The Lost Holiday Drifters
The March Loyals
The May Planners
The November Loyals
The October Loyals
The Pandemic Drop-offs
The Q4 Boomerang
The September Loyals
The Summer-Fall Hybrids
VIPs
merged_labels


LoyaltyStatus Distribution by Merged Segment
Proportion
1.0
0.8
0.6
0.4
0.2
0.0
LoyaltyStatus
Aurora
Nova
Star
Holiday Travelers
New travelers
The Autumn Budget
The December Loyals
The Dormant Browsers
The February Loyals
The January Average
The June Loyals
The Lost Holiday Drifters
The March Loyals
The May Planners
The November Loyals
The October Loyals
The Pandemic Drop-offs
The Q4 Boomerang
The September Loyals
The Summer-Fall Hybrids
VIPs
merged_labels


10
8
6
4
2
0
−2
0
2
4
6
8
10
merged_labels
The March Loyals
The September Loyals
The Lost Holiday Drifters
The January Average
The December Loyals
The Summer-Fall Hybrids
The June Loyals
The February Loyals
The October Loyals
New travelers
The May Planners
The Q4 Boomerang
The November Loyals
The Pandemic Drop-offs
The Dormant Browsers
The Autumn Budget
VIPs
Holiday Travelers

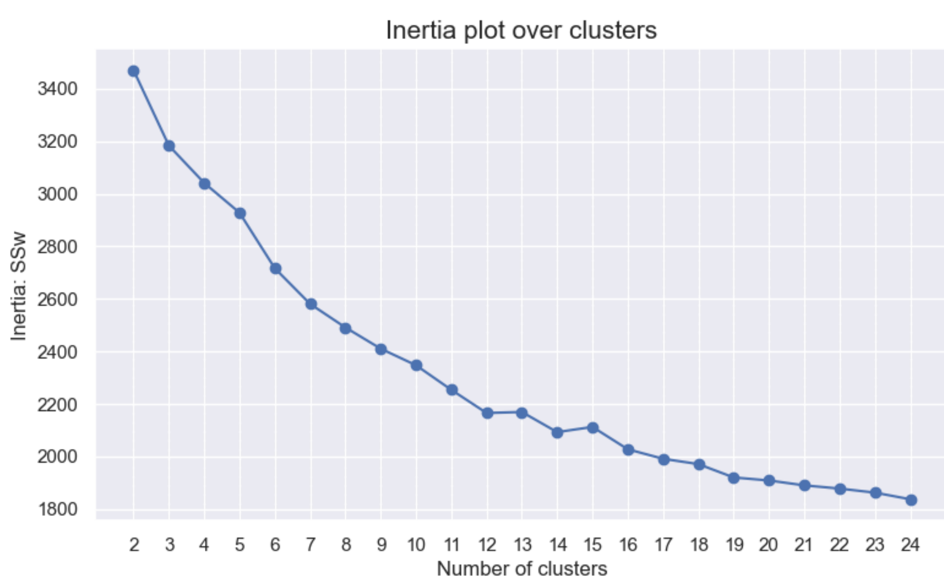




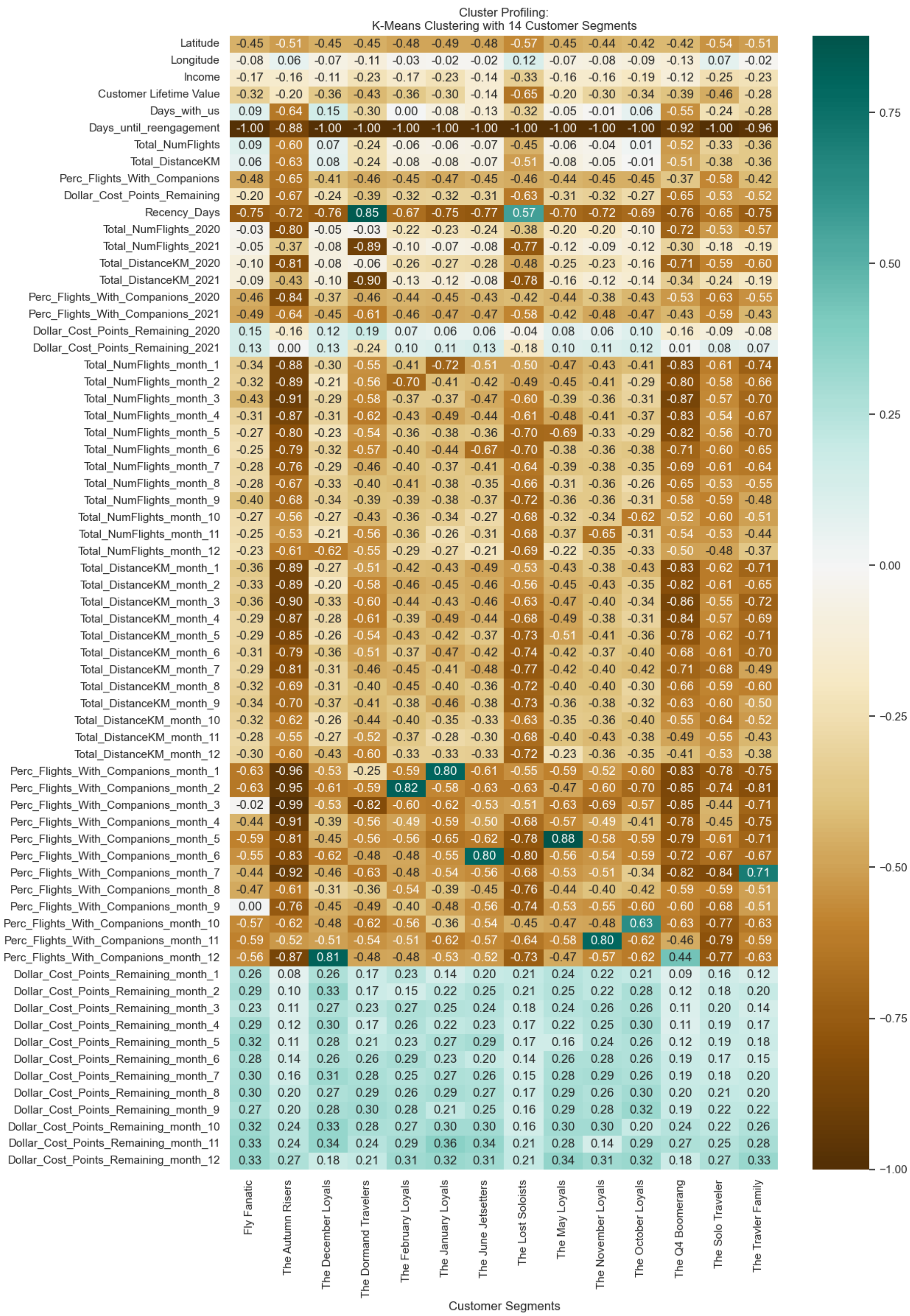
Annex39

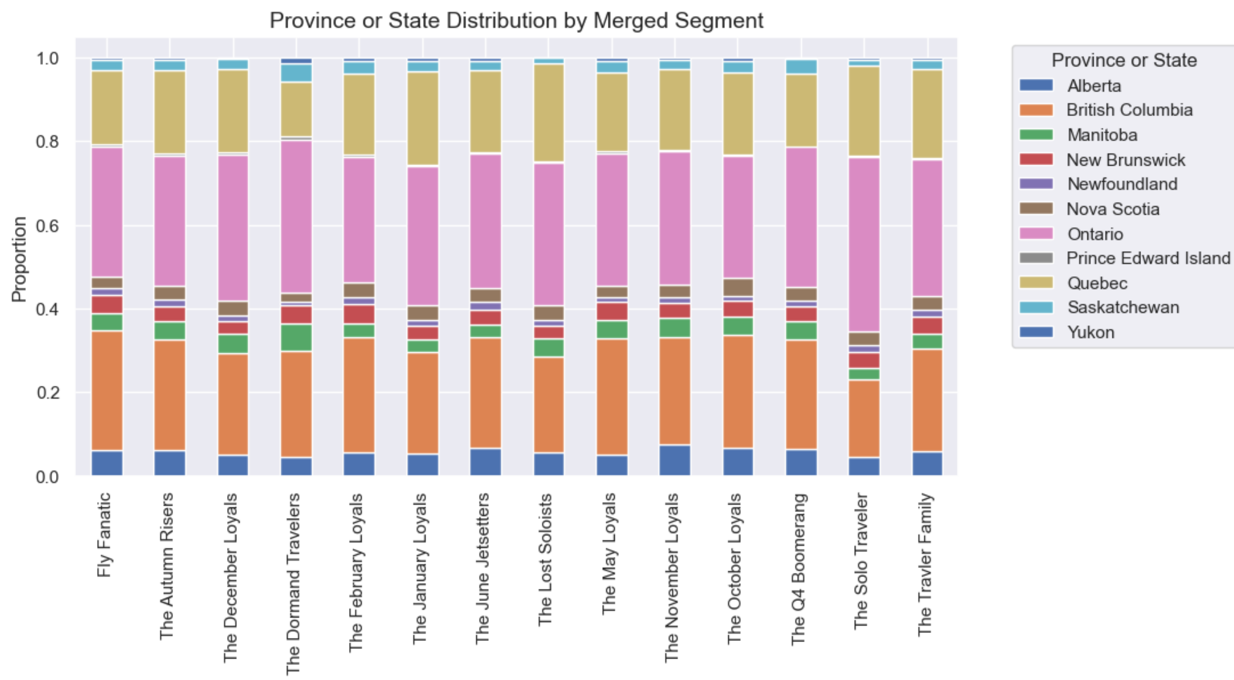
Number of Clusters	SSW (↓ Better)	SSB (↑ Better)	Silhouette Score (↑ Better)	R^2 (↑ Better)
2	3,470.81	477.11	0.148	0.121
3	3,183.76	764.16	0.119	0.194
4	3,041.65	906.26	0.080	0.230
5	2,928.36	1,019.55	0.080	0.258
6	2,716.76	1,231.16	0.099	0.312
7	2,580.83	1,367.09	0.111	0.346
8	2,491.09	1,456.82	0.125	0.369

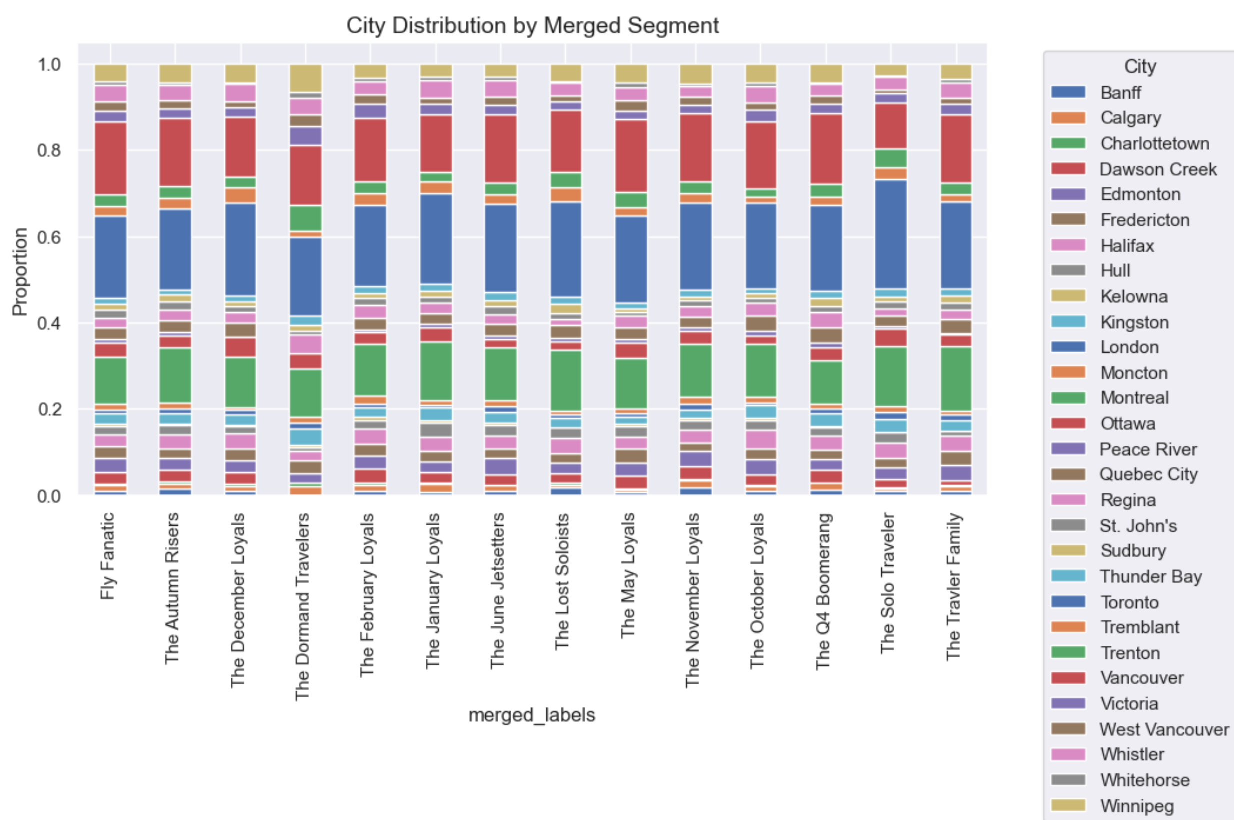
9	2,410.34	1,537.57	0.132	0.389
10	2,347.15	1,600.77	0.129	0.405
11	2,253.64	1,694.27	0.139	0.429
12	2,165.31	1,782.61	0.145	0.452
13	2,169.07	1,778.85	0.137	0.451
14	2,092.44	1,855.48	0.144	0.470
15	2,112.19	1,835.72	0.124	0.465
16	2,027.04	1,920.88	0.134	0.487
17	1,990.92	1,957.00	0.137	0.496
18	1,970.58	1,977.33	0.135	0.501
19	1,919.76	2,028.15	0.139	0.514
20	1,908.25	2,039.66	0.131	0.517
21	1,889.67	2,058.25	0.135	0.521
22	1,877.37	2,070.54	0.134	0.524
23	1,861.98	2,085.93	0.133	0.528
24	1,836.72	2,111.19	0.131	0.535

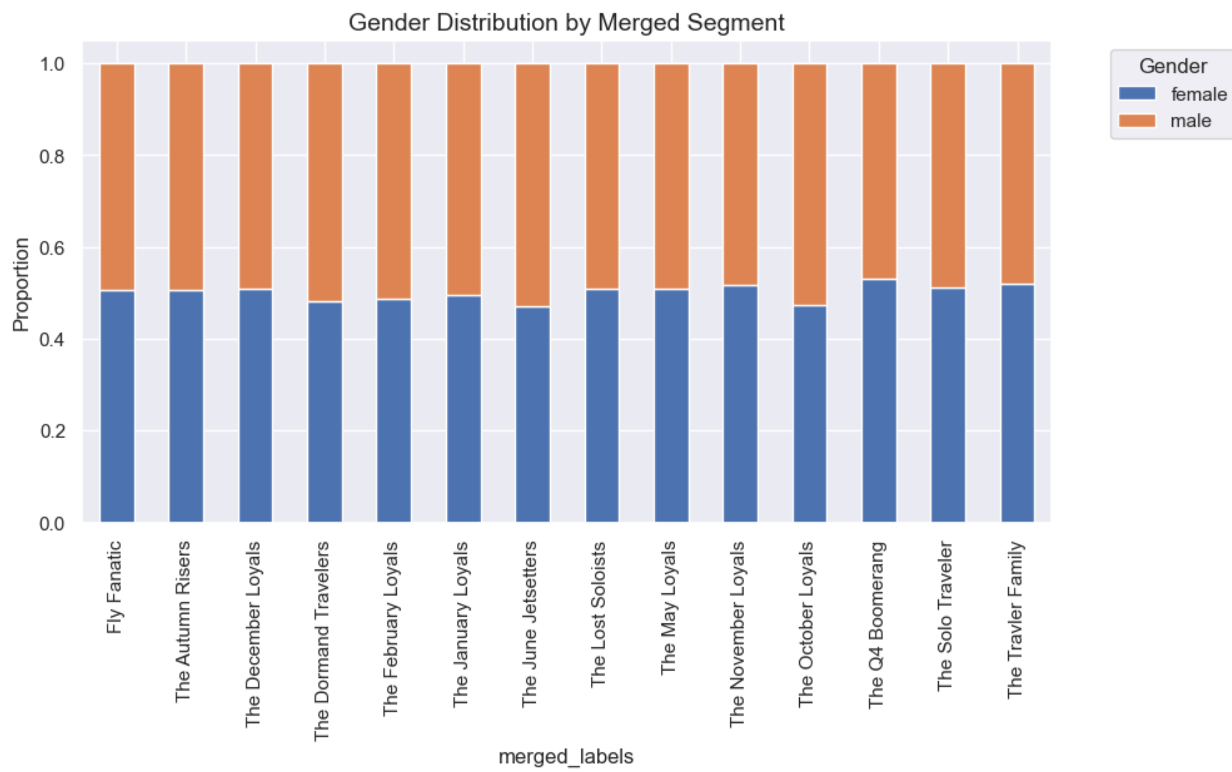


Annex40

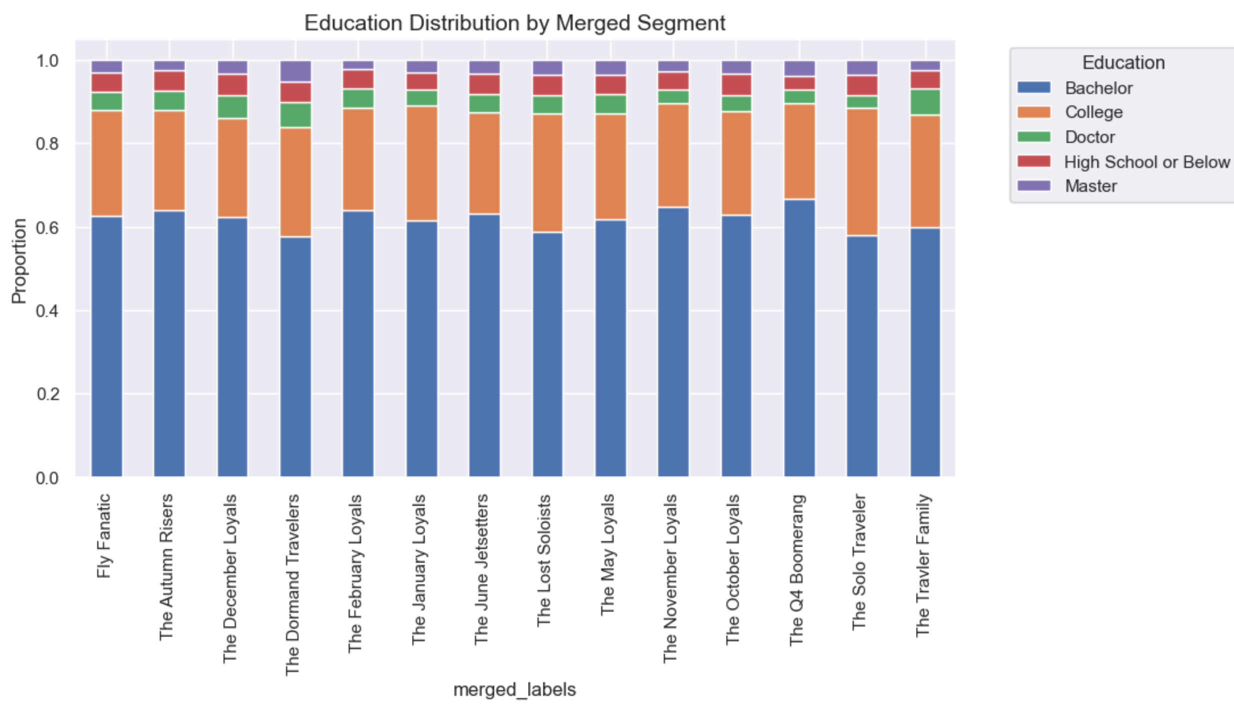

Cluster Profiling:
K-Means Clustering with 14 Customer Segments
Latitude
Longitude
Income
Customer Lifetime Value
Days_with_us
Days_until_reengagement
Total_NumFlights
Total_DistanceKM
Perc_Flights_With_Companions
Dollar_Cost_Points_Remaining
Recency_Days
Total_NumFlights_2020
Total_NumFlights_2021
Total_DistanceKM_2020
Total_DistanceKM_2021
Perc_Flights_With_Companions_2020
Perc_Flights_With_Companions_2021
Dollar_Cost_Points_Remaining_2020
Dollar_Cost_Points_Remaining_2021
Total_NumFlights_month_1
Total_NumFlights_month_2
Total_NumFlights_month_3
Total_NumFlights_month_4
Total_NumFlights_month_5
Total_NumFlights_month_6
Total_NumFlights_month_7
Total_NumFlights_month_8
Total_NumFlights_month_9
Total_NumFlights_month_10
Total_NumFlights_month_11
Total_NumFlights_month_12
Total_DistanceKM_month_1
Total_DistanceKM_month_2
Total_DistanceKM_month_3
Total_DistanceKM_month_4
Total_DistanceKM_month_5
Total_DistanceKM_month_6
Total_DistanceKM_month_7
Total_DistanceKM_month_8
Total_DistanceKM_month_9
Total_DistanceKM_month_10
Total_DistanceKM_month_11
Total_DistanceKM_month_12
Perc_Flights_With_Companions_month_1
Perc_Flights_With_Companions_month_2
Perc_Flights_With_Companions_month_3
Perc_Flights_With_Companions_month_4
Perc_Flights_With_Companions_month_5
Perc_Flights_With_Companions_month_6
Perc_Flights_With_Companions_month_7
Perc_Flights_With_Companions_month_8
Perc_Flights_With_Companions_month_9
Perc_Flights_With_Companions_month_10
Perc_Flights_With_Companions_month_11
Perc_Flights_With_Companions_month_12
Dollar_Cost_Points_Remaining_month_1
Dollar_Cost_Points_Remaining_month_2
Dollar_Cost_Points_Remaining_month_3
Dollar_Cost_Points_Remaining_month_4
Dollar_Cost_Points_Remaining_month_5
Dollar_Cost_Points_Remaining_month_6
Dollar_Cost_Points_Remaining_month_7
Dollar_Cost_Points_Remaining_month_8
Dollar_Cost_Points_Remaining_month_9
Dollar_Cost_Points_Remaining_month_10
Dollar_Cost_Points_Remaining_month_11
Dollar_Cost_Points_Remaining_month_12
Fly Fanatic
The Autumn Risers
The December Loyals
The Dormand Travelers
The February Loyals
The January Loyals
The June Jetsetters
The Lost Soloists
The May Loyals
The November Loyals
The October Loyals
The Q4 Boomerang
The Solo Traveler
The Travler Family
Customer Segments
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00


Province or State Distribution by Merged Segment
Proportion
0.0
0.2
0.4
0.6
0.8
1.0
Fly Fanatic
The Autumn Risers
The December Loyals
The Dormand Travelers
The February Loyals
The January Loyals
The June Jetsetters
The Lost Soloists
The May Loyals
The November Loyals
The October Loyals
The Q4 Boomerang
The Solo Traveler
The Travler Family
merged_labels
Province or State
Alberta
British Columbia
Manitoba
New Brunswick
Newfoundland
Nova Scotia
Ontario
Prince Edward Island
Quebec
Saskatchewan
Yukon

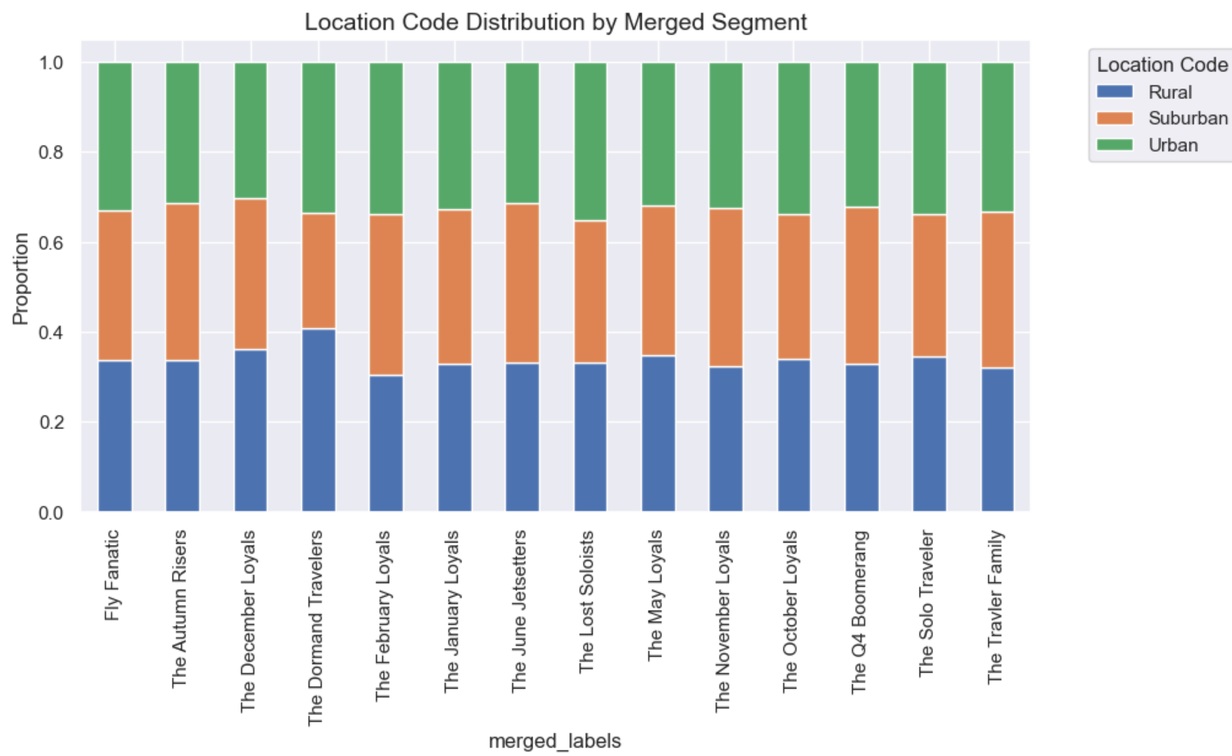

City Distribution by Merged Segment
Proportion
0.0
0.2
0.4
0.6
0.8
1.0
Fly Fanatic
The Autumn Risers
The December Loyals
The Dormand Travelers
The February Loyals
The January Loyals
The June Jetsetters
The Lost Soloists
The May Loyals
The November Loyals
The October Loyals
The Q4 Boomerang
The Solo Traveler
The Travler Family
merged_labels
City
Banff
Calgary
Charlottetown
Dawson Creek
Edmonton
Fredericton
Halifax
Hull
Kelowna
Kingston
London
Moncton
Montreal
Ottawa
Peace River
Quebec City
Regina
St. John's
Sudbury
Thunder Bay
Toronto
Tremblant
Trenton
Vancouver
Victoria
West Vancouver
Whistler
Whitehorse
Winnipeg

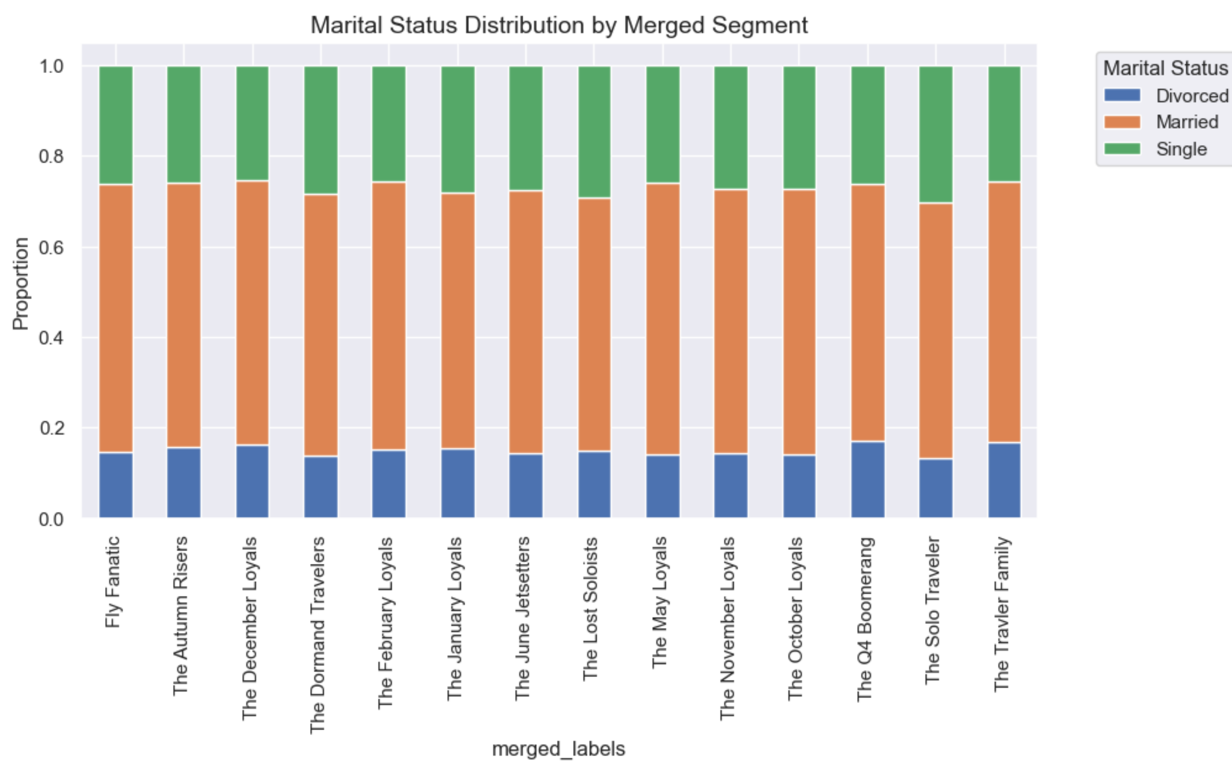


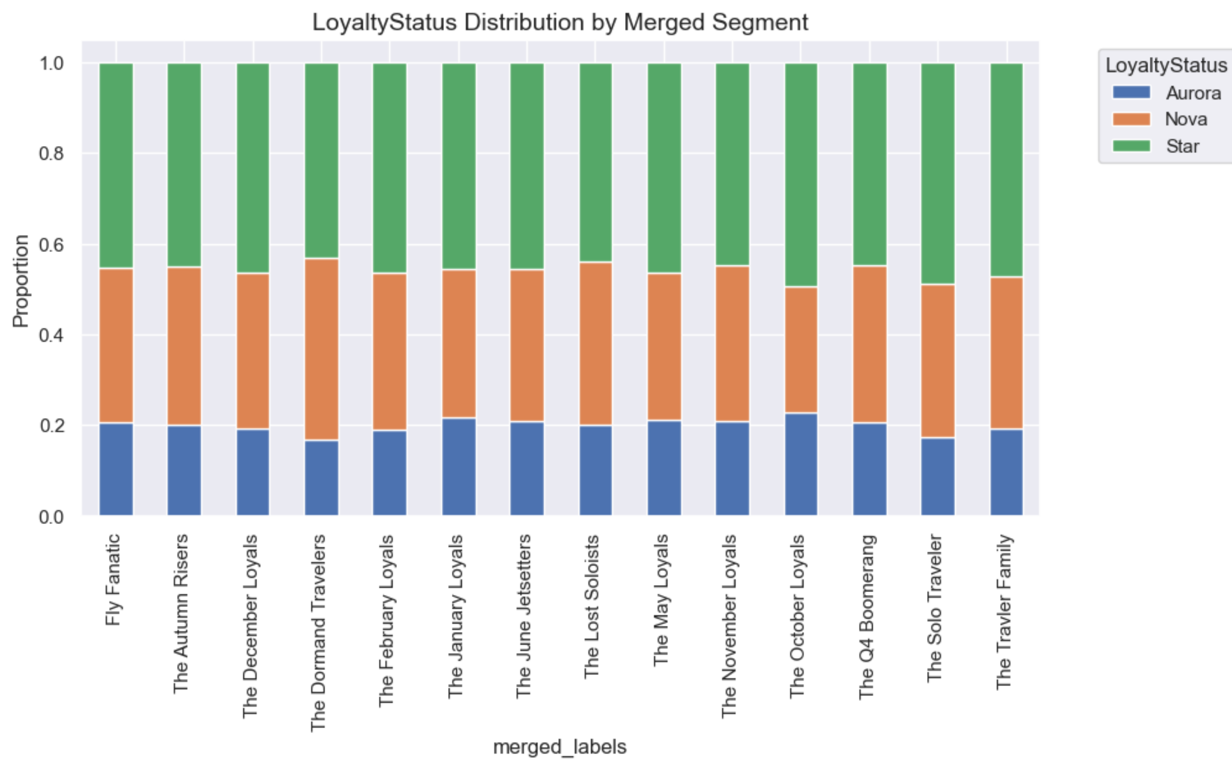
Gender Distribution by Merged Segment
Gender
female
male
Proportion
0.0
0.2
0.4
0.6
0.8
1.0
Fly Fanatic
The Autumn Risers
The December Loyals
The Dormand Travelers
The February Loyals
The January Loyals
The June Jetsetters
The Lost Soloists
The May Loyals
The November Loyals
The October Loyals
The Q4 Boomerang
The Solo Traveler
The Travler Family
merged_labels

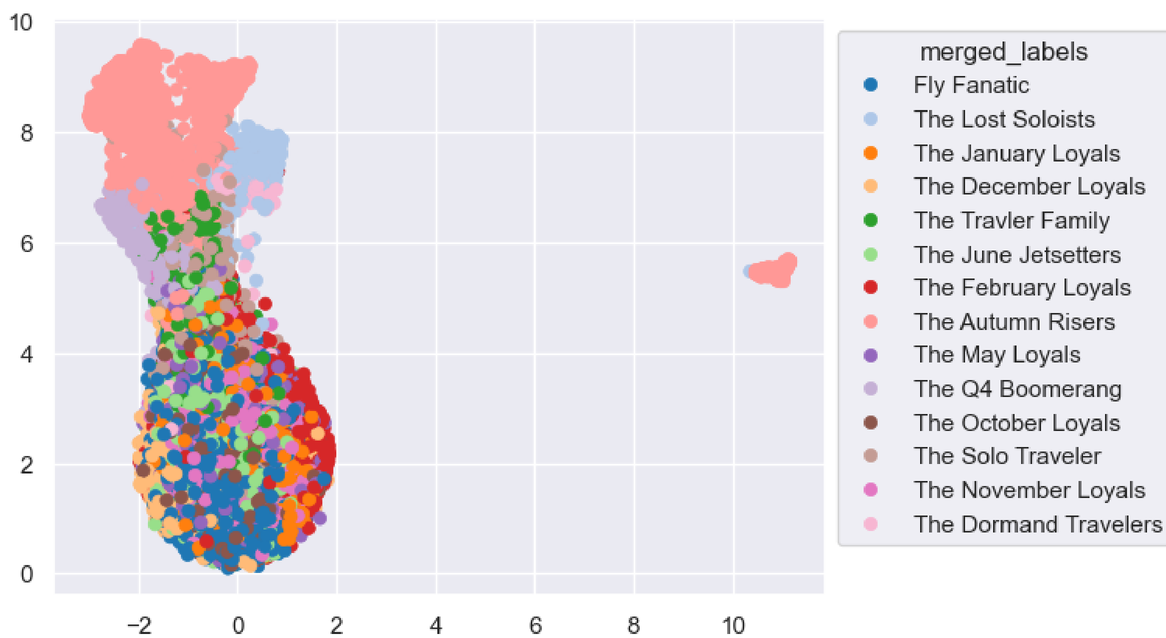


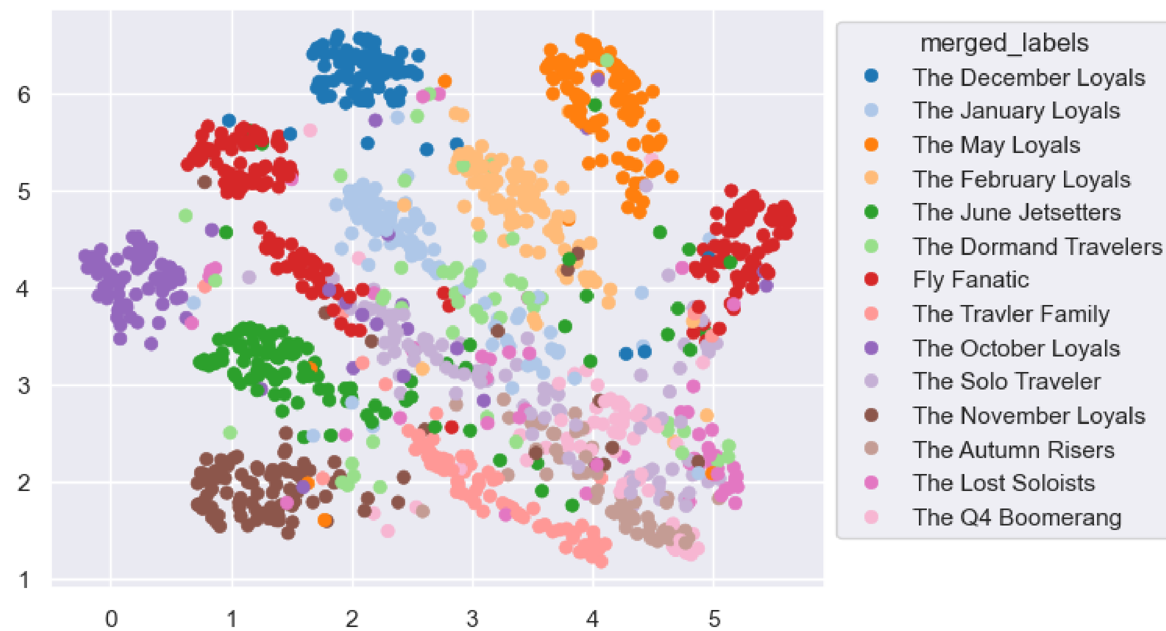
Education Distribution by Merged Segment
Education
Bachelor
College
Doctor
High School or Below
Master
Proportion
0.0
0.2
0.4
0.6
0.8
1.0
Fly Fanatic
The Autumn Risers
The December Loyals
The Dormand Travelers
The February Loyals
The January Loyals
The June Jetsetters
The Lost Soloists
The May Loyals
The November Loyals
The October Loyals
The Q4 Boomerang
The Solo Traveler
The Travler Family
merged_labels


Location Code Distribution by Merged Segment
Location Code
Rural
Suburban
Urban
Proportion
0.0
0.2
0.4
0.6
0.8
1.0
Fly Fanatic
The Autumn Risers
The December Loyals
The Dormand Travelers
The February Loyals
The January Loyals
The June Jetsetters
The Lost Soloists
The May Loyals
The November Loyals
The October Loyals
The Q4 Boomerang
The Solo Traveler
The Travler Family
merged_labels

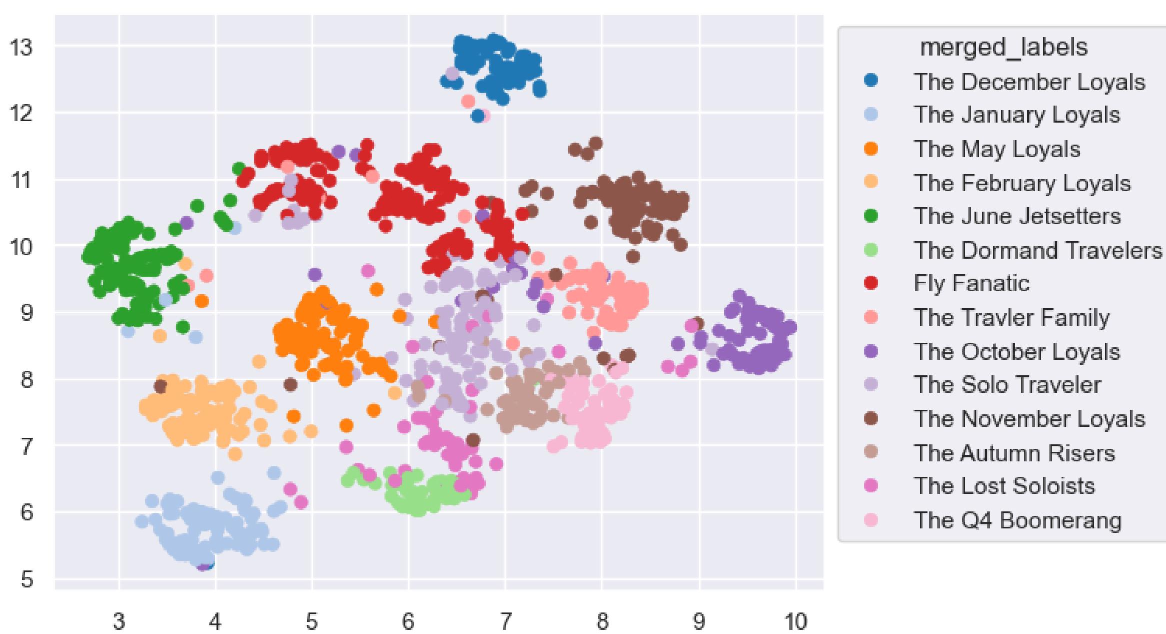

Marital Status Distribution by Merged Segment
Marital Status
Divorced
Married
Single
Proportion
0.0
0.2
0.4
0.6
0.8
1.0
Fly Fanatic
The Autumn Risers
The December Loyals
The Dormand Travelers
The February Loyals
The January Loyals
The June Jetsetters
The Lost Soloists
The May Loyals
The November Loyals
The October Loyals
The Q4 Boomerang
The Solo Traveler
The Travler Family
merged_labels


LoyaltyStatus Distribution by Merged Segment
Proportion
merged_labels
LoyaltyStatus
Aurora
Nova
Star
Fly Fanatic
The Autumn Risers
The December Loyals
The Dormand Travelers
The February Loyals
The January Loyals
The June Jetsetters
The Lost Soloists
The May Loyals
The November Loyals
The October Loyals
The Q4 Boomerang
The Solo Traveler
The Travler Family


merged_labels
Fly Fanatic
The Lost Soloists
The January Loyals
The December Loyals
The Travler Family
The June Jetsetters
The February Loyals
The Autumn Risers
The May Loyals
The Q4 Boomerang
The October Loyals
The Solo Traveler
The November Loyals
The Dormand Travelers

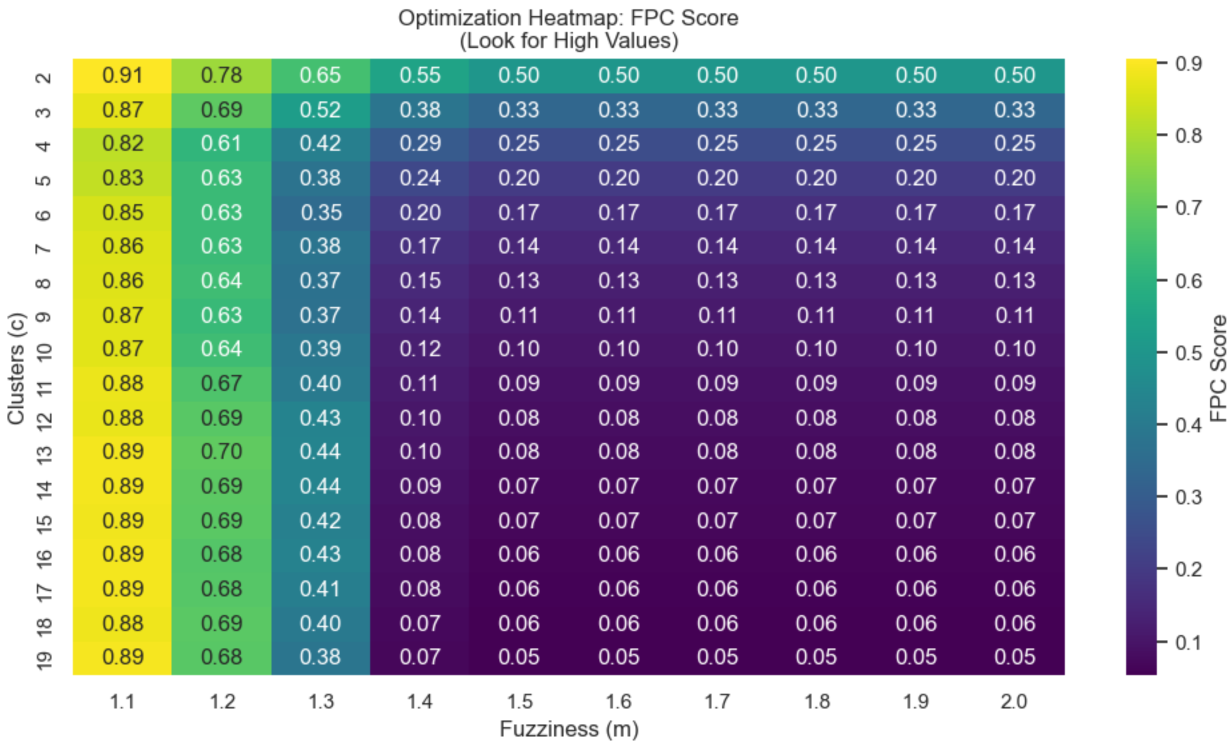


merged_labels
The December Loyals
The January Loyals
The May Loyals
The February Loyals
The June Jetsetters
The Dormand Travelers
Fly Fanatic
The Travler Family
The October Loyals
The Solo Traveler
The November Loyals
The Autumn Risers
The Lost Soloists
The Q4 Boomerang

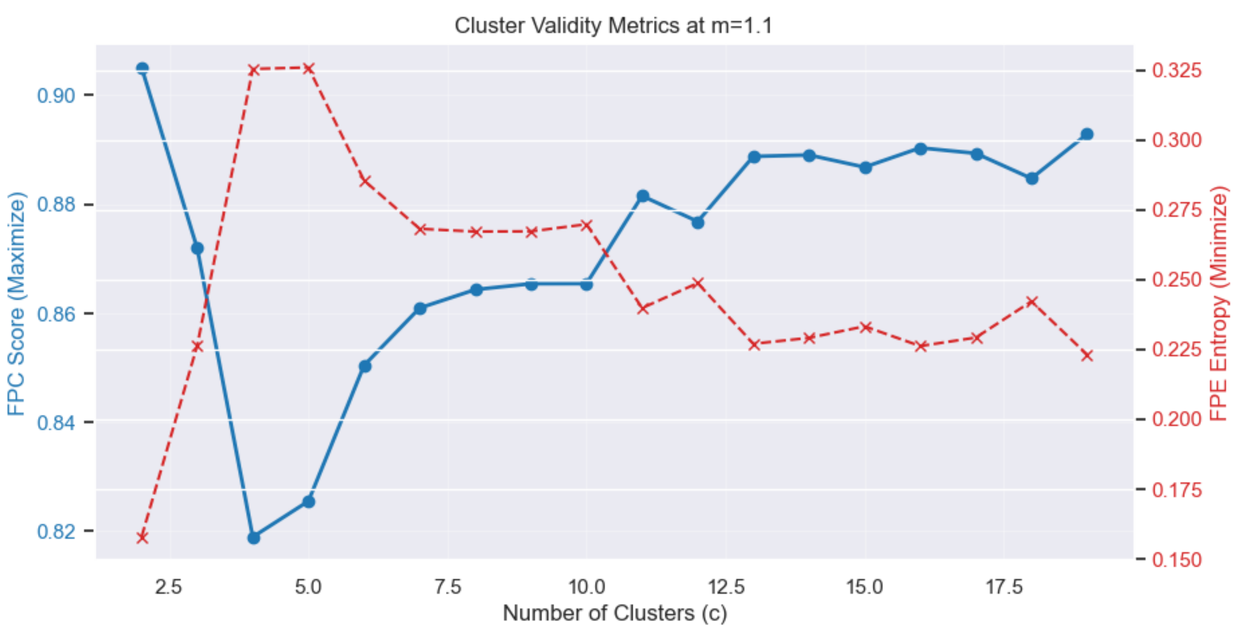


merged_labels
The December Loyals
The January Loyals
The May Loyals
The February Loyals
The June Jetsetters
The Dormand Travelers
Fly Fanatic
The Travler Family
The October Loyals
The Solo Traveler
The November Loyals
The Autumn Risers
The Lost Soloists
The Q4 Boomerang

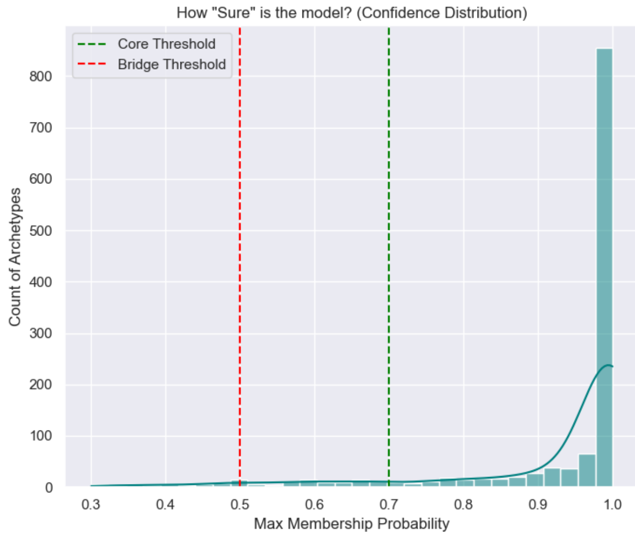
Annex41



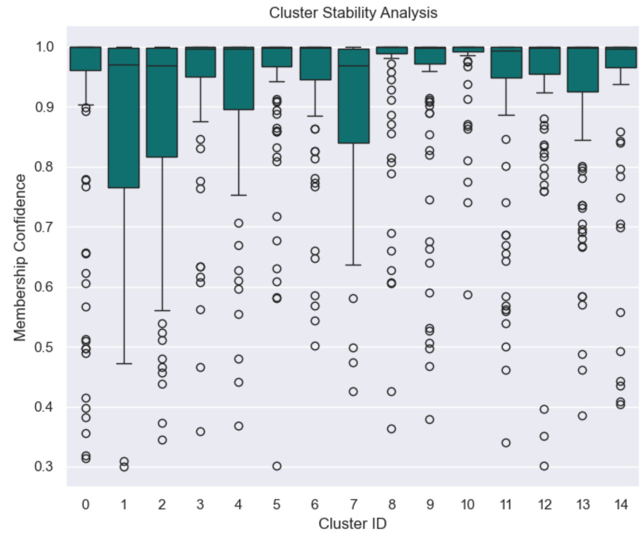
Annex42



Annex43

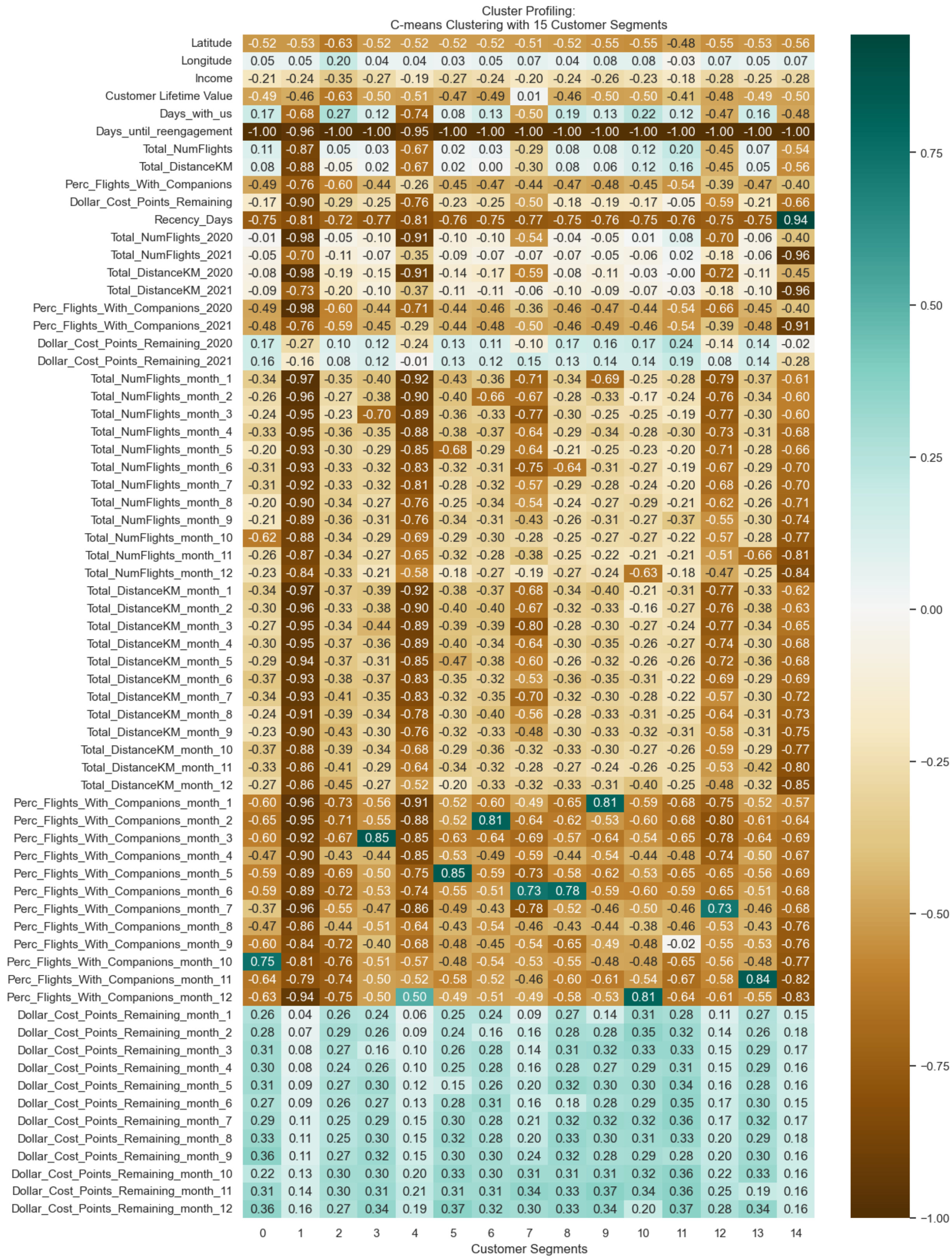


How "Sure" is the model? (Confidence Distribution)
Core Threshold
Bridge Threshold
Count of Archetypes
Max Membership Probability

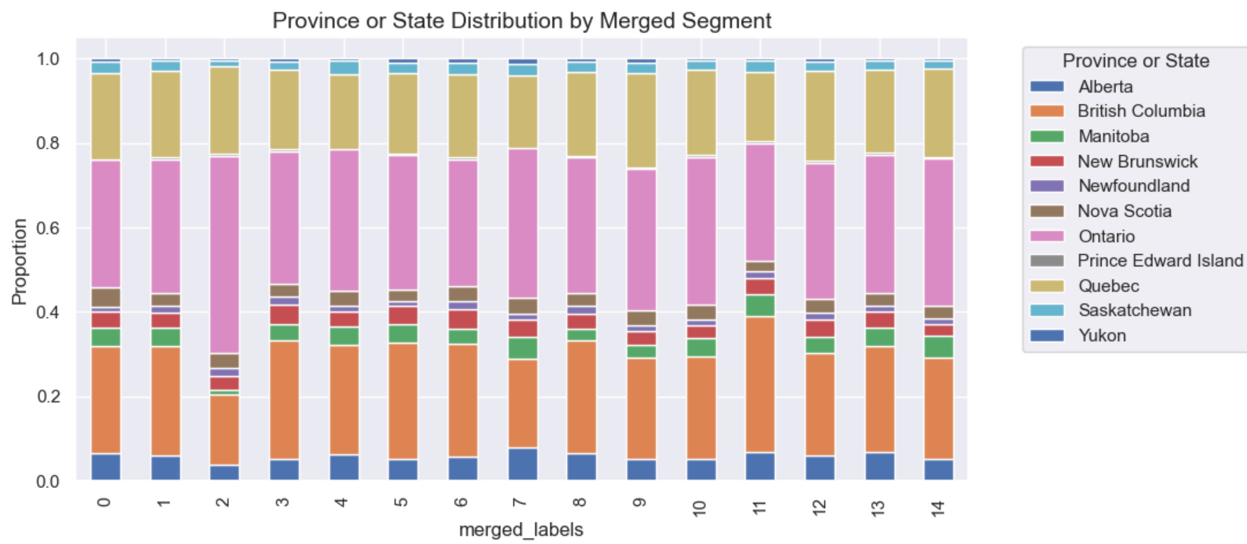


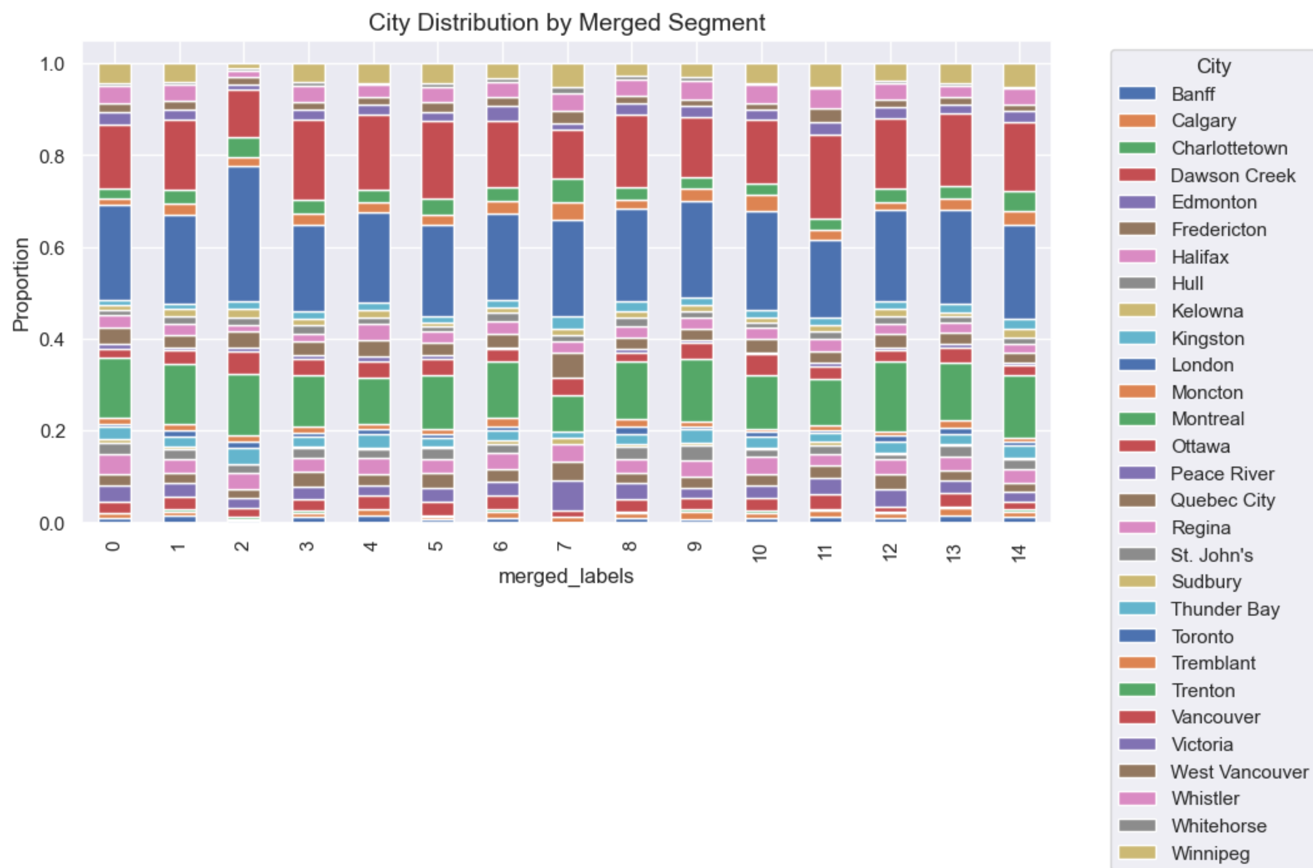
Cluster Stability Analysis
Membership Confidence
Cluster ID

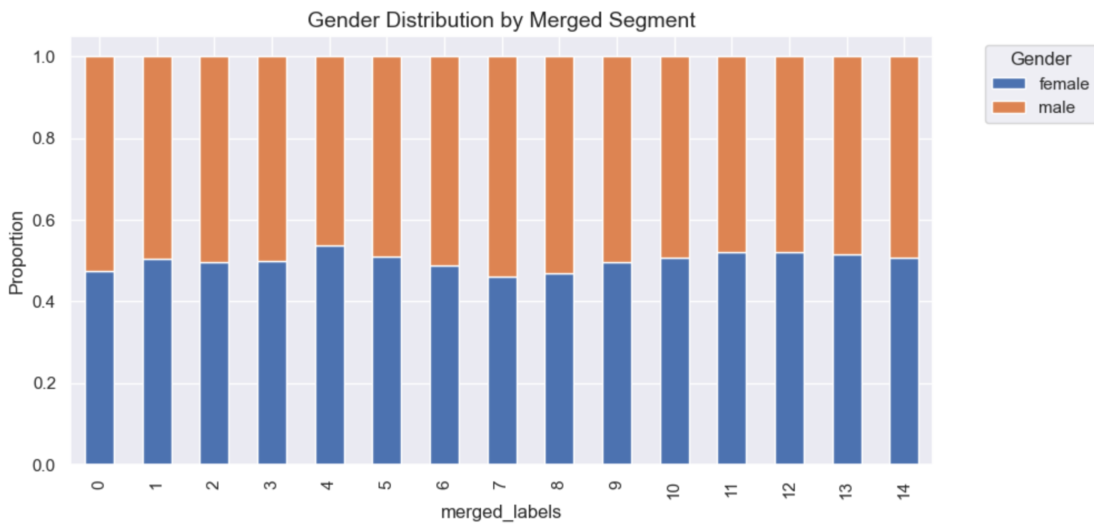
Annex44

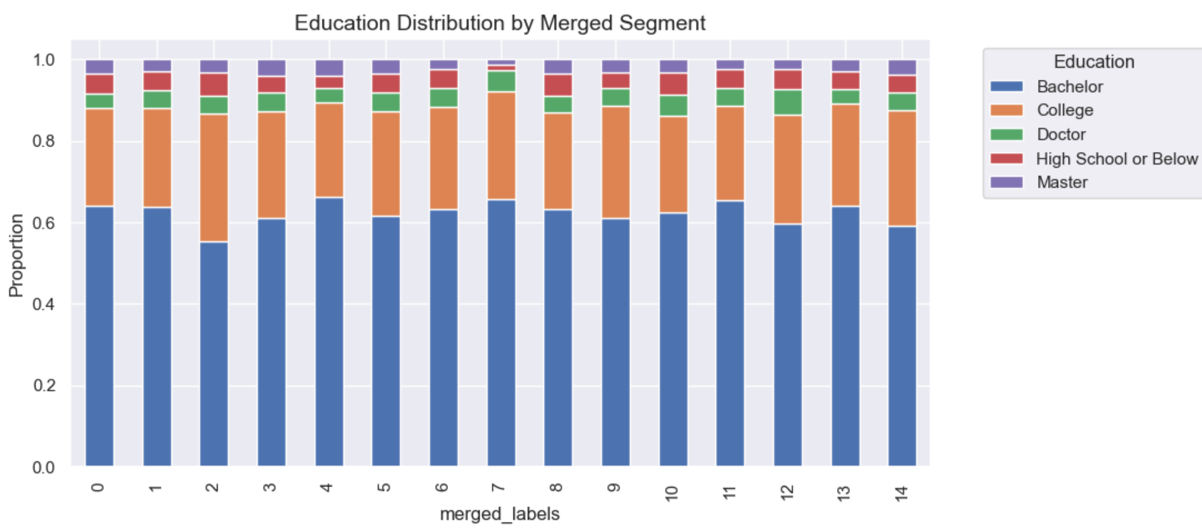


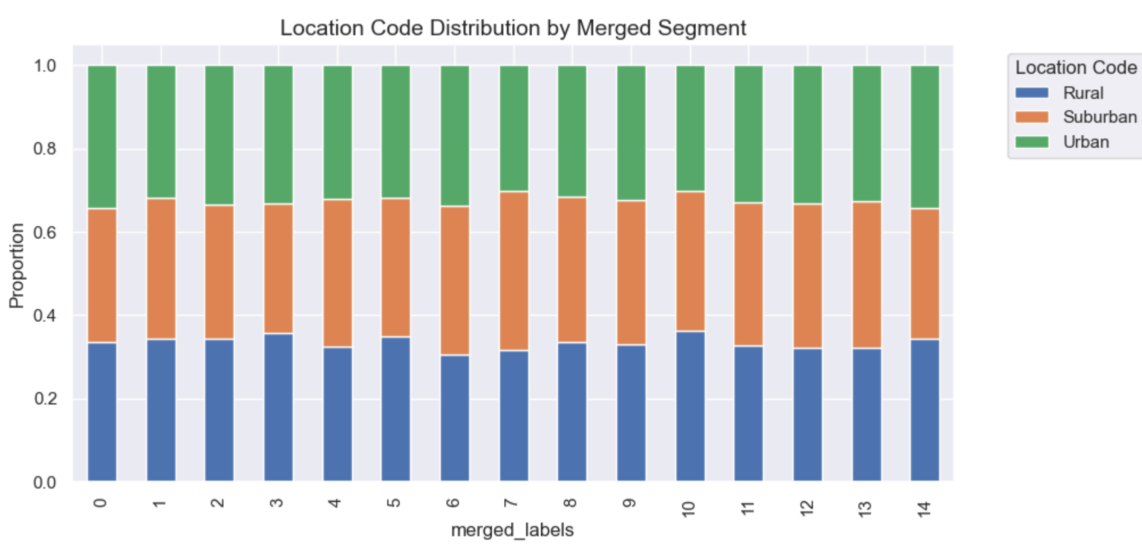
Cluster Profiling:
C-means Clustering with 15 Customer Segments
Latitude
Longitude
Income
Customer Lifetime Value
Days_with_us
Days_until_reengagement
Total_NumFlights
Total_DistanceKM
Perc_Flights_With_Companions
Dollar_Cost_Points_Remaining
Recency_Days
Total_NumFlights_2020
Total_NumFlights_2021
Total_DistanceKM_2020
Total_DistanceKM_2021
Perc_Flights_With_Companions_2020
Perc_Flights_With_Companions_2021
Dollar_Cost_Points_Remaining_2020
Dollar_Cost_Points_Remaining_2021
Total_NumFlights_month_1
Total_NumFlights_month_2
Total_NumFlights_month_3
Total_NumFlights_month_4
Total_NumFlights_month_5
Total_NumFlights_month_6
Total_NumFlights_month_7
Total_NumFlights_month_8
Total_NumFlights_month_9
Total_NumFlights_month_10
Total_NumFlights_month_11
Total_NumFlights_month_12
Total_DistanceKM_month_1
Total_DistanceKM_month_2
Total_DistanceKM_month_3
Total_DistanceKM_month_4
Total_DistanceKM_month_5
Total_DistanceKM_month_6
Total_DistanceKM_month_7
Total_DistanceKM_month_8
Total_DistanceKM_month_9
Total_DistanceKM_month_10
Total_DistanceKM_month_11
Total_DistanceKM_month_12
Perc_Flights_With_Companions_month_1
Perc_Flights_With_Companions_month_2
Perc_Flights_With_Companions_month_3
Perc_Flights_With_Companions_month_4
Perc_Flights_With_Companions_month_5
Perc_Flights_With_Companions_month_6
Perc_Flights_With_Companions_month_7
Perc_Flights_With_Companions_month_8
Perc_Flights_With_Companions_month_9
Perc_Flights_With_Companions_month_10
Perc_Flights_With_Companions_month_11
Perc_Flights_With_Companions_month_12
Dollar_Cost_Points_Remaining_month_1
Dollar_Cost_Points_Remaining_month_2
Dollar_Cost_Points_Remaining_month_3
Dollar_Cost_Points_Remaining_month_4
Dollar_Cost_Points_Remaining_month_5
Dollar_Cost_Points_Remaining_month_6
Dollar_Cost_Points_Remaining_month_7
Dollar_Cost_Points_Remaining_month_8
Dollar_Cost_Points_Remaining_month_9
Dollar_Cost_Points_Remaining_month_10
Dollar_Cost_Points_Remaining_month_11
Dollar_Cost_Points_Remaining_month_12
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14
Customer Segments
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

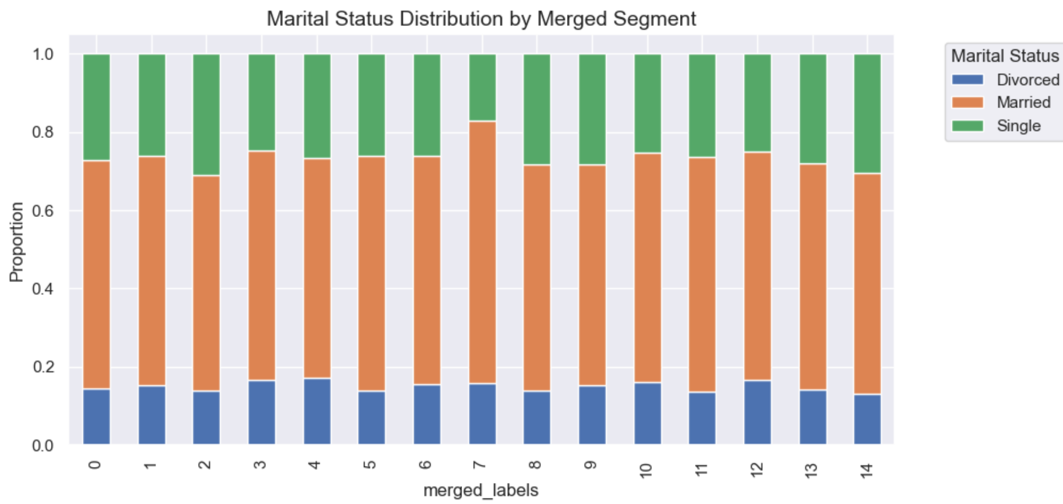

Province or State Distribution by Merged Segment
Proportion
merged_labels
Province or State
Alberta
British Columbia
Manitoba
New Brunswick
Newfoundland
Nova Scotia
Ontario
Prince Edward Island
Quebec
Saskatchewan
Yukon


City Distribution by Merged Segment
Proportion
merged_labels
City
Banff
Calgary
Charlottetown
Dawson Creek
Edmonton
Fredericton
Halifax
Hull
Kelowna
Kingston
London
Moncton
Montreal
Ottawa
Peace River
Quebec City
Regina
St. John's
Sudbury
Thunder Bay
Toronto
Tremblant
Trenton
Vancouver
Victoria
West Vancouver
Whistler
Whitehorse
Winnipeg

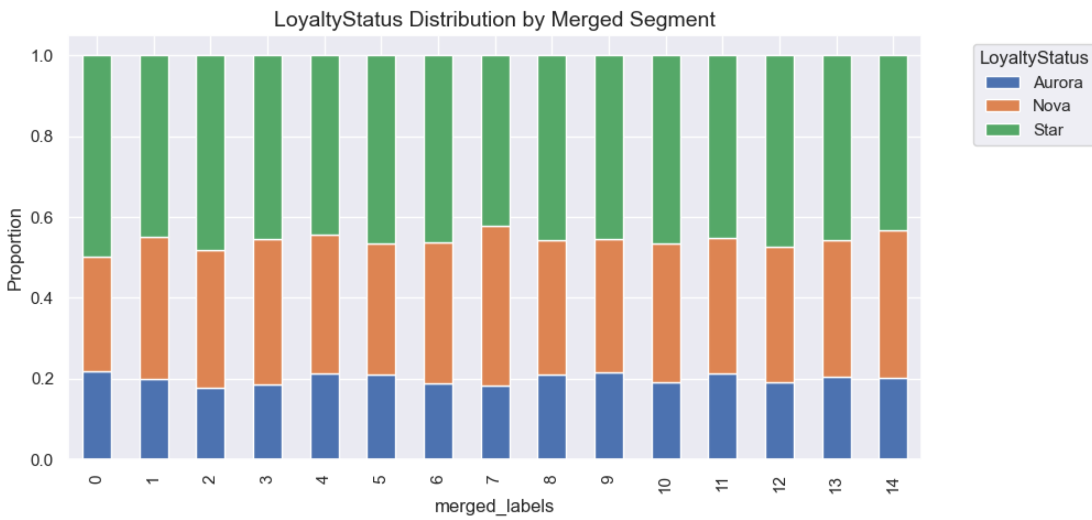

Gender Distribution by Merged Segment
Gender
female
male
Proportion
0.0
0.2
0.4
0.6
0.8
1.0
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
merged_labels


Education Distribution by Merged Segment
Education
Bachelor
College
Doctor
High School or Below
Master
Proportion
0.0
0.2
0.4
0.6
0.8
1.0
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
merged_labels

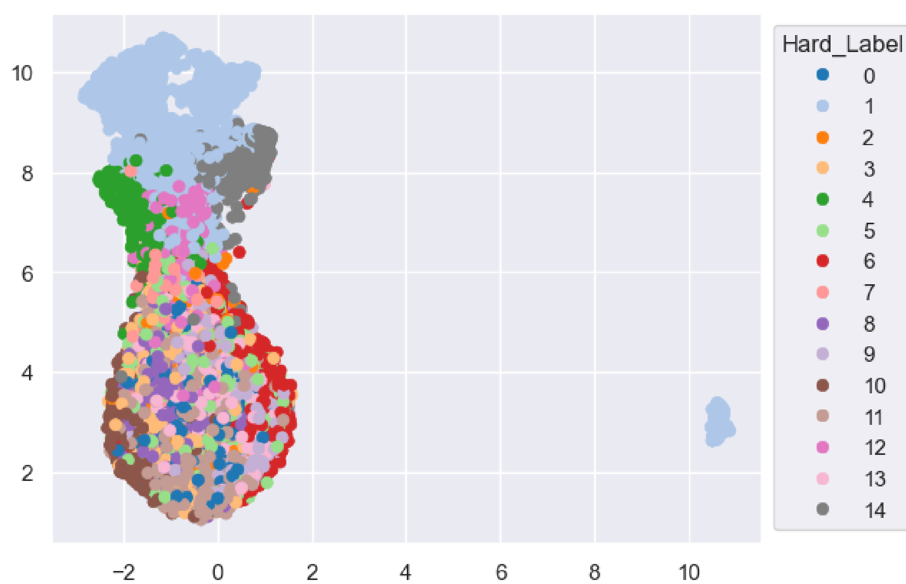

Location Code Distribution by Merged Segment
Location Code
Rural
Suburban
Urban
Proportion
0.0
0.2
0.4
0.6
0.8
1.0
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
merged_labels



Marital Status Distribution by Merged Segment
Proportion
merged_labels
Marital Status
Divorced
Married
Single
0.0
0.2
0.4
0.6
0.8
1.0
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14



LoyaltyStatus Distribution by Merged Segment
Proportion
merged_labels
LoyaltyStatus
Aurora
Nova
Star
0.0
0.2
0.4
0.6
0.8
1.0
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14



Hard_Label
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
−2
0
2
4
6
8
10
2
4
6
8
10

Annex45

Method	Number of Clusters	R^2 (↑ Better)	Silhouette Score (↑ Better)	SSW (↓ Better)	SSB (↑ Better)
(Autoencoder) C-means	15	0.309	0.065	166,360.93	74,492.63
(Autoencoder) K-Means	14	0.294	0.065	175,141.59	73,053.70
(Autoencoder) Hierarchical	**18**	**0.308**	**0.065**	**171,770.31**	**76,424.98**
Hierarchical	15	0.289	0.083	176,506.06	71,689.23
K-Means	16	0.304	0.071	172,719.61	75,475.68

Annex46